Fragile Learning From Others*

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Abstract

Behavioral biases in decision-making are widespread and often persist despite experience and transparent feedback. This paper examines whether exposure to others' optimal decisions can correct initial misconceptions and facilitate learning in an environment where departures from the theoretical benchmark arise from neglecting an informative signal in a worker hiring task. Using a laboratory experiment, I show that such exposure substantially improves the quality of one's own decisions. The analysis of the underlying mechanisms shows that this improvement is not driven by mechanical imitation, as responses to exposure are *asymmetric* in the quality of observed choices, pointing instead to selective adoption of observed behavior. Adopting others' decisions selectively, however, need not reflect an understanding of *why* those decisions are optimal, so I further evaluate the transfer of these learning gains to a modified environment once exposure ends. The main result of the paper is that learning from others is *fragile*: the improvements do not survive a change in the environment's primitives, and the acquired gains from exposure dissipate entirely. Comparing these effects with those of explicit guidance further underscores the limits of social exposure as an effective policy tool. These findings highlight the dual role of observational learning: while it can enhance the incidence of optimal behavior, the improvements it generates often fail to generalize beyond the observed context.

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JEL classification: C91, D83, D91, J71.

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1 Introduction

A large body of work in behavioral economics documents systematic biases in decision-making that arise because agents rely on simple yet incorrect representations of the environment they face.[1] An important question is whether such biases correct with experience: one might expect them to fade as agents repeatedly face the same problem and receive feedback informative of optimal behavior. However, a growing literature on learning under misspecification challenges this expectation, showing that initial misconceptions lead to persistent suboptimal behavior even when corrective evidence is abundant.[2] Notably, the corrective evidence in these settings arrives through individual experience, with agents adjusting in response to their own outcomes.

In practice, decisions are rarely shaped by individual experience alone: people also observe how others behave when facing the same problems. New employees observe how experienced colleagues approach the same tasks, managers watch how their peers screen the same pool of applicants, and farmers learn from how their neighbors cultivate the same crop. This raises a natural question: can exposure to others' decisions accelerate learning beyond what feedback alone achieves? Observing that someone else approaches the same task differently may lead an agent to question her own approach and try the observed alternative. Improvement alone, however, does not reveal what has been learned: does it reflect a genuine understanding of *why* the observed choices are correct, or merely an imitation of *what* those choices are?

In this paper, I study whether observing others' optimal decisions can improve the quality of one's own behavior, and characterize the nature of learning effects beyond the period of exposure. Specifically, I design a laboratory experiment with three key objectives in mind. First, I exogenously vary the presence and optimality of observed choices to evaluate the causal effect of exposure. Second, I examine potential learning mechanisms: an improvement may reflect mechanical *imitation* of observed actions, or a more involved form of learning in which subjects respond to what they observe selectively, testing it against their own experience. Third, I follow subjects beyond the period of observation to assess the *retention* and *transfer* of the learning that exposure generates:

[1] For example, incorrect understanding of the environment can manifest itself in ignoring described correlations (e.g., Eyster & Weizsäcker (2017); Enke & Zimmermann (2019)), failing to consider payoff-relevant contingencies (e.g., Esponda & Vespa (2014, 2024); Martínez-Marquina, Niederle & Vespa (2019)), mistaking one economic mechanism for another (e.g., Cason & Plott (2014); Rees-Jones & Shorrer (2023)), or having difficulty understanding sample selection (e.g., Esponda & Vespa (2018); Enke (2020); Araujo, Wang & Wilson (2021)).

[2] This literature studies the implications of adopting an incorrect model of the environment (e.g., Hanna, Mullainathan & Schwartzstein (2014); Esponda & Pouzo (2016); Spiegler (2020)). Recent evidence in Esponda, Vespa & Yuksel (2024) demonstrates that providing participants with feedback and experience may not always be sufficient to correct initial misconceptions.

gains that rest on an understanding of the optimal decision rule should survive once exposure ends and the environment changes, whereas gains tied to the specific context need not.[3] More broadly, the goal of the paper is to evaluate the *limits* of observational learning as a policy tool: interventions that rely on it are often cost-effective, but their value depends on whether the behavior they induce remains optimal beyond the context in which it was acquired.

In the experiment, participants assume the role of a hiring manager who repeatedly chooses between two workers from two distinct groups with known ability distributions. In every round, one worker is randomly drawn from each group, and participants observe a noisy binary education signal for each of the two workers before choosing which one to hire. The theoretical benchmark pins down the optimal choice, which depends on how the signal differentially screens ability across the two groups, so that in most rounds, conditional on the observed signals, it is optimal to hire the worker from the group with the *lower* average ability.[4] A deviation from this benchmark arises when participants neglect the informational content of the signal and rely instead on prior group averages, leading to suboptimal hiring decisions, a pattern closely related to inaccurate statistical discrimination. After each choice, participants immediately observe the ability of the hired worker and the associated payoff, so that, in principle, they could learn from repeated feedback and experience with the task environment.

The goal of Part 1 is to establish the incidence of optimal behavior at baseline: subjects in the study face the task individually for 20 rounds, and I find that the majority of them depart from the benchmark. Fewer than 40 percent of choices are optimal overall, and even after some experience with the environment, the median participant makes only 2 out of 10 optimal choices.[5] The persistent choice of the dominated option is what gives the question of social exposure its bite: there is substantial room for improvement, and feedback alone delivers it only slowly and partially. The baseline thus serves as a prerequisite result for the rest of the study, establishing that the environment leaves ample room for learning.

[3] Answering these questions with observational data is difficult for at least two reasons: (i) individuals tend to *endogenously select* whom to observe and learn from, so the quality of observed behavior is confounded with the characteristics of those who select it; (ii) most settings capture only the *contemporaneous effects* of observing others, and the same individuals are rarely tracked after the environment changes, thus limiting the potential to distinguish between imitation and understanding.

[4] In the experiment, low-ability workers are never educated, so observing that a worker is educated rules out low ability. Conditional on both workers being educated, the comparison reduces to the chances of the worker being of high rather than medium ability, which favor the group with the lower average ability. The experimental design section presents the full derivations.

[5] Throughout the paper, reported optimality rates refer to the *key rounds* of the experiment: rounds in which the optimal approach and naive reliance on group averages prescribe opposing hiring choices (71 percent of all rounds). In the rest of the rounds, one option is transparently dominant, and subjects pick it in 96 percent of cases.

The first main result of the paper is that exposure to others' optimal choices substantially improves the quality of subjects' own decisions. In Part 2 of the experiment, which lasts for 40 rounds, participants are randomly assigned to one of two main treatment groups: subjects in the EXPOSURE condition additionally observe the hiring decisions of another participant who took part in one of the past study sessions, saw the exact same information in every round, and chose optimally throughout. In each round, the choice of another participant is presented to subjects as part of the interface before they make their own decisions, and that participant's payoffs are never displayed. Importantly, observed choices are not labeled as optimal, given that recognizing the quality of observed behavior is an integral part of the design. Participants assigned to the NOEXPOSURE group continue facing the decision problem on their own, just as in Part 1. I document that, by the end of Part 2, the two treatment groups diverge significantly: 67 percent of EXPOSURE subjects make the optimal choice in the final round, compared with 46 percent in NOEXPOSURE, and the median EXPOSURE subject makes 8 out of 10 optimal choices, compared with 5 out of 10 in NOEXPOSURE.[6]

I then turn to the mechanisms behind this effect, beginning with a natural question: do subjects exhibit more optimal behavior simply because they directly imitate the decisions they observe? To address it, I run a diagnostic treatment called EXPOSURENEG in which, instead of optimal choices, subjects observe another participant whose decisions are consistently *suboptimal*. If mechanical imitation is driving the effect, then negative exposure should in principle produce a comparable decline in the optimality rate, most clearly among subjects who start out choosing optimally. The data reject this prediction: performance in EXPOSURENEG remains virtually indistinguishable from that in NOEXPOSURE. Instead, looking at transition patterns between consecutive rounds, I find evidence that subjects adopt observed decisions *selectively*: those in EXPOSURE are over 80 percent more likely to switch to the optimal choice after their own suboptimal decision yields a low payoff than after it yields a high one, a sensitivity significantly stronger than in NOEXPOSURE.[7] For a subject who has settled on the suboptimal option, observing someone else repeatedly choose differently makes the alternative salient, and a low realized payoff of her own is what triggers the switch. Payoff dynamics thus govern the timing of behavioral change and explain the strongly

[6] Moreover, the improvement reflects a broad shift in the distribution of behavior: the cumulative distribution of optimal choices in EXPOSURE first-order stochastically dominates that in NOEXPOSURE. The improvement is also unlikely to reflect the informational advantage of observing optimal choices: adding counterfactual payoff information on top of exposure yields performance similar to EXPOSURE alone (Appendix C.1).

[7] This pattern is consistent with the reinforcement learning idea (Erev & Roth, 1998) that adoption of observed behavior is governed by the subject's own payoff experience. A simple Q-learning model, in which subjects update the value of each choice based on realized payoffs, captures the qualitative features of this transition behavior well (Appendix E).

asymmetric response to the quality of observed behavior. Rather than copying whatever they see, subjects test what they observe against their own experience and adopt it when it pays off.

Before turning to the durability of these learning gains, I distinguish several learning types based on subjects' responses to exposure in the first place. Specifically, I classify Exposure participants by their behavioral trajectories between the individual learning and social exposure phases of the experiment (Parts 1 and 2). Approximately 25 percent of subjects behave suboptimally at baseline but exhibit optimal behavior by the end of the exposure phase, and I refer to them as *compliers*.[8] Compliers are the central group of interest, since they respond to the treatment by improving on their baseline performance. Another 30 percent make optimal choices both before and after exposure (*always-takers*), while about 42 percent continue to choose suboptimally despite observing optimal decisions for 40 rounds (*never-takers*). In policy terms, exposure thus shifts roughly a third of its natural target group, the subjects who behave suboptimally at baseline, toward the optimal approach, and I focus on these compliers in the subsequent analysis of learning gains.

The result that subjects adopt observed choices selectively does not necessarily imply that they also understand *why* those choices are optimal, and the next two parts of the experimental design examine whether learning gains survive beyond exposure. I begin with a simple test of *retention*: in Part 3 of the experiment, which lasts for 20 rounds, exposure is removed and subjects return to facing the decision problem on their own. This design separates learning induced by exposure from contemporaneously following the presented choices. Under exposure to optimal choices, the subgroup of compliers reaches a high optimality level of 93 percent by the end of Part 2. I find that retention is substantial but not complete: the average optimality rate in this group declines by 11 percentage points to 82 percent once exposure ends. The result thus has two sides: some of the improvement erodes, but the rate remains high in absolute terms and well above the 53 percent in the NoExposure group.

The central finding of the paper is that learning from others is *fragile*: the substantial gains that survive the removal of exposure fail to generalize to a modified environment. In Part 4 of the experiment, which also lasts for 20 rounds, subjects continue facing the task on their own, but under modified environment primitives.[9] As before, making the correct inference requires integrating the

[8] A subject is classified as exhibiting optimal behavior in a given part of the experiment if her choice is optimal in at least 8 of the final 10 key rounds of that part. This threshold is high enough to indicate systematic reliance on the optimal approach while still allowing for occasional mistakes. The main results are qualitatively robust to using alternative complier classification metrics, as discussed in Appendix D.

[9] Specifically, the new primitives involve a slightly different distribution of abilities in the two groups and a different probability that a medium-ability worker is educated, while all other features of the task remain unchanged.

information embedded in the education signal, which again implies hiring the worker from the group with the lower average ability. The principle behind the optimal choice is thus exactly the same, and only the parameters of the environment differ. I find that the learning gains among compliers largely *do not transfer*: their average optimality rate falls to roughly 60 percent, statistically indistinguishable from the 61 percent attained by NoExposure subjects, who were never exposed to others' choices in the first place. The optimality gap of more than 40 percentage points that compliers had built over the NoExposure group by the end of Part 2 is entirely gone in Part 4. While one might attribute this decline to the task simply becoming more difficult under the new primitives, the learning dynamics in the NoExposure group, which faces the same environment in Part 4, point in the opposite direction: its optimality rate in fact rises from 53 to 61 percent.[10] For most compliers, exposure thus seems to have taught *what* to choose in a particular environment rather than *why* that choice was optimal.[11]

Noting that a substantial share of Exposure subjects do not respond to mere exposure to others' choices, I use the final Part 5 of the experiment to compare the effects of social exposure with those of explicit guidance. In this part, subjects in the Exposure group receive short written advice on how to approach the task, composed by the same participant whose choices they had observed earlier, while NoExposure subjects continue facing the task on their own.[12] Among compliers, the optimality rate rebounds to 88 percent, close to its original level under exposure, and it does so in the same modified environment in which their gains had just dissipated. It is more instructive, however, to look at the response of never-takers, who were unaffected by exposure alone and whose optimality rate never exceeded 33 percent in the first four parts of the experiment. Under optimal advice, their rate more than doubles to 70 percent, statistically indistinguishable from that of the NoExposure group, which itself improves to 74 percent.[13] Advice thus reaches subjects whom observation alone does not: adding it to exposure raises the share of initially suboptimal

[10] Always-takers show no comparable decline either: their optimality rate stays at 97 percent through Parts 2 and 3 and at 89 percent in Part 4.

[11] A benchmark within the control group points in the same direction. Thirteen of the 80 NoExposure subjects switch from suboptimal behavior in Part 1 to optimal behavior in Part 2 through individual learning alone, making them the natural comparison group for Exposure compliers. Their gains hold up notably better under the new primitives: their average optimality rate in Part 4 is 77 percent, significantly above the 60 percent among compliers. Although this comparison is descriptive and the group is small, it is consistent with the fragility documented here being specific to learning from others rather than to learning in this environment in general.

[12] The advice, written for the environment of Parts 4 and 5, reads: *"Hire Orange worker. Both are uneducated, so the worker cannot be high skilled. There are 2 medium workers in orange as opposed to only 1 in green, so orange has a 50% chance of being medium while green only has a 33% chance."*

[13] In the ExposureNeg group, where subjects receive advice from the participant whose suboptimal choices they had observed, the optimality rate does not decline (62 percent), although it ends up 12 percentage points below the NoExposure rate, a marginally significant difference.

subjects who attain optimal behavior from roughly one third to about two thirds.

The main takeaway of the paper is that observing others helps people make better decisions, but it often does not teach them *why* those decisions are better. This distinction matters in practice, as organizations routinely lean on observational learning, whether through onboarding, shadowing, or the sharing of best practices, in part because it demands far fewer resources than many other types of training. The evidence in this paper points to both the promise and the limits of such practices. They can improve behavior substantially, yet the choices they instill remain tied to the environment in which they were acquired: a hiring manager who imitates a successful colleague may, for example, choose well only for as long as the applicant pool resembles the one she learned on. The findings therefore caution against evaluating such programs by contemporaneous performance alone, since strong behavior during the period of observation is not, by itself, evidence of durable learning. Instead, programs that rely on demonstration should be judged by whether the acquired behavior withstands a change in circumstances. Where persistence of learning gains matters, demonstration is best complemented with an explanation of why the observed behavior works: in this experiment, explanation reaches many subjects for whom observation alone had no effect, although whether its gains generalize more readily than those from exposure is a question the design leaves open. Observational learning, in short, is a powerful way to spread behavior, and a far weaker way to spread understanding.

Connections to the Literature

First, this paper contributes to the large literature on social learning and learning through imitation.[14] A classical strand studies sequential settings in which agents observe predecessors' actions and must infer the private information behind them, with theoretical foundations in A. V. Banerjee (1992), Bikhchandani, Hirshleifer & Welch (1992), and Smith & Sørensen (2000), and a substantial body of experimental work documenting deviations from Bayesian aggregation (Anderson & Holt, 1997; Duffy et al., 2019; Hung & Plott, 2001; Weizsäcker, 2010). Within this paradigm, Eyster,

[14] The theoretical foundations for imitation as a learning strategy include Schlag (1998), who shows that the proportional imitation rule is the unique rule that is improving and monotone under minimal information requirements, and related work on imitation dynamics in games (Björnerstedt & Weibull, 1996; Schlag, 2010; Vega-Redondo, 1997). Experimental evidence confirms that "imitate the best" is a robust behavioral tendency (Apesteguia, Huck & Oechssler, 2007; Huck, Normann & Oechssler, 1999; Offerman & Schotter, 2009). In psychology, Bandura's (1977) social learning theory identifies four processes (attention, retention, reproduction, and motivation) governing whether exposure translates into behavioral change, and Bandura, Ross & Ross (1963) show that imitation is modulated by the perceived payoff consequences of the modeled actions. In field settings, Conley & Udry (2010) show that farmers adjust inputs to align with those of successful peers, and Rendell et al. (2010) highlight that social learning strategies tend to outperform individual learning because demonstrators inadvertently filter information by displaying their highest-payoff behaviors.

Rabin & Weizsäcker (2018) find that subjects imitate predecessors even when their actions are redundant with public information. A more recent strand considers repeated settings in which agents observe others over time (Evdokimov & Garfagnini, 2020; Merlo & Schotter, 2003), and Agranov et al. (2026) show that observing others can improve performance even when the observed actions carry no additional payoff-relevant information, while Agranov & Nielsen (2026) study endogenous transmission, examining how much people learn from the information others choose to share with them.[15]

The present experiment shares this repeated structure and extends it in three directions: I exogenously vary the quality of observed choices, I test whether improvements persist once exposure ends, and I assess whether learning transfers to a modified environment. Relatedly, Conlon et al. (2026) document that participants underweight a partner's signals relative to equivalent signals of their own, even when the partner's information is perfectly communicated or directly observed. While they study how people weight others' *signals* when forming beliefs, I study whether observing others' *choices* can shift behavior, and the two sets of findings are consistent: people discount the information others hold, and even when they respond to what they observe, the learning that results is often shallow.

Second, this paper connects to a growing body of work on learning under misspecified mental models, which shows that agents who interpret feedback through a flawed representation of the environment converge to systematically biased steady states (Bohren & Hauser, 2021; Esponda & Pouzo, 2016; Fudenberg, Romanyuk & Strack, 2017; Heidhues, Kőszegi & Strack, 2018).[16] A complementary set of experiments studies how such models are formed from data in the first place, showing that extracting the correct model is often difficult and that the models people settle on can be systematically distorted (Fan, 2026; Fréchette, Vespa & Yuksel, 2025; Kendall & Oprea, 2024). Feedback may have limited corrective power if agents remain confident in incorrect models, selectively attend to confirming information, or lack confidence in the mapping from signals to optimal actions (Enke, Graeber & Oprea, 2023; Gagnon-Bartsch, Rabin & Schwartzstein, 2026; Hanna, Mullainathan & Schwartzstein, 2014; Spiegler, 2020; Szkup & Treviño, 2025). Esponda, Vespa & Yuksel (2024) document persistent suboptimal behavior under extensive feedback in a

[15] Relatedly, Lahno & Serra-Garcia (2015) decompose the imitation of peers' risky choices into payoff-based and action-based motives, a distinction that resonates with the payoff-anchored adoption documented here.

[16] Esponda & Pouzo (2016) show that agents who maximize expected utility under a misspecified model converge to biased steady states; Heidhues, Kőszegi & Strack (2018) demonstrate that even small belief departures can sustain overconfident behavior indefinitely; Fudenberg, Romanyuk & Strack (2017) show that incorrect priors can prevent agents from experimenting sufficiently to encounter corrective data; and Bohren & Hauser (2021) extend these results to social settings where agents also draw inferences from others' behavior.

statistical inference task similar to the one studied here.[17] I contribute to this line of work by studying a different channel through which mental models may be challenged: rather than receiving corrective information from the experimenter, participants observe the choices of another agent facing the same task, an external reference point that may prompt reassessment of how they engage with and interpret feedback.

Third, a distinctive feature of this paper is its focus on the durability and generalizability of learning gains. While most of the social learning literature documents contemporaneous effects of observation, substantially less is known about what happens once the model is removed. In psychology, Bandura's (1977) theory emphasizes that behavioral gains from observation may dissipate if retention relies on superficial features rather than deeper understanding of why the modeled behavior is effective. The question of whether learning transfers to novel settings dates back to Thorndike & Woodworth (1901), who proposed that transfer depends on shared elements between training and test environments.[18] In economics, field experiments on technology adoption provide some evidence on persistence (Dupas, 2014). This paper contributes controlled evidence on both margins, and the mechanism it uncovers speaks to why observation-based gains can be fragile in the first place. Personal experiences disproportionately shape beliefs and decisions even when more complete data are readily available (Malmendier & Nagel, 2016), and people respond more strongly to outcomes they experience themselves than to equivalent ones they observe (Simonsohn et al., 2008), with realized outcomes carrying more weight than paper ones (Imas, 2016). The behavior of Exposure subjects fits this pattern: they never see the observed participant's payoffs, try the observed alternative primarily after their own choice delivers a low payoff, and keep it when their own experience turns favorable. Once exposure ends, the external cue around which this own-payoff experimentation was organized is gone, and part of the improvement erodes with it. When the environment itself changes, the remaining gains largely vanish.

Fourth, this paper contributes to understanding the relative effectiveness of passive exposure to others' actions and explicit guidance. In Part 5 of the experiment, subjects receive written advice from the same agent whose choices they previously observed, providing a within-design comparison of the two channels. A related body of work studies intergenerational transmission of knowledge: Schotter & Sopher (2003) show that free-form advice can establish and sustain coordi-

[17] In one of their diagnostic treatments, the authors directly challenge participants' confidence by informing them that their initial choice was incorrect, which alters subsequent behavior and sheds light on how confidence in the initial response can reduce engagement and result in slow and partial learning.

[18] See Barnett & Ceci (2002) for a taxonomy of near and far transfer, and Singley & Anderson (1989) for an extension to cognitive skills, arguing that transfer depends on the degree to which underlying production rules, rather than surface features, are shared across tasks.

nation conventions, and Chaudhuri, Graziano & Maitra (2006) find that the effectiveness of such advice depends on its publicity.[19] A recent line of work studies guidance designed around human reasoning errors, from algorithmic feedback that improves judgment (Serra-Garcia & Gneezy, 2025) to information design for AI-assisted decisions (Hoong & Dreyfuss, 2026). In this paper, advice produces improvements that are more broadly distributed than those achieved through exposure alone: compliers recover most of their earlier gains, and never-takers, who show little response to observation, improve substantially, indicating that explicit guidance can reach agents whom passive exposure does not move. Suboptimal advice, by contrast, produces no comparable decline, mirroring the asymmetry observed under exposure itself.

Finally, given that participants face a repeated inference task framed as a hiring decision, this paper also connects to the literature on inaccurate statistical discrimination. The classic theory of statistical discrimination assumes that decision-makers hold accurate beliefs about group-level productivity and condition on them rationally (Arrow, 1973; Phelps, 1972), and several empirical tests find behavior consistent with such accurate beliefs (Knowles, Persico & Todd, 2001; List, 2004). A more recent literature shows, however, that discrimination is often *inaccurate*, rooted in miscalibrated beliefs about groups (Bohren, Imas & Rosenberg, 2019). Subsequent experimental work has sharpened these findings: Esponda, Oprea & Yuksel (2023) document representative signal distortion, Campos-Mercade & Mengel (2024) attribute approximately 40 percent of the hiring gap in their experiment to non-Bayesian statistical discrimination driven by signal neglect, and Bohren et al. (2025) show that ignoring inaccurate beliefs can lead to erroneous attributions of taste-based discrimination. This paper contributes by examining whether exposure to another agent's optimal behavior can serve as a debiasing mechanism, finding that it partially corrects signal neglect but that the resulting improvements are fragile and largely do not transfer to a modified environment.

2 Experimental Design

The experimental design must accomplish three objectives that follow directly from the questions posed in the introduction. First, it requires an environment with a clear theoretical benchmark in which a systematic bias arises naturally and persists despite repeated, informative feedback, so that there is genuine scope for learning from others. Second, it must vary exposure to others' choices

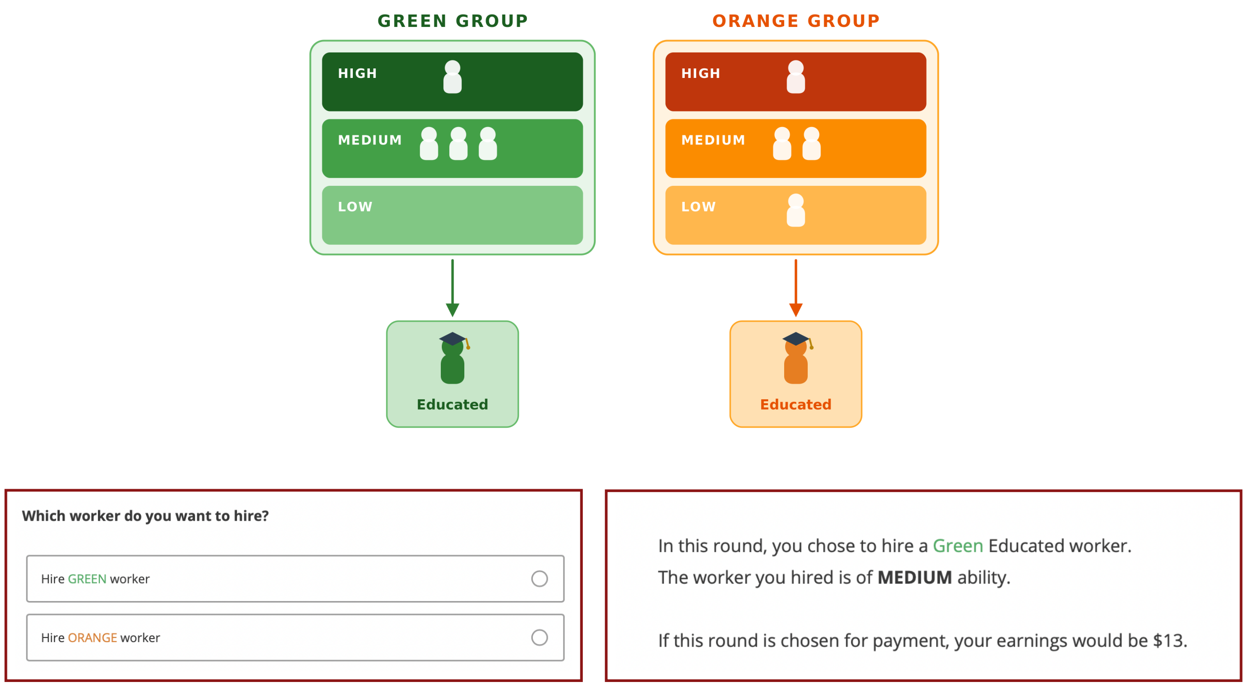
[19] Nyarko, Schotter & Sopher (2006) study experimental markets in which subjects can purchase either raw data about predecessors' choices or synthesized advice, finding that subjects bid significantly more for data. In field settings, A. Banerjee et al. (2013) show that the network position of initially informed individuals shapes diffusion through social ties; Cai, De Janvry & Sadoulet (2015) demonstrate that peer effects on insurance adoption operate through knowledge diffusion rather than imitation; and Duflo & Saez (2003) find that peer exposure to information sessions increases retirement plan enrollment among untreated colleagues.

exogenously, including the *quality* of the observed behavior, in order to identify the causal effect of observation and distinguish it from indiscriminate imitation. Third, it must follow the same subjects after exposure ends and after the environment's primitives change, so that the retention and the transfer of any learning gains can be measured directly.[20]

2.1 Inference Task

Throughout the experiment, all participants assume the role of a hiring manager who must choose between two workers: one from a group of Green workers ($k = green$) and one from a group of Orange workers ($k = orange$). Each group consists of four workers, and each worker can be of either low, medium, or high ability: $a \in \{l, m, h\}$. In every round, one worker is randomly drawn from each group, and participants must choose which of the two they would like to hire. Participants can observe the distribution of ability levels in both groups, but do not observe the abilities of the two selected workers.[21]

Figure 1: Group Composition and Decision Screen Interface



Instead, they observe the selected workers' binary education status (*educ* or *not educ*). Partic-

[20]Full experimental instructions can be found in Appendix H.

[21]In the experiment, one of two ability distributions is randomly selected in each round. One distribution is shown in Figure 1, and the other is nearly identical except that one medium-ability worker in each group is replaced with a high-ability worker. Theoretical predictions are identical for the two distributions.

ipants are also informed about the relationship between ability level and education status – they know that high-ability workers are educated with certainty, $\mathbb{P}(educ \mid h) = p_h = 1$, medium-ability workers are educated with a 90 percent chance ($p_m = 0.9$), and low-ability workers are *not* educated with certainty ($p_l = 0$). The interface of the experiment is presented in Figure 1. After learning the education status of two selected workers, participants choose either to hire a Green worker or an Orange worker.

Benchmark. Given the environment described above, it is straightforward to identify the theoretical benchmark. Consider the modal scenario in which both the green and the orange worker are educated. It is useful to contrast explicitly the two approaches a participant may take to this decision.

Suboptimal approach. A participant may neglect the informational content of the education signal (after all, both workers display the *same* signal, which makes it appear uninformative for the comparison) and instead rank the two groups by their *average* ability. Because the green group has higher average ability ex ante, this reasoning leads to hiring the **green** worker whenever both candidates are educated.

Optimal approach. The education signal, however, screens ability differently across the two groups. Since low-ability workers are never educated ($p_l = 0$), observing that the selected orange worker is educated *eliminates* the possibility that he is of low ability. Conditional on both workers being educated, the relevant comparison is therefore the ratio of high- to medium-ability workers, which is higher in the orange group – so the payoff-maximizing choice is to hire the **orange** worker, despite the green group's higher average ability.[22]

This logic is confirmed by the Bayesian posterior. Let k_a represent the proportion of workers of group identity $k \in \{green, orange\}$ who have ability $a \in \{l, m, h\}$. Using Bayes' rule, the posterior probability that an educated worker from group k is of high ability is given by:

$$\mathbb{P}(h \mid educ) = \frac{\mathbb{P}(educ \mid h) \times \mathbb{P}(h)}{\mathbb{P}(educ)} = \frac{p_h \times k_h}{p_h \times k_h + p_m \times k_m + p_l \times k_l} = \frac{k_h}{k_h + 0.9k_m}$$

Noting that $green_h = 0.25, green_m = 0.75$, $orange_h = 0.25$ and $orange_m = 0.5$, we have

[22]The inference task in this experiment is adapted from Campos-Mercade & Mengel (2024), who study how difficulties with Bayesian updating interfere with hiring decisions in a setting that speaks to the literature on statistical discrimination. While their focus is on documenting non-Bayesian behavior in individual decision-making, the present paper uses a similar environment to study whether exposure to others' choices can mitigate such biases.

$\mathbb{P}(h \mid educ) \approx 0.27$ for the green worker and $\mathbb{P}(h \mid educ) \approx 0.36$ for the orange worker. Importantly, participants only need to make a binary hiring choice, which does not require computing these conditional probabilities: recognizing that education rules out low ability, and comparing the resulting high-to-medium ratios, suffices. The focus in this paper is therefore on a type of learning that is more general than computing the correct Bayesian posterior belief: the key element is to not neglect the education signal but rather extract information from it, which is consistent with Bayesian reasoning.

Observation: *In the experiment, the payoff-maximizing choice conditional on both workers being educated is to hire the* ***Orange*** *worker. A naive participant, seeing that both workers are educated, would underinfer from the education signal and instead hire the* ***Green*** *worker, who comes from the group with higher average ability ex ante.*

2.2 Experiment Structure

Below I describe each of the five parts of the experiment and the between-subject treatment design, summarized in Figure 2.

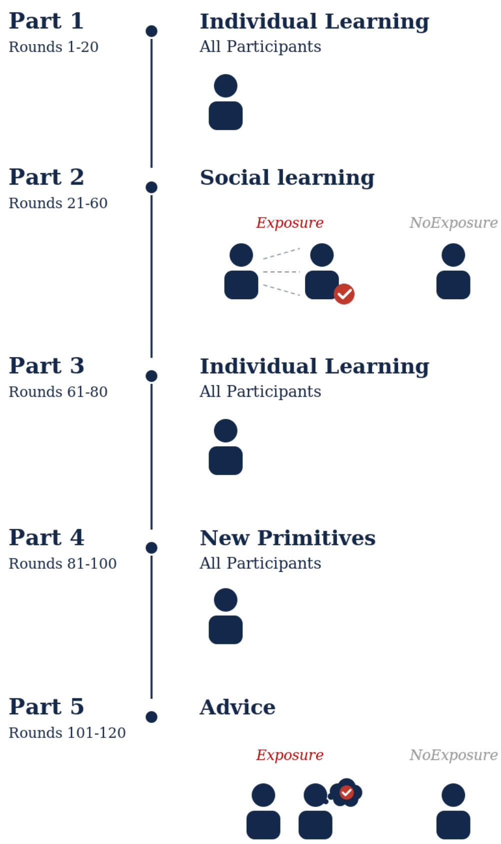
Part 1: Baseline Behavior (20 rounds). In the first part, all participants face the decision problem individually, making every hiring choice on their own. After each round, participants receive feedback about the ability of the worker they hired, along with the earnings for that round. Participants' earnings in the experiment depend on both the ability level and education status of the worker they hire. In particular, they earn more if the hired worker is educated, and conditional on an education status, they earn more if the hired worker is of higher ability.[23] Note that the ability distributions and the choice of environment primitives result in most rounds having both workers be educated, which is useful because I am specifically interested in subjects' behavior in these "rounds of interest," referred to as *key rounds* throughout the paper.[24] In the remaining rounds, in which at least one worker is not educated, one option is transparently dominant, and 96 percent of subjects' choices are optimal. The goal of Part 1 is to establish the share of optimal choices at baseline and to give subjects an opportunity to practice with the experimental interface.

Part 2: Exposure to External Choices (40 rounds). Subjects continue making hiring decisions between a green worker and an orange worker, exactly as in Part 1. The central ma-

[23]Specifically, if they hire an Educated worker their payoff is either \$18 (high ability) or \$13 (medium ability). If they hire a Not educated worker, their payoff is either \$10 (medium ability) or \$5 (low ability).

[24]For the distribution of abilities in Figure 1, the probability of being educated is 0.925 for the green worker and 0.7 for the orange worker, so the probability that both workers are educated in a given round is approximately 0.65. In the experiment, around 71 percent of rounds were rounds of interest.

Figure 2: Experiment Structure and Treatment Design



nipulation of the experiment occurs in this part: to exogenously vary exposure to others' choices, subjects are randomly assigned to one of two main treatment conditions, depicted in Figure 3.

- **Exposure** (**Exp**). Subjects in this group are additionally presented, in each round, with the hiring choice made by another participant ("participant X") who faced the exact same inference task in one of the previous study sessions and always made *optimal* decisions.

- **NoExposure** (**NoExp**). Subjects in this group continue to make all decisions independently, in an environment identical to Part 1. This group serves as the reference condition.

Participant X is an actual subject rather than a computerized agent. From the pool of participants in earlier sessions who completed the task individually, I select an individual whose decision history is optimal in every round of the experiment, and Exposure subjects observe her choices round by round.[25] Subjects in the Exposure group, and only they, are introduced to this partici-

[25]In every key round, in which both workers are educated, this participant hires the orange worker; the key rounds

pant at the start of Part 2. Their instructions read:

> "In rounds 21–60, you will also be provided with the *hiring choice of another study participant* who took part in the exact same study in one of the past sessions, we will call this participant *Participant X*. In each round, participant X saw the exact same information as you will (distribution of abilities and education statuses of the selected workers)."[26]

Figure 3: Treatment Conditions in Part 2

Although subjects know that all the hiring choices they see were made by the same person, they are not told whether these choices are optimal or not – in fact, the experiment is designed to better understand subjects' ability to recognize the (sub)optimality of the choices they are exposed to. In particular, the hope is that subjects who initially make suboptimal hiring decisions will begin to reconsider their approach when presented with optimal choices. Notably, this environment differs from more common social learning settings in the literature, where others' choices are designed to convey payoff-relevant information and guide learning.[27] In contrast, in my setting, others' decisions are only useful to the extent that a subject does not already know the optimal approach; if they do, observing others should in theory provide no additional benefit.

The main hypothesis in Part 2 is that subjects in the Exposure treatment would exhibit more optimal behavior relative to the control condition with no exposure. Observing optimal choices could especially be helpful for participants who demonstrate higher subjective uncertainty about their approach to the task.[28]

are the margin on which optimal and suboptimal behavior are defined throughout the paper. In the remaining rounds of Part 2, in which one option is transparently dominant, her choices coincide with the dominant option, as do the choices of the overwhelming majority of subjects, so the informational content of the treatment is concentrated in the key rounds.

[26] Subjects are specifically informed: "For example, if in round 30 you see that the abilities draw is Draw #1, the selected Green worker is educated, and the selected Orange worker is educated, it means that participant X in round 30 saw that the abilities draw is Draw #1, the selected Green worker is educated, and the selected Orange worker is educated. In rounds 21-60, you will be provided with this hiring choice and the interface will, for instance, say 'Participant X chose to hire the Green worker' or 'Participant X chose to hire the Orange worker' ".

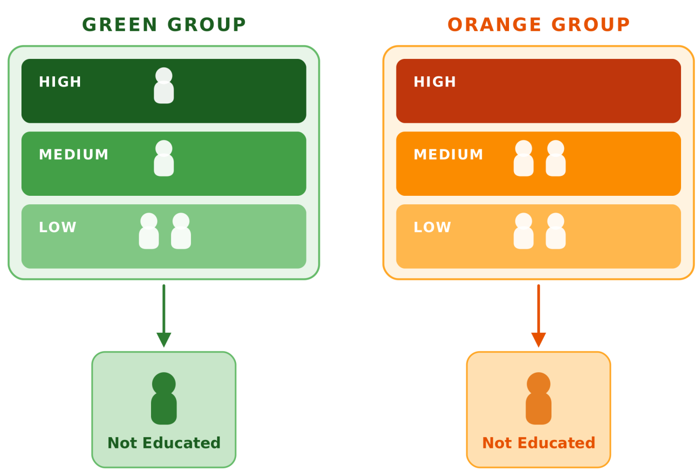
[27] For example, by observing someone's action, a subject can often infer the private signal that person received, which can then inform their own decision.

[28] Presumably, feeling certain or confident about one's own approach is associated with lower susceptibility to being influenced by the observed choices of another subject, regardless of whether one's own choices are optimal. I test this prediction in a later section.

Part 3: Learning Retention (20 rounds). This part follows the same structure as Part 1: all participants, regardless of their treatment group, make decisions independently. The goal of including this part is to examine whether the participants have retained what they learned from being exposed to another subject's choices in Part 2.

If behavior in Part 3 returns to its Part 1 pattern, the improvement observed under exposure did not outlast the presence of the observed choices; if it persists, the change in behavior does not depend on those choices remaining on display. Retention alone, however, cannot distinguish a subject who has internalized the optimal decision rule from one who continues to repeat the specific choices observed in Part 2, since the two prescribe identical behavior as long as the environment is unchanged. Separating them requires modifying the environment, which motivates the next part.

Figure 4: Distribution of Abilities in Parts 4 and 5



Part 4: New Environment Primitives (20 rounds). This part examines whether the behavior acquired in the earlier parts extends to a modified environment. Subjects in all treatments complete 20 rounds independently, similar to Part 1 and Part 3. The main difference is that in Part 4 the inference task uses a *different distribution of abilities* depicted in Figure 4. In this new setup, the green group consists of two low-ability workers, one medium-ability worker, and one high-ability worker, while the orange group has two low-ability and two medium-ability workers. Subjects are also informed that a medium-ability worker is educated with a 10 percent chance ($p_m = 0.1$) under the new primitives of the environment. The remaining features of the hiring task stay intact: high-ability workers are always educated, low-ability workers are never educated, and the green group has a higher average ability.

Crucially, most of the rounds will now feature both workers being *Not Educated*, and conditional on this, the optimal choice remains *hiring an orange worker*.[29] The objective of Part 4 is straightforward: to establish whether subjects would exhibit similar treatment effects under the new environment primitives, i.e., whether they would continue making significantly more optimal choices after exposure to optimal choices. In particular, comparing the prevalence of optimal behavior among Exposure subjects across the two environments reveals the extent to which behavior acquired under exposure extends beyond the environment in which it was acquired; the interpretation of these comparisons is discussed in Section 4.

Figure 5: Advice in Exposure

Part 5: Direct Advice (20 rounds). In the last part of the experiment, subjects complete 20 more rounds under the new environment primitives. In every round, subjects in the exposure treatments are shown a written recommendation by the same participant whose choices they were observing in Part 2, while NoExposure subjects continue facing the task on their own. In particular, subjects in the Exposure treatment read advice written by the participant making 100 percent optimal choices, which explains how to optimally approach the task and extract useful information from education signals (see Figure 5).[30] The goal here is to see how exposure to an elaborate written recommendation to make a particular choice performs relative to observing others' choices.

[29] If both workers are not educated, then neither can be of high ability, since high-ability workers are always educated. The relevant comparison is therefore the probability of being of medium ability conditional on not being educated, which equals approximately 0.31 for the green worker and 0.47 for the orange worker. Since the orange worker is about one and a half times as likely to be of medium ability conditional on being not educated, the optimal choice is again to hire the orange worker. The stakes on this margin are, if anything, larger than in Parts 1–3: the expected per-round payoff difference between the two options in the key rounds is \$0.82 in Parts 4–5, compared with \$0.43 in Parts 1–3, so the payoff feedback available for learning the optimal choice is stronger in the new environment. As in Parts 1–3, subjects need only make a binary hiring choice rather than compute these posterior probabilities explicitly.

[30] The numerical rationale in the participant-written advice is approximate: for the both-not-educated scenario it computes the chance of a medium-ability worker from worker counts after excluding high-ability workers (2 of 4 for orange; 1 of 3 for green), rather than from the full posterior that also weights education probabilities ($\approx 47\%$ versus $\approx 31\%$; see the derivation in the Part 4 description above). The recommendation and the ranking of the two options are correct. If anything, this feature reinforces the interpretation developed later: the advice that moves subjects is an accessible, directionally correct rationale rather than exact Bayesian arithmetic.

2.3 Implementation Procedures

The experiment was conducted in person at the University of California San Diego Economics Laboratory (EconLab). Prior to the main part of the experiment, all subjects were required to correctly answer six comprehension questions to proceed. After that, participants could move at their own pace but they were informed that they would not be able to leave early. A total of 303 subjects participated in the two main treatments: 223 in EXPOSURE and 80 in NOEXPOSURE.[31] On average, subjects earned $14.4 in the experiment. Final payment was based on the outcome of one randomly selected round for the main part, together with a separately drawn round that determined the chance of winning an additional $2 bonus. At the end of the experiment, subjects completed a short demographic survey.

Preregistration and empirical strategy. The study was pre-registered in the AEA RCT Registry under AEARCTR-0013607. The registered primary outcome is the round-level binary indicator of whether a subject's choice coincides with the Bayesian benchmark, to be used both for round-by-round dynamics and aggregated across rounds and treatment groups; the registered secondary outcomes are the incentivized confidence elicitations and subjects' end-of-experiment strategy descriptions. Each subject was randomly assigned to one of the treatment groups; recruitment materials and sign-up procedures were identical across conditions, so subjects had no information about treatment when enrolling, and no subject who completed the experiment is excluded from the analysis. Sessions took place in May and December 2024, May–June 2025, and May 2026; all four treatments were fielded from the first wave in May 2024 onward, consistent with the preregistration, and the instructions and procedures remained identical across waves.[32]

In implementing the registered aggregate comparisons, the paper reports the share of optimal choices in the last ten key rounds of each part as its principal summary, alongside the last-round indicator and a binary indicator for making at least eight of ten optimal choices; where these endpoints differ in statistical precision, all are reported.[33] The randomized treatment comparisons

[31] An additional 53 subjects were assigned to a diagnostic treatment in which the observed participant always made *suboptimal* choices (EXPOSURENEG), and 44 subjects to a treatment that supplements exposure with counterfactual payoff information (EXPOSURE (FULL INFO)). These treatments are introduced where they enter the analysis, in Section 3.3, and analyzed there and in Appendix C.1.

[32] The later sessions served to increase the overall sample size and, in particular, to over-recruit the EXPOSURE arm: because much of the analysis characterizes heterogeneity in individual responses within this group, a larger sample in this arm was needed for adequately powered subgroup comparisons.

[33] Because participants make all decisions autonomously, with no communication, no shared feedback, and no group-level payoffs, there is no mechanical channel for within-session correlation of errors; standard errors are accordingly clustered at the subject level throughout.

of Sections 3.2 and 3.3 implement the registered confirmatory analyses, and the remaining analyses of the paper characterize the heterogeneity and mechanisms behind these comparisons.[34]

3 Results

I organize the main experimental results as follows: In Section 3.1, I report the incidence of optimal behavior at baseline and show that a majority of subjects choose a suboptimal option. In Section 3.2, I then present the main treatment effects of exposure to external choices in Part 2 of the experiment, establishing that exposure to optimal decisions facilitates learning.[35] I postpone the analysis of learning robustness and retention to the subsequent section.

3.1 Baseline Suboptimal Behavior

In the initial rounds of the experiment in Part 1, all subjects complete the tasks individually regardless of their treatment assignment and receive feedback after each round. At baseline, fewer than two in five of subjects' choices in the key rounds of Part 1 are optimal (37 percent), and the share remains at only 40 percent in the last ten rounds; that is, when both the Green and the Orange worker are educated, most subjects fail to hire the Orange worker.[36] In fact, even after some experience with the environment more than half of subjects stick to the suboptimal Green worker option in the last round of Part 1.

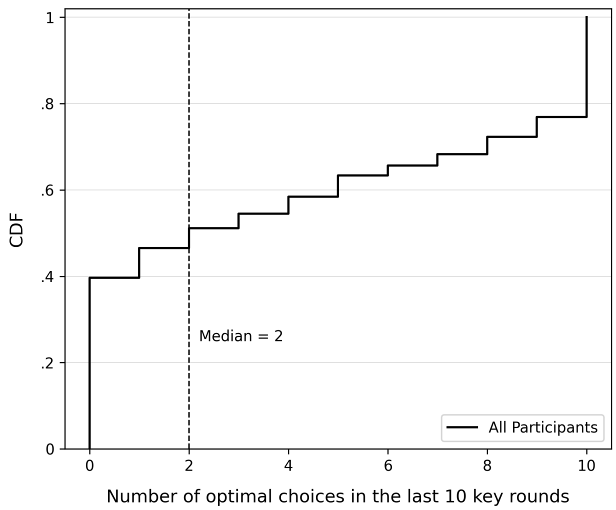
The bimodal nature of baseline behavior is further illustrated in Figure 6, which plots the cumulative distribution of optimal choices in the last ten rounds of Part 1 pooled across the Exposure and NoExposure groups. It can be seen that the median subject makes only two optimal choices out of ten. While approximately 23 percent of subjects are fully optimal with all ten choices correct, 40 percent of subjects make zero optimal choices. The overall shape confirms that subjects are not clustered around an intermediate performance level; rather, the distribution is spread across the entire range, with pronounced concentrations at the two extremes.

[34] In particular, the 80 percent threshold used in the learning-type classification was not part of the pre-registration. The appendix on alternative complier specifications therefore examines a range of alternative thresholds and classification criteria and shows that the composition of learning types and their dynamics are qualitatively unchanged.

[35] As defined in Section 2, the analysis throughout focuses on the key rounds: rounds in which both workers are Educated in Parts 1–3, and rounds in which both workers are Not Educated in Parts 4–5. Out of 120 rounds across five parts of the experiment, 85 constitute key rounds (71 percent). In the remaining 35 rounds, 96 percent of subjects' choices are optimal and there is no statistically significant difference between treatments based on a joint test ($p = 0.755$).

[36] Departure from the optimal benchmark is costly. The gap in earnings for the average subject is statistically significant ($p < 0.001$), demonstrating that suboptimal behavior translates into measurable monetary losses.

Figure 6: Cumulative Distribution of Optimal Choices in Part 1



Notes: This figure plots the cumulative distribution of the number of optimal choices in the last ten rounds of Part 1 for all 303 participants, pooled across the Exposure and NoExposure groups. All subjects complete Part 1 individually, so no treatment differences are expected at baseline. The vertical dashed line denotes the median.

Given that all subjects complete the tasks individually in Part 1, irrespective of treatment assignment, no differences across treatment groups are expected at baseline. Consistent with this prediction, a test of equality of baseline optimality rates across the Exposure and NoExposure groups fails to reject the null hypothesis ($p = 0.88$). Throughout the paper, NoExposure serves as the reference group and represents the counterfactual of completing the inference task individually in every part of the experiment. Because all subjects receive immediate feedback on the outcome of their hiring choice in each round, it is natural to expect gradual learning even in the absence of social exposure.[37]

While details on the distribution of subject types are presented in subsequent sections, the key takeaway from this section is that the study's baseline condition holds: in the first part of the experiment, most subjects choose a dominated option, indicating substantial scope for learning. This result is summarized below:

Prerequisite Result: *At baseline, fewer than two in five choices that subjects make are optimal, and the median subject chooses optimally in only two of the last ten rounds of Part 1.*

Next, I study subjects' behavior in Part 2 to evaluate to what extent exposure to others' choices can influence their own decisions. The goal of the next section is, therefore, to assess whether

[37]Consistent with this intuition, the share of optimal choices in the NoExposure group increases modestly over time, reaching 49 percent in the last ten rounds of Part 2.

observing decisions made by another subject can accelerate learning relative to the individual-learning control group.

3.2 Learning from Exposure to Others' Decisions

In Part 2 of the experiment, which lasts for 40 rounds, subjects in the Exposure group additionally observe choices made by another participant who approached the task optimally. Given that subjects are not informed whether the observed choices are optimal or not, the goal of Part 2 is to assess whether participants can recognize the quality of these decisions and learn from them.

By the end of Part 2, optimal behavior in the Exposure group is substantially higher than in NoExposure. In the last round of Part 2, 67 percent of Exposure subjects make the optimal choice, compared with 46 percent in NoExposure, a 21 percentage point difference that is statistically significant ($p = 0.001$). The treatment effect is especially pronounced at the median: a quantile regression reveals that the median Exposure subject makes 8 out of 10 optimal choices in the last ten rounds, compared with 5 out of 10 in NoExposure ($p < 0.001$).[38]

To explore the dynamics of these treatment effects, I deploy the following measure that captures the notion of switching to optimal behavior: for each treatment group and for each of the 40 rounds in Part 2, I compute the cumulative percentage of subjects who choose the optimal alternative in a given round and continue to do so in at least 80 percent of remaining rounds in this part.[39] As Figure 7a shows, about 40 percent of NoExposure subjects choose optimally in most of the remaining rounds from the start, and this number rises only to 43 percent by round 55, the end of the displayed horizon. In contrast, the share of Exposure subjects switching to optimal behavior starts to grow steadily from round 35 onward, gradually widening the gap between the two groups.[40]

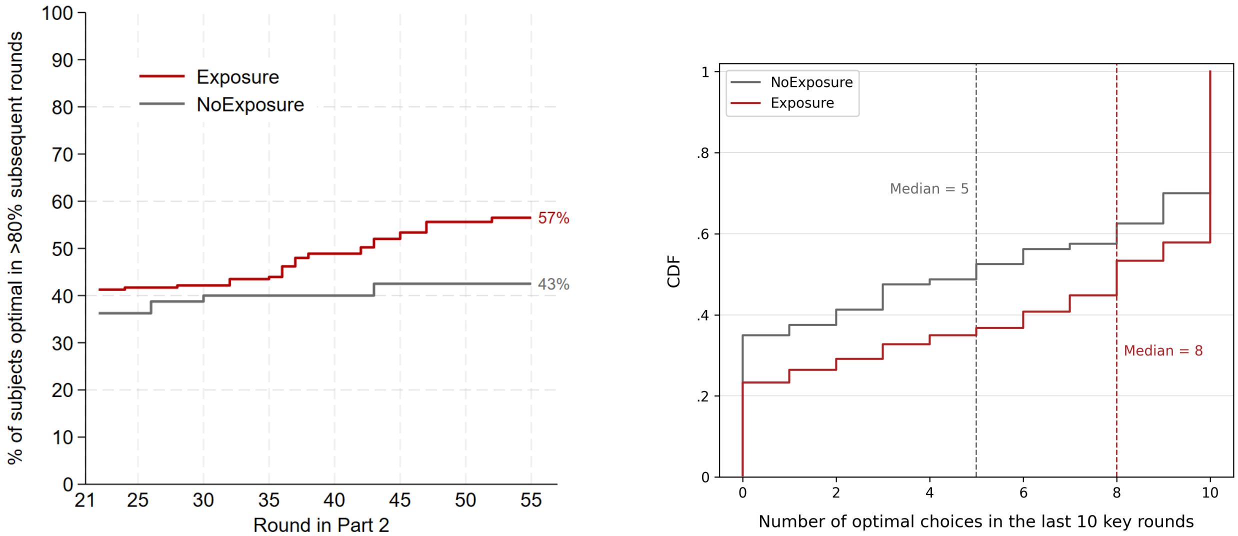
The cumulative distribution of optimal choices in Part 2, plotted in Figure 7b, confirms that the improvement in Exposure reflects a broad shift in behavior: the distribution of optimal choice

[38] Detailed statistical tests at multiple time horizons are reported in Table 7. The treatment effect estimates over the last ten rounds are somewhat attenuated relative to the last-round estimates, reflecting the fact that many Exposure subjects switch to optimal behavior gradually rather than all at once. The treatment effect remains significant at multiple time horizons and under alternative specifications reported in the appendix.

[39] This measure requires that a subject who chooses the optimal option in a given round continues to behave mostly optimally thereafter, while allowing for occasional mistakes. The threshold is justified by the empirical transition probability from optimal choice in round $t-1$ to suboptimal choice in round t of around 0.13, estimated for the Exposure group. In later sections of the paper, I continue to use this threshold for other types of analyses; qualitatively, the results are robust to alternative specifications.

[40] In Section 3.3, I provide further evidence of empirical patterns consistent with payoff-driven selective adoption: subjects in the Exposure group are over 80 percent more likely to switch to the optimal choice after their own suboptimal choice yields a low payoff than after a high one.

Figure 7: Effects from Exposure to Optimal Choices



(a) Dynamics of Optimal Behavior (b) Cumulative Distribution of Optimal Choices

Notes: The red line corresponds to Exposure and the gray line corresponds to NoExposure. Panel (a) displays the percentage of subjects choosing the optimal alternative in a given round of Part 2 and then continuing to do so in at least 80% of subsequent rounds; the lines end at round 55, beyond which too few rounds remain for the criterion to be meaningful, and the numbers displayed next to each line report the value of this measure at round 55. Panel (b) plots the cumulative distribution of the number of optimal choices in the last ten rounds of Part 2 for each group. The vertical dashed lines denote the median for each group. The distribution of optimal choices in Exposure first-order stochastically dominates that in NoExposure; details of statistical tests are provided in the text.

shares across subjects first-order stochastically dominates that in NoExposure.[41] Another notable feature is that the Exposure group tends to split into three broad clusters: roughly 42 percent of subjects make all 10 optimal choices, 23 percent make none, and the remaining 35 percent fall between these two extremes.

The results on learning under exposure to external decisions are summarized as follows:

Result #1: *By the end of Part 2, there are significantly more optimal choices in Exposure than in NoExposure. The median Exposure subject makes 8 out of 10 optimal choices in the last ten rounds, compared with 5 out of 10 in NoExposure.*

[41] I test for first-order stochastic dominance using the two-stage procedure in Barrett & Donald (2003). I first test the null hypothesis that the distribution in the NoExposure treatment either first-order stochastically dominates or is equal to the distribution in the Exposure treatment. I reject this null hypothesis at the 1 percent level ($p < 0.001$). I then test the null hypothesis that the distribution in the Exposure treatment first-order stochastically dominates the distribution in the NoExposure treatment. The null in this case is not rejected ($p = 0.539$).

Table 1: Candidate Learning Channels

Channel	Description	Observable prediction
1. Mechanical imitation	Subjects copy observed behavior, regardless of its quality	Symmetric effects: exposure to suboptimal choices lowers performance as much as exposure to optimal choices raises it
2. Selective adoption	Subjects test the observed alternative against their own payoff experience	Asymmetric, gradual adoption that tracks own payoffs; gains tied to the original context
3. Model revision	Subjects infer *why* the observed choice is optimal	Gains persist once exposure ends and transfer to related environments

3.3 Understanding Learning Mechanisms

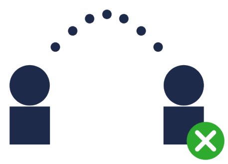
The results so far show that subjects perform substantially better when exposed to optimal behavior, even though the observed choices are never labeled as such. A natural question is what drives this positive effect. A higher optimality rate relative to the individual-learning control group is consistent with several distinct underlying processes. On the one hand, it could mean that subjects blindly imitate the observed behavior without internalizing why it is optimal to do so; on the other hand, it could mean that exposure to an alternative approach challenges subjects to reconsider what the optimal decision rule is and why hiring the orange worker is indeed the payoff-maximizing option. In this and the subsequent section, I show that in the aggregate it is neither, and that the operative channel lies between these two extremes. There is, of course, heterogeneity across subjects (some are pure imitators and some are deep learners, as the classification in Section 3.4 makes precise), but the goal here is to explain what drives the treatment effect in the aggregate. To this end, I consider three potential learning channels, summarized together with their observable predictions in Table 1.

This subsection evaluates the first two channels: a diagnostic treatment with exposure to *suboptimal* choices tests mechanical imitation, and the timing of subjects' switching behavior provides direct evidence on selective adoption. The third channel, model revision, requires observing behavior once exposure ends and the environment changes; it is the subject of Section 4.

Exposure to Suboptimal Behavior

If subjects simply replicate observed behavior regardless of its content, then exposure to suboptimal decisions should produce a decline in performance comparable in magnitude to the improvement generated by exposure to optimal decisions. To test this prediction, the experimental design includes an additional treatment, ExposureNeg (**ExpNeg**), depicted in Figure 8. A total of 53 subjects were assigned to this treatment. In Part 2, ExposureNeg subjects observe the hiring choices of a participant from a prior session who consistently made *suboptimal* decisions, always hiring the green worker whenever both candidates were educated. As in Exposure, the observed choices are never labeled, and the instructions are identical up to the identity of participant X. In Part 5, subjects in this group read the written recommendations of the same participant. In all other parts, ExposureNeg subjects make decisions individually, identical to all other participants.

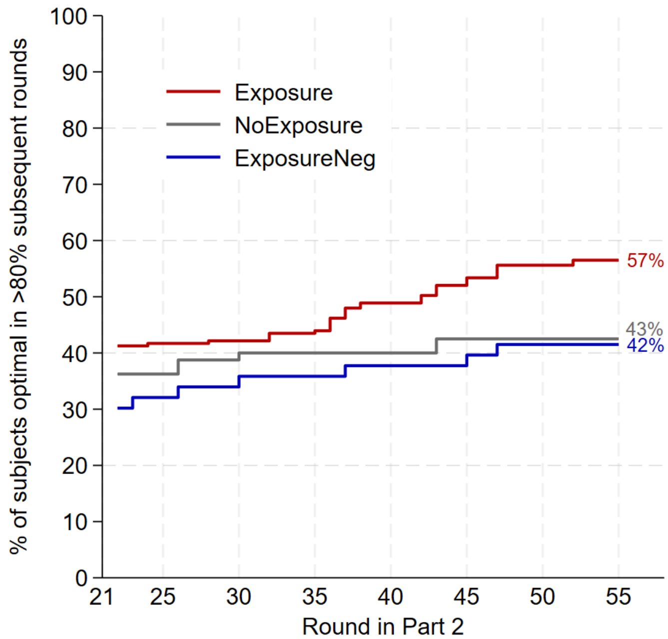
Figure 8: The ExposureNeg Treatment



The data reject the mechanical imitation hypothesis. In the last ten rounds of Part 2, the share of optimal choices in ExposureNeg is 48 percent, compared with 49 percent in NoExposure, a difference that is not statistically significant ($p = 0.935$). By the final round of Part 2, 47 percent of ExposureNeg subjects make the optimal choice, again indistinguishable from the control group ($p = 0.918$). In contrast, the Exposure group reaches 62 percent in the last ten rounds and 67 percent in the last round. The gap between Exposure and ExposureNeg is statistically significant in the last ten rounds ($p = 0.046$) and especially pronounced in the last round ($p = 0.008$), confirming that observing optimal choices generates substantially larger behavioral gains than observing suboptimal ones. The null for ExposureNeg is also informative about magnitudes: with the baseline control, the 95 percent confidence interval for its effect on the last-ten optimal choice share spans -0.10 to $+0.09$ (Table 7), so the data rule out deterioration larger than about ten percentage points, though not more moderate adverse effects, and the confidence heterogeneity reported in Appendix F indicates where such effects concentrate.

The dynamics of these treatment effects are illustrated in Figure 9. As in Figure 7a, the figure plots the cumulative percentage of subjects who choose the optimal alternative in a given round and continue to do so in at least 80 percent of subsequent rounds. The Exposure line rises steadily from round 35 onward, reaching 57 percent by round 55, while the ExposureNeg trajectory closely tracks the NoExposure control group throughout (42 percent versus 43 percent at round 55). The same ordering holds for the simple last-round choice rates: in the final round of Part 2, 67 percent of

Figure 9: Dynamics of Optimal Behavior: Exposure to Optimal and Suboptimal Choices



Notes: Red, blue, and gray lines correspond to Exposure ($N = 223$), ExposureNeg ($N = 53$), and NoExposure ($N = 80$), respectively. The figure displays the percentage of subjects choosing the optimal alternative in a given round of Part 2 and then continuing to do so in at least 80% of subsequent rounds; the lines end at round 55, beyond which too few rounds remain for the criterion to be meaningful, and the numbers displayed next to each line report the value of this measure at round 55.

Exposure subjects choose optimally, compared with 46 percent in NoExposure and 47 percent in ExposureNeg. The visual pattern confirms that the treatment effects documented in the main analysis are specific to exposure to optimal choices and cannot be attributed to a generic effect of observing any external decisions. Regression evidence underlying these patterns is reported in Table 2.

The asymmetry is sharpest when subjects are split by their own baseline behavior. Two groups in the design observe a peer who systematically *contradicts* their own choices: baseline-suboptimal subjects in Exposure, whose observed peer always chooses optimally, and baseline-optimal subjects in ExposureNeg, whose observed peer always chooses suboptimally. Mechanical imitation predicts that both groups should drift toward the observed behavior. As Table 3 shows, only the first group moves: baseline-suboptimal Exposure subjects raise their share of optimal choices from 0.13 to 0.46 (+33 percentage points, against +14 for comparable NoExposure subjects), while baseline-optimal ExposureNeg subjects do not move at all (0.94 to 0.94, against −2 points for comparable control subjects). Subjects adopt observed behavior only when it outperforms their own approach, a first indication of the selective adoption channel developed next. The baseline split also locates the causal effect where randomization identifies it: within the baseline-suboptimal

Table 2: Exposure to Optimal vs. Suboptimal Choices

	Part 1 (Individual)		**Part 2 (Social)**	
	Last 10	**Last 1**	**Last 10**	**Last 1**
	(1)	(2)	(3)	(4)
Exposure	0.008 (0.053)	0.023 (0.065)	0.128** (0.056)	0.210*** (0.064)
ExposureNeg	0.004 (0.073)	0.028 (0.088)	-0.006 (0.078)	0.009 (0.089)
NoExposure Mean	0.398	0.425	0.491	0.463
Observations	3,560	356	3,560	356
H_0: Exp = ExpNeg	$p = 0.957$	$p = 0.954$	$p = 0.046$	$p = 0.008$

Notes: Each column reports a separate OLS regression of a round-level optimal choice indicator on treatment dummies (Exposure and ExposureNeg), with the NoExposure group as the reference category. In Part 1, all subjects ($N = 356$) make individual choices, so no treatment differences are expected. In Part 2, Exposure subjects ($N = 223$) observe another participant's optimal choices, while ExposureNeg subjects ($N = 53$) observe suboptimal choices; NoExposure subjects ($N = 80$) continue independently. Standard errors are clustered at the subject level and shown in parentheses. NoExp Mean denotes the average share of optimal choices among NoExposure subjects. The bottom row reports the p-value from a test of equality between the Exposure and ExposureNeg coefficients. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

stratum, regressing the Part-2 outcome on treatment with a baseline control yields an Exposure effect of 17.9 percentage points ($p < 0.001$), free of the post-treatment selection that complicates within-group classifications.[42]

Payoff-Driven Selective Adoption

The asymmetry documented above rules out mechanical imitation but does not by itself establish how exposure to optimal choices produces learning. This part provides direct evidence on the selective adoption channel by characterizing *when* subjects switch to the observed alternative: transitions between consecutive choices reveal what triggers the decision to try the observed choice, and what sustains it once adopted.

For each treatment, Table 4 reports the probability of choosing optimally in round t conditional

[42] Baseline confidence, measured through an incentivized elicitation at the end of Part 1 in which subjects bet on the optimality of their own choices, provides a complementary margin of susceptibility. While exposure to suboptimal choices has no effect on average, its interaction with baseline confidence is statistically significant: point estimates indicate an adverse effect on the quality of own decisions concentrated among the least confident subjects, whereas the choices of more confident subjects appear unaffected. The beneficial effect of exposure to optimal choices, by contrast, is stable across the entire confidence distribution, consistent with confident subjects evaluating and filtering the behavior they observe rather than copying it. Appendix F describes the elicitation and reports the full analysis.

Table 3: Who Moves? Baseline Behavior and Contradicting Observations

	Part 1	**Part 2**	**Change**
Subjects suboptimal at baseline ...			
... observing **optimal** choices (EXPOSURE)	0.13	0.46	+33 pp
... observing no one (NOEXPOSURE)	0.18	0.32	+14 pp
Subjects optimal at baseline ...			
... observing **suboptimal** choices (EXPOSURENEG)	0.94	0.94	0 pp
... observing no one (NOEXPOSURE)	0.94	0.92	−2 pp

Notes: Entries are shares of optimal choices in the last ten key rounds of Parts 1 and 2. Subjects are optimal at baseline if they make at least eight optimal choices in the last ten key rounds of Part 1 ($y_1 \geq 0.8$), the threshold used throughout the paper. Group sizes: 150 (EXPOSURE, suboptimal), 57 (NOEXPOSURE, suboptimal), 18 (EXPOSURENEG, optimal), 23 (NOEXPOSURE, optimal).

on the subject's own choice in round $t-1$ and the payoff that choice delivered. Aggregate behavior exhibits substantial inertia: in EXPOSURE, subjects who chose optimally in round $t-1$ continue to do so with probability 0.87, and suboptimal choices persist at a similar rate of 0.82. The informative variation lies in how transitions respond to the subject's own realized payoff, on two margins. The first margin is *trying* the observed alternative. Conditional on a suboptimal choice in round $t-1$, subjects in EXPOSURE switch to the optimal choice with probability 0.22 when their own choice yielded a low payoff ($13), but only 0.12 when it yielded a high payoff ($18): a low own payoff raises the propensity to try the observed alternative by over 80 percent.[43] A weaker version of the same response is present in NOEXPOSURE (0.18 versus 0.13), reflecting exploration driven by own feedback alone, and it is entirely absent in EXPOSURENEG (0.13 versus 0.13), where the observed peer *agrees* with the subject's own suboptimal choice. The second margin is *keeping* the adopted choice. Conditional on an optimal choice in round $t-1$, EXPOSURE subjects persist with probability 0.90 after a high payoff and 0.84 after a low one, whereas persistence in NOEXPOSURE is payoff-insensitive (0.83 versus 0.83). This contrast is consistent with the composition of the two groups: optimal choosers in the control group predominantly follow a rule they understand, while the EXPOSURE group includes many recent adopters whose new behavior is still sustained by favorable payoff realizations.[44]

These contrasts are statistically distinguishable, which separates selective adoption from generic

[43]Throughout, I describe this as a response to a *low payoff* rather than to "disappointment": the data record payoffs, not emotions, though it is natural to conjecture that a low realized payoff is accompanied by disappointment about not having earned more.

[44]The within-EXPOSURE type split, using the classification introduced in Section 3.4, confirms this reading: always-takers are payoff-insensitive on the keeping margin (0.98 versus 0.97), while compliers are strongly payoff-sensitive on both margins (0.92 versus 0.84 for keeping; 0.47 versus 0.26 for trying).

Table 4: Choice Transitions by Own Payoff Experience, Part 2

	Probability of choosing optimally in round t			
	Trying: switch to optimal (suboptimal at $t-1$)		**Keeping**: stay optimal (optimal at $t-1$)	
Own payoff at $t-1$:	High ($18)	Low ($13)	High ($18)	Low ($13)
Exposure	0.12	**0.22**	**0.90**	0.84
NoExposure	0.13	0.18	0.83	0.83
ExposureNeg	0.13	0.13	0.86	0.84

Notes: The table reports transition probabilities between consecutive rounds of interest in Part 2, conditional on the subject's own choice in round $t-1$ and the payoff it delivered. In rounds of interest both workers are educated, so realized payoffs equal $18 if the hired worker is of high ability and $13 if of medium ability. All conditioning variables are observable to the subject. Trying-margin cells pool 1,029 (high) and 1,701 (low) transitions from 145 and 164 subjects in Exposure, 442 and 699 transitions from 59 and 62 subjects in NoExposure, and 313 and 492 transitions from 38 and 42 subjects in ExposureNeg.

win-stay/lose-shift exploration. Estimating a linear probability model for switching to the optimal choice on the trying margin (with indicators for a low own payoff in round $t-1$, treatment, and their interactions, and standard errors clustered at the subject level) yields a low-payoff response of 10.8 percentage points within Exposure ($p < 0.001$), of which 5.9 points exceed the own-feedback exploration response in NoExposure ($p = 0.013$ for the interaction). In ExposureNeg, where the observed peer agrees with the subject's own suboptimal choice, the response is absent (-0.6 points, $p = 0.76$) and significantly smaller than in Exposure ($p < 0.001$). A low payoff on one's own choice thus triggers movement specifically toward an observed alternative that contradicts the subject's current approach.

These patterns characterize the *timing* of adoption. A lower-than-expected payoff from one's own choice opens the window in which the observed alternative is tried: subjects act on the contrast between their own approach and the observed one primarily after their own approach delivers a low payoff. Favorable payoff realizations then sustain the new behavior once adopted. The comparison across treatments underscores that both ingredients matter: absent an observed alternative (NoExposure), a low own payoff produces only weak undirected exploration; and when the observed peer confirms the subject's own suboptimal choice (ExposureNeg), the low-payoff response disappears altogether. In Appendix E, I show that a simple Q-learning model in which observation expands the state space reproduces these dynamics.

The evidence in this subsection thus points to selective adoption as the operative channel on aggregate: observation supplies a concrete alternative, and subjects' own payoff experience governs whether and when it is adopted and retained. The first two candidate channels are summarized in the following result:

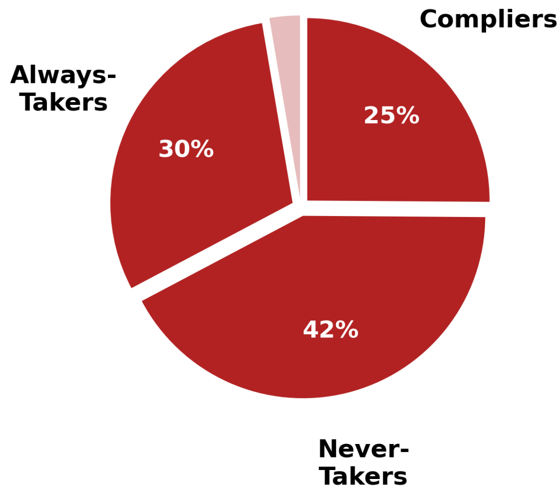
Result #2: *The effects of exposure are asymmetric in the quality of observed choices: exposure to optimal behavior substantially improves the quality of choice, while exposure to suboptimal behavior leaves performance statistically indistinguishable from the control group. Consistent with selective adoption, subjects in Exposure are much more likely to switch to the optimal choice after their own suboptimal choice yields a low payoff, a response that is weaker without exposure.*

The mechanism documented in this subsection also clarifies what Exposure subjects actually receive: they observe another participant's *actions*, never her payoffs, and this exposure to choices alone generates a substantial treatment effect. A natural question is whether the effect would be even larger if observation were accompanied by richer payoff information. The question relates to the literature on learning from endogenously selected samples (Esponda & Vespa, 2018): because subjects' own choices determine the feedback they generate, a subject who adopts the observed optimal action begins to experience its payoff distribution directly, and part of the measured improvement could in principle reflect this endogenously enriched feedback rather than the observation of choices per se.

To address this question, the experimental design includes a further diagnostic treatment, Exposure (Full Info), in which subjects observe the same optimal choices as in Exposure and are additionally shown, in every round, the payoff that would have resulted from the alternative hiring decision. A subject whose own choice is suboptimal therefore sees both that the observed participant chose differently and what that choice would have paid. By construction, this treatment removes any informational advantage of adopting the observed action, since counterfactual payoffs are available regardless of one's own choice. Behavior in this group is statistically indistinguishable from Exposure: 70 percent versus 62 percent optimal choices over the last ten key rounds ($p = 0.203$) and 66 percent versus 67 percent in the final round ($p = 0.863$). Exposure to choices alone thus captures the bulk of the treatment effect, and supplementing it with counterfactual payoff information yields no significant further improvement. Given the size of this arm, the comparison rules out large incremental gains from richer feedback, though not moderate ones; Appendix C.1 reports the full analysis and the associated caveats.

What this evidence cannot yet establish is the depth of the resulting learning: whether adoption remains tied to the context in which it was reinforced, or whether some subjects go further and revise their model of the task. Distinguishing these possibilities is the purpose of the classification introduced next and of the analysis in Section 4.

Figure 10: Exposure Learning Type Classification



Notes: The Exposure treatment includes 223 participants. Among them, 56 are classified as Compliers, 67 as Always-Takers, and 94 as Never-Takers. Compliers are participants who performed suboptimally at baseline in Part 1 and optimally in Part 2 when exposed to optimal choices; Always-Takers are those who performed optimally in both parts; and Never-Takers are those who performed suboptimally in both parts. The remaining 6 subjects performed optimally in Part 1 and suboptimally in Part 2 (defiers); they are excluded from the analysis.

3.4 Classification of Exposure Subjects By Learning Effects

To assess learning gains from observing others' optimal decisions, it is helpful to compare subjects' baseline behavior, when they confronted the task independently, with their choices at the end of the social-learning component. In principle, three learning trajectories may arise. The first consists of subjects who start with a low share of optimal choices but reach high optimality rates once they observe others' decisions; these subjects are referred to as *compliers*. The second consists of subjects who make optimal choices regardless of social exposure; these are labeled *always-takers*. The third consists of subjects who show little or no response to exposure and continue to choose suboptimally even when presented with optimal decisions; these are labeled *never-takers*.[45] To operationalize these learning types, I rely on the following performance measure:

Definition: *Let $y_t \in [0, 1]$ denote the share of optimal choices a subject makes in the final ten* ***key*** *rounds of Part t. A subject is said to exhibit* ***optimal behavior*** *in Part t if $y_t \geq 0.8$.*

Using this criterion, Exposure subjects can be grouped into three learning types based on

[45]There is also a fourth possibility in which subjects begin with a high share of optimal choices but later move to a low optimality rate after observing others' optimal decisions. These subjects can be viewed as *defiers*. In this sample, only six out of 223 subjects fall into this category, so I exclude them from subsequent analysis and focus on the three primary groups.

how their choices respond to observing optimal decisions: *compliers* satisfy $y_1 < 0.8$ and $y_2 \geq 0.8$, *always-takers* satisfy $y_1 \geq 0.8$ and $y_2 \geq 0.8$, and *never-takers* satisfy $y_1 < 0.8$ and $y_2 < 0.8$. This classification provides a direct mapping between subjects' observed choices and the learning trajectories described above.[46] As Figure 10 shows, approximately 25 percent of Exposure subjects are classified as compliers, 30 percent as always-takers, and 42 percent as never-takers. The main focus of the analysis in the remainder of the paper is on the roughly one-quarter of subjects classified as compliers, whose realized trajectories shift from suboptimal to optimal behavior during the exposure phase, while the remaining subjects either were already making optimal choices or did not cross the threshold. Because the classification conditions on realized Part-2 outcomes within the treated group, it describes realized trajectories rather than a causally identified stratum: in NoExposure, 13 of 80 subjects (16 percent) cross the same threshold through individual learning alone, compared with 25 percent under exposure. The causal statements in this paper accordingly rest on the randomized comparisons of Sections 3.2 and 3.3, with the taxonomy serving as a lens on the heterogeneity behind them.

Observation: *Exogenously varying exposure to optimal behavior has its largest impact on the one quarter of participants who comply with the treatment and shift from suboptimal to optimal choices, the group I refer to as compliers.*

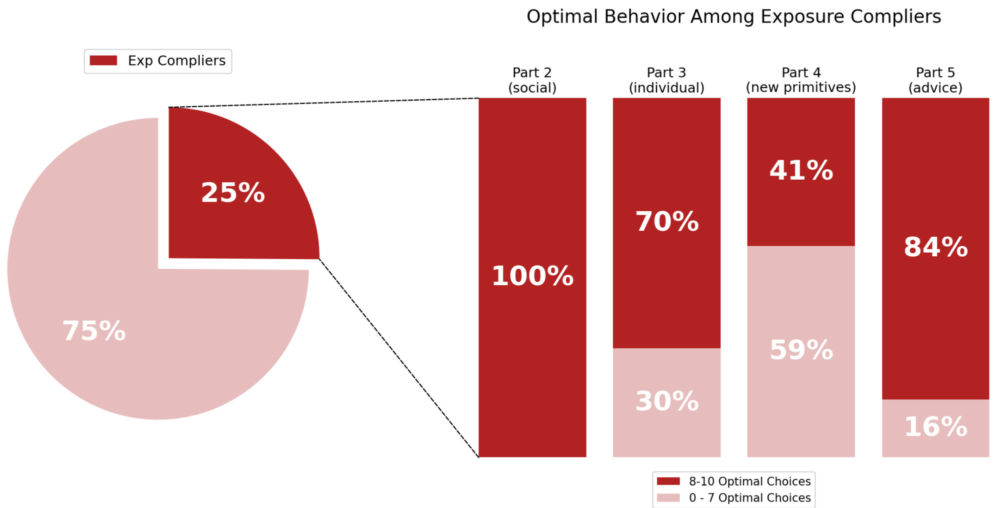
The goal of the next section is to evaluate how much of this treatment effect persists once social exposure is removed, providing insight into the robustness of the learning gains.

4 Learning Gains Beyond Exposure

The analysis so far shows that the positive effect of exposure is not driven by pure imitation: subjects distinguish the quality of the choices they observe, successfully adopting them when exposed to optimal behavior while disregarding external choices when exposed to suboptimal behavior. The dynamics of their behavior point in the same direction: the payoffs from subjects' own choices predict the likelihood of switching to the observed alternative, and that likelihood differs sharply across treatment groups, indicating that subjects engage in selective adoption of observed behavior. The question I tackle next is whether exposure can accomplish more than that: is it powerful enough that participants internalize not only the behavioral pattern but the underlying decision rule, in the sense that their learning gains would be retained beyond exposure and transfer to similar but

[46] The threshold of 0.8 aligns with the median optimality rate of 80 percent for Exposure subjects in Part 2 (y_2). Importantly, the qualitative conclusions from the complier analysis in the next section are robust to using alternative thresholds.

Figure 11: Share of EXPOSURE Compliers Exhibiting Optimal Behavior



Notes. Compliers in the EXPOSURE group are subjects who performed suboptimally in Part 1 but switched to optimal decisions in Part 2, where optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part. Out of 223 subjects in this group, 56 (25%) meet this criterion. Each bar reports the percentage of these original compliers who continue to exhibit optimal behavior in subsequent parts. Part 3 (Individual) returns subjects to independent decision-making; Part 4 (New Primitives) introduces a change in the environment's underlying primitives; and Part 5 (Advice) provides subjects with written guidance.

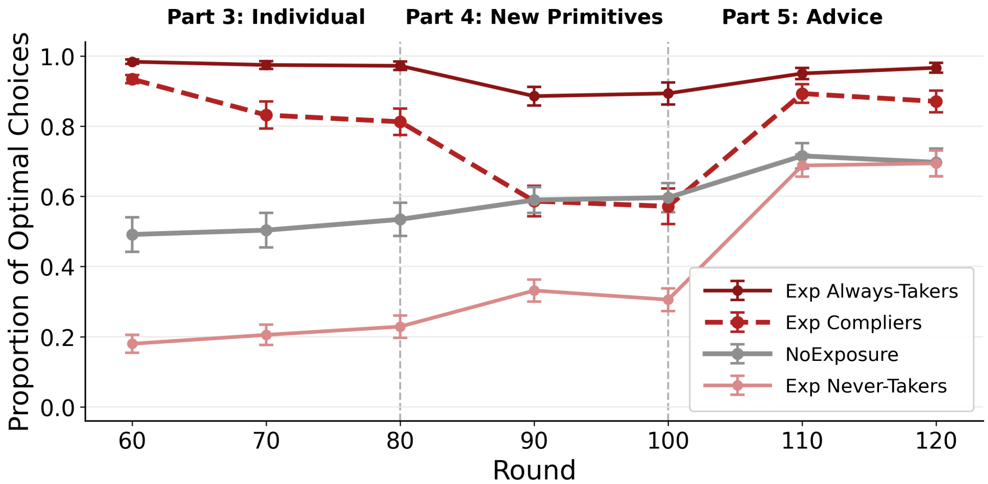
new decision environments, consistent with the third channel proposed earlier, model revision?

The learning induced by exposure may range from mechanical imitation of observed choices, through rules that remain bound to the context in which they were acquired, to revision of the underlying model. The main goal of this section is to provide further insights into what subjects actually learn from exposure and to assess whether this learning can be transferred when the primitives of the environment change. In Section 4.1, I examine subjects' decisions once they return to making choices independently in Part 3 of the experiment. In Section 4.2, I study whether subjects continue to behave optimally when they encounter a new hiring task in Part 4, in which the primitives of the environment differ from those they experienced previously. Based on the classification described above, I estimate that roughly 30 percent of the learning observed among EXPOSURE compliers reflects shallow imitation, another 30 percent reflects context-bound imitation, and the remaining 40 percent reflects deeper learning.

4.1 Retention of Learning

The simplest test of the stability of learning gains is to keep the task environment fixed and remove only the exposure to others' decisions. This is precisely what subjects face in Part 3, where they return to making choices independently under the same conditions as in Part 1. For

Figure 12: Incidence of Optimal Behavior Beyond Exposure



Notes. This figure displays the evolution of the average proportion of optimal choices made by subjects across Parts 3–5 for the three subgroups within the Exposure treatment, as well as for the NoExposure control group. Each line tracks the average performance from the last 10 rounds by aggregating individual choices across subjects in a given group. The initial level corresponding to round 60 represents the proportion of optimal choices in the last ten rounds of Part 2 (not shown in the graph). Compliers are subjects who performed suboptimally in Part 1 but switched to optimal decisions in Part 2; Always-Takers are those who performed optimally in both parts; and Never-Takers are those who performed suboptimally in both parts. Part 3 (Individual) returns subjects to independent decision-making; Part 4 (New Primitives) introduces a change in the environment's underlying primitives; and Part 5 (Advice) provides subjects with written guidance. Markers denote group means at each ten-round mark; error bars indicate ± 1 standard error around each estimate.

compliers in Exposure, Figure 11 reports the share who continue to follow the optimal decision rule. By construction, all compliers choose optimally by the end of the social-learning phase in Part 2. Once external choices are removed, however, 17 of the 56 compliers (30 percent) revert to the dominated option. This evidence suggests that once exposure ends, a substantial share of the learning gains dissipates. Because compliers are selected for crossing a threshold in a short window, part of the measured reversion reflects sampling noise rather than behavior: the null benchmarks in Appendix D.4 attribute roughly one fifth of observed reverters to chance threshold-crossing, so the 30 percent figure is best read as an upper bound on shallow imitation. For most of these compliers, however, the improvements observed during social learning appear to rest on imitation of others' choices, leading to gains that are sizable but ultimately fragile.

To shed light on the intensive margin, Figure 12 plots the share of optimal choices for each Exposure subgroup alongside the NoExposure benchmark. Although compliers reached an average optimality rate of 93 percent at the end of Part 2 under exposure, this falls to roughly 82 percent by the end of Part 3. Despite this decline, their performance remains well above that of the control group (53 percent), though markedly below the level observed among always-takers, whose average remains extremely high at 97 percent ($p < 0.001$, see Table 5). A further noteworthy pattern in Figure 12 is that while Exposure never-takers are defined as having fewer than 80 percent optimal choices in Parts 1 and 2, their actual average optimality rate in Parts 1–2 is below 20 percent. For this group, returning to independent decision-making produces little change, consistent with the fact that they did not respond to social exposure in the first place.

In summary, a simple test of learning retention, implemented by returning subjects to individual decision-making once exposure is removed in Part 3, reveals a notable pattern. Among compliers in the Exposure group, who initially switched to optimal behavior, 30 percent (17 of 56) revert to suboptimal behavior. Correspondingly, their share of optimal choices declines from 93 percent to 82 percent. This group is the only one to experience such a decline, while optimal choice proportions among Exposure always-takers and never-takers, as well as NoExposure subjects, remain essentially unchanged. This suggests that for a substantial subset of subjects, improved performance depends on shallow imitation through continued exposure, and that once this exposure is removed, these subjects revert toward suboptimal choices rather than internalizing the optimal strategy.

4.2 Transfer of Learning

The results in the previous section show that 70 percent of Exposure compliers retain their learning gains even after exposure to optimal behavior ends. However, this evidence does not distinguish between two underlying forms of learning. One possibility is that subjects learn to recognize the optimality of observed choices and continue to imitate them during and after exposure, as long as the task environment remains unchanged. A second possibility is that subjects acquire a deeper understanding of why those choices are optimal, allowing them to apply what they have learned to a related task with different underlying primitives. In this section, I examine whether the learning gains observed in the Exposure group transfer to a related but distinct environment, in which subjects face new primitives and therefore cannot directly rely on previously accumulated experience.

Beginning in Part 4 of the experiment, subjects switch to a new hiring task with a different set of primitives. The ability distributions change so that the green group now includes two low-, one medium-, and one high-ability worker, while the orange group includes two low- and two medium-ability workers. In addition, the probability that a medium-ability worker is educated, p_m, falls from 0.9 to 0.1. Together, these changes flip the environment: whereas in earlier parts both workers

Table 5: Incidence of Optimal Behavior by Exposure Learning Type

	Part 3 (Individual)		Part 4 (New Primitives)		Part 5 (Advice)	
	Last 10	**Last 1**	**Last 10**	**Last 1**	**Last 10**	**Last 1**
Compliers	0.289***	0.237***	-0.014	0.025	0.145***	0.286***
	(0.059)	(0.081)	(0.060)	(0.085)	(0.045)	(0.067)
Always-Takers	0.443***	0.443***	0.283***	0.296***	0.226***	0.375***
	(0.048)	(0.062)	(0.048)	(0.067)	(0.038)	(0.054)
Never-Takers	-0.304***	-0.310***	-0.280***	-0.291***	-0.033	0.024
	(0.056)	(0.070)	(0.049)	(0.073)	(0.048)	(0.074)
NoExposure Mean	0.529	0.512	0.610	0.600	0.735	0.625
Observations	2,970	297	2,970	297	2,970	297
H_0: Compl. = AT	$p < 0.001$	$p = 0.001$	$p < 0.001$	$p < 0.001$	$p = 0.009$	$p = 0.021$
H_0: Compl. = NT	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$

Notes: Each column reports a separate OLS regression of a round-level optimal choice indicator on Exposure behavior type dummies, with the NoExposure group ($N = 80$) as the reference category. The regression sample consists of 297 subjects: 80 from NoExposure and 217 from Exposure ($223 - 6$ defiers). Compliers are Exposure subjects who performed suboptimally in Part 1 but switched to optimal decisions in Part 2, where optimality is defined as making at least 80% optimal choices in the final ten rounds; Always-Takers performed optimally in both parts; and Never-Takers performed suboptimally in both parts. The Exposure group is classified into 56 Compliers, 67 Always-Takers, 94 Never-Takers, and 6 Defiers (excluded). Part 3 (Individual) returns subjects to independent decision-making; Part 4 (New Primitives) introduces a change in the environment's underlying primitives; and Part 5 (Advice) provides subjects with written guidance. The reported tests assess whether mean optimal choice rates differ across behavioral types within each part. Standard errors are clustered at the subject level and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

were educated in most rounds, in Part 4 both workers are now *not* educated in most rounds.[47] All other features remain the same: $p_h = 1$ and $p_l = 0$, and the green group still has higher average ability. Because a not-educated worker cannot be of high ability, the relevant comparison is the posterior probability of medium ability conditional on not being educated, which is higher for the orange group ($\mathbb{P}(m \mid not\ educ) \approx 0.47$ versus ≈ 0.31 for green; see Section 2.2). As a result, conditional on the modal signal, the optimal choice is still to hire an orange worker.

One can see from Figure 11 that among the original Exposure compliers, only 41 percent continue to exhibit optimal behavior under the new task primitives in Part 4. Their share of optimal choices drops further to 60 percent, falling below the level observed in NoExposure, as Figure 12 shows. In other words, despite having reached near-fully optimal performance under exposure,

[47]The analysis for Parts 4 and 5 focuses on these key rounds where both workers are not educated. In the remaining rounds, where at least one worker is educated, nearly all subjects choose the dominant option.

the compliers' optimal choice rate reverts to that of the control group, whose members were never exposed to others' decisions. Because compliers are selected on realized Part-2 outcomes while NoExposure is an unselected mixture of types, this equality is a descriptive benchmark rather than a complier-specific causal contrast; the causally identified counterpart is the disappearance of the aggregate treatment advantage: taking the Exposure group as a whole, its share of optimal choices in the last ten rounds of Part 4 (57 percent) is statistically indistinguishable from NoExposure (61 percent, $p = 0.344$).

Could the decline simply reflect a harder environment? The structure of the task argues otherwise: the decision problem is unchanged (conditional on the modal signal, the same one-step inference identifies the optimal group) and the new primitives are close to a mirror image of the original ones. More importantly, the groups that never relied on observed choices show no deterioration at all. NoExposure subjects *improve* from 53 percent to 61 percent across the change in primitives (Table 5), never-takers improve from 22 percent to 33 percent, and always-takers dip only modestly, from 97 percent to 89 percent. A general increase in difficulty would depress all of these trajectories; instead, the sharp decline is specific to compliers. The always-takers' modest dip does suggest that re-mapping a familiar rule onto new signals is not entirely costless, and the possibility that such adjustment costs bind more tightly for recent adopters cannot be fully excluded; but adjustment costs alone cannot explain why compliers fall all the way to the level of a control group that is simultaneously improving. Nor can weaker feedback explain it: the expected per-round payoff difference between the two options is nearly twice as large in Part 4 as in Parts 1–3 ($0.82 versus $0.43; see Section 2.2). On balance, the evidence indicates that learning gains from exposure are largely tied to the context in which the observed choices were adopted, such that the average treatment effect dissipates once the environment primitives change.

Below, I summarize the main result from the analysis of learning gains beyond direct exposure using learning retention and learning transfer as natural benchmarks:

Result #3: *In Part 3, most Exposure compliers continue to behave optimally when returning to facing the task individually, and their average optimality rate remains high. In Part 4, under new task primitives, the average optimality rate among Exposure compliers falls substantially, to a level statistically indistinguishable from NoExposure subjects, who were never exposed to external choices. The advantage of more than 40 percentage points that compliers had built over the control group during exposure is entirely gone, pointing to the fragility of learning from others.*

4.3 Direct Advice

To contrast the effect of exposure to others' decisions with the explicit provision of recommendations, Part 5 of the experiment keeps the new environment primitives but introduces written advice on how to approach the task. This advice is provided by the same participant whose choices subjects observed in Part 2. The purpose of this section is twofold. First, since learning gains among Exposure compliers disappear under the new primitives, as shown in Section 4.2, this part examines whether explicit advice can restore optimal choice rates to previously high levels. Second, the largest subgroup within Exposure consists of never-takers (94 out of 223 subjects), who did not respond to exposure to optimal behavior and maintained optimality rates below 40 percent. This section therefore examines whether explicit recommendations have a stronger effect than exposure alone for this subgroup.

Recall what this part entails. Before the final 20 rounds, each treated subject reads a written recommendation composed by participant X, the same individual whose round-by-round choices she observed in Part 2. For Exposure subjects, the advice explains how to approach the task and how to extract information from the education signals; it communicates the *why* that observing choices alone could not.[48] The comparison between passive exposure and explicit guidance also anticipates the policy discussion in Section 4.4: the two interventions differ not only in how strongly they move behavior, but in *whom* they reach.

For the Exposure group, the provision of explicit optimal guidance has a substantial effect on both margins. Among compliers, a large fraction respond to the advice: 84 percent exhibit optimal behavior by making at least eight optimal choices in the last ten rounds of Part 5, and their optimal choice rate rises to 88 percent, significantly above the NoExposure level, as Table 5 shows ($p = 0.001$).[49] The response among Exposure never-takers is even more striking. On the extensive margin, the share of never-takers exhibiting optimal behavior rises from 0 percent in Part 2 to 53 percent in Part 5. On the intensive margin, their optimality rate, which never exceeded 33 percent in the first four parts of the experiment, more than doubles to 70 percent, statistically

[48] ExposureNeg subjects instead read the recommendation of the participant whose *suboptimal* choices they observed. Explicitly suboptimal advice does not produce a symmetric decline: the ExposureNeg optimality rate in Part 5 (62 percent) is comparable to its Part 4 level (58 percent), while NoExposure continues to improve from 61 percent to 74 percent, generating a marginally significant 12 percentage point gap ($p = 0.056$), the only part of the experiment in which the two groups diverge. Suboptimal advice thus operates by failing to support the continued individual learning observed in the control group, rather than by steering subjects toward the dominated choice. In all other post-exposure parts, ExposureNeg behavior remains statistically indistinguishable from NoExposure (Part 3: $p = 0.837$; Part 4: $p = 0.602$).

[49] Always-takers also respond to the advice, reaching a 96 percent optimality rate in Part 5 compared to 89 percent in Part 4.

indistinguishable from the control group level, as Figure 12 illustrates. Following the provision of optimal advice, the Exposure group as a whole achieves a 9 percentage point higher optimal choice rate than the control group ($p = 0.020$).[50]

The picture that emerges is consistent across subgroups: never-takers, whom mere exposure to optimal behavior left unmoved, respond strongly once the justification behind the optimal choice is provided. Two features of the design qualify how these gains should be read. Advice arrives in Part 5 for every treated subject (there is no continuation arm that withholds it), and NoExposure itself improves from 61 percent to 74 percent over the same rounds, so the Part-5 changes identify the effect of adding advice to each group's accumulated history rather than the standalone effect of advice. The never-takers' jump from at most 33 percent in earlier parts to 70 percent is nonetheless far larger than any natural trajectory observed in the data, consistent with a substantial contribution of the explanation itself. The results on explicit guidance are summarized as follows:

Result #4: *Following explicit advice in Part 5, the optimal choice rate among Exposure compliers returns to its original high level. The largest response comes from Exposure never-takers: for this subgroup, whom exposure to optimal behavior alone failed to move, a direct explanation of the optimal approach more than doubles the share of optimal choices.*

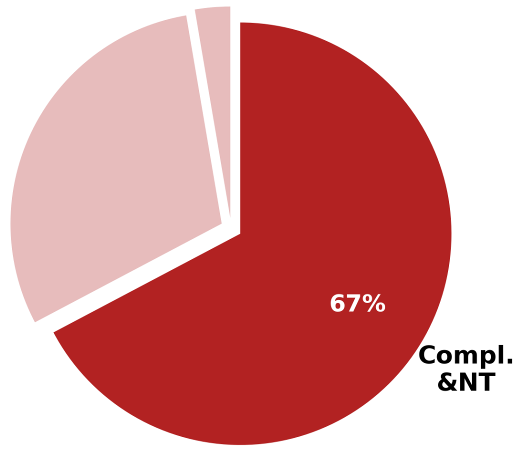
4.4 Implications for Policy

The preceding analysis reveals substantial heterogeneity in how subjects respond to exposure and advice. In this section, I synthesize these findings to draw out their implications for intervention design. I begin by defining the population that is plausibly amenable to policy and then develop a taxonomy of learning types that distinguishes subjects by the depth and durability of the behavioral change they exhibit. I then examine how each type responds across the sequence of interventions and discuss what these patterns imply for the design of policies aimed at reducing statistical discrimination.

Of the 223 subjects in the Exposure group, 67 are always-takers who already behave optimally before any social exposure, and 6 are defiers who move in the opposite direction; both groups are set aside, the former because they do not require intervention and the latter because of their negligible number. The remaining 150 subjects, 56 compliers and 94 never-takers, constitute the *treatable*

[50] The overall effect of positive recommendation is relatively moderate despite improvements across all Exposure subgroups. This is partly driven by a gradual increase in the NoExposure group to a 74 percent optimality rate in Part 5, even though these subjects continue to face the task individually and learn only from feedback.

Figure 13: Treatable Sample within the EXPOSURE Group



Notes. This figure highlights the treatable sample within the EXPOSURE group ($N = 223$). Always-takers (67 subjects) are those who performed optimally before exposure and are excluded from the policy-relevant population. The treatable sample consists of 150 subjects: 56 compliers, who switched from suboptimal to optimal behavior under exposure, and 94 never-takers, who did not respond to exposure. Optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part.

sample: individuals who begin the experiment with suboptimal decision-making and are therefore plausible targets of policy interventions (Figure 13). This group exhibits considerable heterogeneity in the depth of learning from exposure and in its responsiveness to explicit guidance, which I now characterize in detail.

Complier subtypes. Among the 56 compliers, the experimental design allows me to distinguish three subtypes based on how their behavior evolves across Parts 3–5. The subtypes are the behavioral footprints of the three learning channels introduced in Section 3.3: deep learners correspond to model revision, context-bound learners to selective adoption that remains anchored to the environment in which it was reinforced, and shallow imitators to adoption sustained only by contemporaneous observation. The first subtype, which I label *Deep Learners* (C1), consists of subjects who retain optimal behavior both after exposure ends in Part 3 and when the environment primitives change in Part 4. This pattern is consistent with having internalized the underlying decision rule, namely, that hiring from the minority group maximizes expected payoff conditional on the modal signal, rather than merely replicating the observed choices. Because the optimal action remains Orange in the key rounds of the new environment, however, continued optimal behavior is also compatible with a simpler carried-over rule, such as choosing Orange whenever the two signals coincide; the Deep Learner label is therefore a provisional name for the strongest observed behavioral pattern, with the response to advice and subjects' own reflections providing complementary, though not decisive, evidence of understanding. With this qualification, Deep Learners represent the strongest form of behavioral change induced by exposure. Of the 56 compliers, 22 subjects (39 percent) fall into this category.

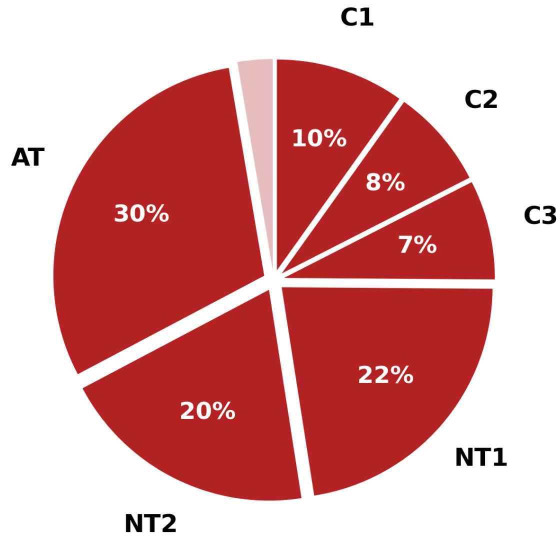
The second subtype, *Context-Bound Learners* (C2), comprises subjects whose learning persists when the task environment is unchanged (Part 3) but who revert to suboptimal behavior when the underlying primitives change (Part 4). These subjects appear to have learned a strategy that works well in the original context but remains anchored to the specific features of that environment. When the environment shifts, their understanding does not transfer. This group contains 17 subjects (30 percent of compliers).

The third subtype, *Shallow Imitators* (C3), consists of subjects who revert to suboptimal behavior as soon as direct exposure ends in Part 3. For these individuals, the improvement observed during social learning appears to rest entirely on contemporaneous reinforcement of observed choices, with no durable change in decision-making. This group contains 17 subjects (30 percent of compliers). Given the size of these subgroups, the subtype-level analysis is best read as a descriptive characterization of the heterogeneity behind the average effects, rather than as a basis for formal between-subtype inference; Appendix D shows that the composition of the taxonomy is stable across alternative classification thresholds and quantifies how much of it could arise from noise alone.

Never-taker subtypes. Among the 94 never-takers, the introduction of explicit advice in Part 5 reveals a further distinction. *Eventual Responders* (NT1) are subjects who do not respond to passive exposure but reach optimality once they receive explicit guidance explaining the rationale behind the optimal strategy. This group comprises 50 subjects (53 percent of never-takers). Notably, while NT1 subjects are classified as never-takers because they do not cross the optimality threshold in Part 2, they show gradual signs of improvement even before advice is introduced: 12 percent exhibit optimal behavior in Part 3 and 24 percent in Part 4. The large jump to 100 percent optimality in Part 5, however, occurs only after explicit guidance is provided, suggesting that these subjects benefit from a combination of accumulated experience and a clearly articulated rationale. The remaining 44 subjects are *Non-Responders* (NT2), who remain suboptimal throughout all five parts of the experiment despite receiving both exposure and advice. Among NT2 subjects, the share exhibiting optimal behavior never exceeds 2 percent in any part. These subjects may require fundamentally different forms of intervention, such as changes to the choice architecture itself.

The full taxonomy is summarized in Figure 14. To visualize how each learning type responds across the sequence of experimental interventions, Figure 15 plots the share of subjects exhibiting

Figure 14: Full Taxonomy of Learning Types in the EXPOSURE Group



Notes. This figure displays the complete classification of subjects in the EXPOSURE group ($N = 223$) into six learning types. Always-Takers (67 subjects) performed optimally before exposure. The remaining 150 subjects form the treatable sample, further classified by their response to exposure and advice. Among compliers (56 subjects): C1 Deep Learners retain optimal behavior across all parts including under new primitives; C2 Context-Bound Learners retain optimality in the original context but revert under new primitives; C3 Shallow Imitators revert as soon as exposure ends. Among never-takers (94 subjects): NT1 Eventual Responders reach optimality only after receiving explicit advice in Part 5; NT2 Non-Responders remain suboptimal throughout. Optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part.

optimal behavior in each part, separately for each subtype within the treatable sample. The heatmap reveals sharply divergent trajectories. Deep Learners (C1) reach 100 percent optimality in Part 2 and maintain it throughout all subsequent parts, including under the new primitives in Part 4. Context-Bound Learners (C2) follow the same path through Part 3 but drop to 0 percent under new primitives, before recovering to 82 percent once explicit advice is provided. Shallow Imitators (C3) revert immediately in Part 3 and remain at or near 0 percent until advice restores two-thirds of them to optimality. Among never-takers, Eventual Responders (NT1) show a gradual trajectory, from 0 percent in Parts 1 and 2, to 12 percent in Part 3 and 24 percent in Part 4, culminating in 100 percent optimality once advice is introduced. Non-Responders (NT2), by contrast, remain at or near 0 percent throughout. These patterns underscore that exposure and advice operate through distinct channels and that the same intervention can produce qualitatively different responses depending on the depth of a subject's underlying learning process.

Open-ended reflections by subtype. At the end of the experiment, subjects answered two unincentivized open-ended questions: the first asked them to describe the strategy they employed and whether it evolved over the 120 rounds; the second asked specifically how they used participant X's hiring choices in Rounds 21–60 and X's recommendations and explanations in Rounds 101–120

Figure 15: Response to Interventions by Learning Type

Notes. This figure displays the share of subjects exhibiting optimal behavior in each part of the experiment, separately for each learning type within the treatable sample of the Exposure group. Rows correspond to the five subtypes: C1 Deep Learners, C2 Context-Bound Learners, and C3 Shallow Imitators among compliers; NT1 Eventual Responders and NT2 Non-Responders among never-takers. Columns correspond to the five experimental parts: Part 1 (Individual Learning), Part 2 (Social Exposure), Part 3 (Individual Learning), Part 4 (New Primitives), and Part 5 (Direct Advice). Cell color reflects the share behaving optimally, ranging from dark red (0%) to dark green (100%). Optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part.

when making their own decisions. The second question is particularly informative, and reading these responses by learning type reveals a striking correspondence between what subjects *say* they did with the observed choices and how their behavior classifies them.

Deep Learners (C1) describe observation as a prompt for understanding, using the language of insight rather than compliance: one writes that following X's choices "*made me realize how the probability worked, and that it was better to choose out of three educated vs four educated*," and another that X's choices "*made me realize that when the given information says educated, it rules out low skill workers, and that when the given information says uneducated, it rules out high skill workers*," a complete statement of the screening logic at the heart of the task. Their answers to the strategy question tell the same story: one deep learner articulates the full contingent rule, "*When worker is educated I chose the color with more workers in high and medium abilities. When worker was uneducated I chose the color that had least workers as low ability.*" Context-Bound Learners (C2) describe adopting what they saw without extracting a general principle; one reports switching away from green "*after studying participant X's choices*," yet continuing to select green workers under the new primitives "*in the scenario of uneducated workers for both*" until X's written explanation corrected them, precisely the on-support adoption without transfer that defines this subtype. Shallow Imitators (C3) use the language of deference: "*I follow the suggestions*," "*it was the same as participant X's*," or "*I went with participant X's choices and*

Table 6: Cumulative Reach of Interventions in the Treatable Sample

Intervention	Reach	Composition
Exposure alone	37% (56/150)	Compliers: C1 + C2 + C3
Exposure + Advice	65% (97/150)	Adds all NT1, most C2 & C3
Not optimal in Part 5	35% (53/150)	NT2 + some C2, C3

Notes: This table summarizes the cumulative share of the treatable sample (150 subjects: 56 compliers and 94 never-takers) that exhibits optimal behavior under each intervention. "Exposure alone" counts all subjects who reach optimality during the social learning phase in Part 2 (i.e., all compliers). "Exposure + Advice" counts all subjects who exhibit optimal behavior in Part 5, after receiving both exposure and explicit guidance. "Not optimal in Part 5" counts subjects who do not exhibit optimal behavior in the final part: all 44 NT2 subjects, together with 9 compliers who reached optimality earlier but do not return to it under advice. Optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part.

recommendations a lot since it seems like they know the strategy." Among never-takers, Eventual Responders (NT1) report discounting the observed actions but being persuaded by the explanation: "*I use X's recommendation more in 101–120 as the explanation make sense but not a lot in 21–60 rounds. Evidence can persuade people more.*" Non-Responders (NT2), finally, express deliberate disregard for both channels: "*I kept my same strategy of picking green workers no matter what*," or "*I looked at participant X's choices and it didn't seem right so I just continued with my strategies.*"

While these responses are self-reported and collected after all decisions were made, their alignment with the behavioral classification is reassuring: the depth of learning inferred from choices alone is mirrored in how subjects themselves describe what they took from observing, and later being advised by, participant X.

Aggregate reach of interventions. Table 6 summarizes the cumulative reach of the two interventions studied in this experiment. Under exposure to the optimal choices of others, roughly 37 percent of the treatable sample reaches optimal behavior (56 out of 150 subjects), corresponding to all complier subtypes combined (C1, C2, and C3). However, as the analysis above demonstrates, fewer than half of these subjects exhibit learning that is both durable and transferable. When exposure is supplemented with explicit advice in Part 5, the share of the treatable sample exhibiting optimal behavior rises to nearly two-thirds (97 out of 150, or 65 percent): a net increase of 41 subjects, as 50 never-takers newly reach optimality while 9 compliers fall back. Counting subjects who reach optimality at any point during exposure or advice, the two interventions together reach 106 of 150 (71 percent). The largest gains come from two sources: all 50 Eventual Responders (NT1), by construction the group that responds to advice but not to exposure (so the informative quantity is its size, over half of all never-takers), and the majority of Context-Bound Learners and Shallow

Imitators (C2 and C3), who had reverted under changed conditions but return to optimality once guidance is provided. The remaining 53 subjects (35 percent of the treatable sample), namely all 44 Non-Responders (NT2) together with 9 compliers who do not return to optimality under advice, are not at optimality at the end of the experiment.

Implications for intervention design. These findings carry practical implications for policies aimed at improving the quality of decision-making through observational learning. One important application concerns inaccurate statistical discrimination. Consider, for example, a hiring manager who systematically favors candidates from a particular educational background, even when applicants are equally qualified, thereby making a mistake by failing to leverage useful information about candidate quality *conditional on qualifications*. The results suggest that exposure to optimal behavior can serve as an effective, low-cost first intervention, similar to letting a manager observe the hiring decisions of a better-performing peer. However, the gains from such exposure are often fragile and context-specific: a manager may revert to old hiring practices once the peer observation ends, or struggle to apply the lesson when encountering a different pool of applicants.

Supplementing peer exposure with explicit guidance substantially extends the reach of the intervention. In the experimental setting, explaining *why* a particular choice is optimal (how conditioning on the common signal reverses the apparent ranking of the two groups) raises the share of the treatable sample at optimality by 27 percentage points, largely by reaching never-takers whom exposure alone had left unmoved; whether advice-induced gains would themselves survive a further change of environment is a question this design leaves open. At the same time, a nontrivial fraction of the population (35 percent) is not moved to optimality by either passive exposure or explicit recommendations, pointing to the potential value of complementary approaches, such as decision aids or algorithmic evaluation, that embed the conditional reasoning rather than relying on individual learning.

5 Discussion

Observational learning is widespread in workplaces: managers observe how colleagues screen the same job applicants, farmers watch how neighbors cultivate the same crop, doctors see how peers treat the same condition. Organizations rely on it because it is cost-effective and scalable, yet we know surprisingly little about what it actually teaches. If observers internalize *why* the observed decisions are correct, they can draw on that knowledge in new situations; if they only learn *what* to do in a particular setting, the gains can vanish as soon as the context changes. The two are indistinguishable while the environment is fixed. This paper separates them experimentally, in a setting where individual feedback alone corrects a systematic bias in statistical inference only slowly and partially.

The experiment delivers three main findings. First, exposure to others' optimal choices substantially improves the quality of subjects' own decisions relative to the individual-learning control group, even though the observed choices are never labeled as optimal. Second, the mechanism is selective adoption rather than imitation: exposure to suboptimal choices leaves behavior statistically indistinguishable from the control group, and within EXPOSURE, subjects try the observed alternative primarily after their own choice delivers a low payoff and retain it when it pays off. Subjects filter what they observe through their own payoff experience, adopting the observed behavior selectively rather than copying it. Third, the resulting gains are *fragile*. The subjects who respond to exposure reach near-fully optimal behavior, yet when the environment's primitives change (the same underlying logic, different parameters), their performance reverts to the level of a control group that was never exposed to optimal behavior at all. Exposure changes what subjects do, not their model of why it works. Because the design measures choices rather than beliefs, this conclusion about subjects' models is necessarily indirect: it is inferred from transfer behavior, the response to advice, and subjects' own end-of-experiment reflections rather than from elicited posteriors. Explanation reaches further: a short written rationale from the same observed peer, added in this design on top of each group's accumulated experience, returns compliers to nearly their earlier performance and substantially raises the optimality rate of never-takers, the group that observation alone left unmoved.

These findings carry a practical message for organizations and policies that lean on observational learning, such as shadowing, onboarding, or sharing of best practices. Such interventions are attractive and the evidence in this paper shows that they can change behavior. But the same evidence implies that the gains may not survive a change in the environment. In the context of inaccurate statistical discrimination, for instance, letting a hiring manager observe a better-performing peer is a sensible first intervention, but explaining why the peer's decisions are correct (how conditioning on qualifications reverses the apparent ranking of backgrounds) is what extends the reach to decision-makers whom observation alone leaves unmoved. Whether explanation also produces gains that generalize to new applicant pools is a natural conjecture that this design does not test. For the remainder who respond to neither channel, complementary approaches such as decision aids or algorithmic evaluation that embed the conditional reasoning may be required.

The results also open questions that reach beyond this experiment. Real decision-makers rarely observe a uniformly optimal peer: they watch many imperfect ones, and they choose whom to pay attention to, so how endogenous observation shapes what is adopted remains largely unexplored.

Environments also differ in how often their primitives shift, and the results suggest that this rate of change is a central determinant of how much observational learning is worth investing in. Finally, as advice increasingly arrives from algorithmic sources that can explain as well as demonstrate, the distinction documented here, between transmitting behavior and transmitting the reasoning behind it, is likely to become more consequential rather than less. Understanding when watching others produces knowledge that outlasts the moment of observation is a promising direction for future work.

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Online Appendix for

Fragile Learning From Others

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Contents:

A. Additional Analysis: Treatment Differences

This appendix provides a comprehensive set of statistical tests complementing the main treatment comparisons discussed in Section 3.2. I present results for all four treatment groups, with NoExposure ($N = 80$) as the omitted category, alongside Exposure ($N = 223$), ExposureNeg ($N = 53$), and Exp+F ($N = 44$). Throughout, I focus on optimal choice behavior in Part 2 of the experiment, measured over the last 10 rounds of interest and in the last round.[51]

A.1. OLS Regression Estimates

Estimates of treatment effects on optimal choice in Part 2 under several specifications are reported in Table 7. Columns (1)–(3) use the round-level binary optimal choice indicator pooled across the last 10 rounds of Part 2, with standard errors clustered at the subject level; Columns (4)–(6) use the subject-level binary optimal choice in the last round, with heteroskedasticity-robust standard errors.

The baseline specification in Column (1) confirms the main finding from Section 3.2: Exposure subjects make 12.8 percentage points more optimal choices than NoExposure subjects ($p = 0.022$), while ExposureNeg subjects perform indistinguishably from NoExposure ($p = 0.935$). The Exp+F group shows an even larger improvement of 20.6 percentage points ($p = 0.005$), although the difference between Exposure and Exp+F is not statistically significant ($p = 0.203$), consistent with the interpretation that additional feedback does not substantially augment the effects of social exposure alone.

These results are robust to controlling for baseline behavior: adding the Part 1 optimal choice share in Column (2) explains substantial additional variation (R^2 increases from 0.021 to 0.334) but leaves the treatment effect estimates virtually unchanged. The strong predictive power of baseline behavior indicates substantial persistence in individual performance (whether rooted in prior understanding, stable heuristics, or attention), but it does not confound the treatment comparisons. Adding demographic controls in Column (3) for gender, age, and major further confirms this stability: none of the demographic variables are individually significant, and the treatment coefficients remain nearly identical.

[51] The four treatment groups are balanced on observable pre-treatment characteristics. Baseline performance in Part 1 is nearly identical across groups: 39.8 percent (NoExposure), 40.5 percent (Exposure), 40.2 percent (ExposureNeg), and 39.5 percent (Exp+F); a Kruskal-Wallis test fails to reject equality ($p = 0.989$). The groups are also balanced on gender ($p = 0.392$) and major ($p = 0.584$). Age shows a statistically significant difference ($p = 0.028$), driven by a slightly higher mean in Exp+F (20.6 years) relative to the other groups (19.7–19.9 years); however, as shown in Table 7, adding age as a control does not affect the treatment effect estimates.

Table 7: OLS Estimates of Treatment Effects on Optimal Choice in Part 2

	Last 10 Rounds			Last Round		
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure	0.128**	0.123***	0.122***	0.210***	0.205***	0.203***
	(0.056)	(0.038)	(0.038)	(0.064)	(0.051)	(0.051)
ExposureNeg	−0.006	−0.009	−0.007	0.009	0.006	0.011
	(0.078)	(0.048)	(0.048)	(0.089)	(0.063)	(0.063)
Exp+F	0.206***	0.208***	0.212***	0.197**	0.198***	0.209***
	(0.073)	(0.056)	(0.057)	(0.091)	(0.076)	(0.076)
Part 1 Optimal Share		0.659***	0.650***		0.660***	0.655***
		(0.033)	(0.035)		(0.039)	(0.040)
Female			0.011			0.050
			(0.032)			(0.040)
Age			−0.008			−0.013
			(0.007)			(0.010)
Econ Major			−0.044			−0.043
			(0.032)			(0.040)
Constant	0.491***	0.229***	0.416***	0.463***	0.200***	0.451**
	(0.049)	(0.036)	(0.145)	(0.056)	(0.046)	(0.201)
R^2	0.021	0.334	0.337	0.039	0.357	0.364
Observations	4,000	4,000	4,000	400	400	400
Equality Tests (p-values)						
Exp = Exp+F	0.203	0.108	0.092	0.863	0.919	0.926
ExpNeg = NoExp	0.935	0.849	0.891	0.918	0.921	0.860
Exp = ExpNeg	0.046	0.003	0.004	0.008	<0.001	<0.001

Notes: This table reports OLS estimates of the effect of treatment assignment on the share of optimal choices in Part 2 of the experiment, with NoExposure as the omitted category. In Columns (1)–(3), the dependent variable is the round-level binary optimal choice indicator pooled across the last 10 rounds of Part 2, with standard errors clustered at the subject level. In Columns (4)–(6), the dependent variable is the binary optimal choice in the last round of Part 2, with heteroskedasticity-robust (HC1) standard errors. *Part 1 Optimal Share* is the proportion of optimal choices in the last 10 rounds of Part 1. *Econ Major* equals one if the subject's major is Economics, Business Economics, or Joint Mathematics-Economics. The equality tests report p-values from F-tests of the stated null hypotheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The last-round estimates in Columns (4)–(6) are qualitatively similar but larger in magnitude, reflecting the fact that many Exposure subjects switch to optimal behavior gradually over Part 2, so that the last-round comparison captures the full cumulative treatment effect. The treatment effect of Exposure in the last round is 21.0 percentage points ($p = 0.001$), and the effect is robust to the inclusion of baseline and demographic controls. The equality test confirms that Exposure and ExposureNeg produce significantly different outcomes ($p = 0.008$), while Exposure and Exp+F do not ($p = 0.863$).

A.2. Alternative Specifications and Nonparametric Tests

The robustness of the main findings to alternative definitions of the dependent variable and to distributional considerations is examined in Table 8. Columns (1)–(2) use a binary indicator that equals one if a subject makes at least 8 optimal choices out of 10 in the last 10 rounds of Part 2, capturing whether a subject achieves a high level of optimality rather than the continuous share. Under this binary measure, 55.2 percent of Exposure subjects achieve high optimality compared with 42.5 percent in NoExposure, a 12.7 percentage point difference ($p = 0.051$). The ExposureNeg rate (41.5 percent) is virtually identical to NoExposure ($p = 0.910$), and Exp+F (54.5 percent) is indistinguishable from Exposure ($p = 0.941$). These patterns are unchanged when controlling for baseline behavior in Column (2).

Table 8: Alternative Specifications: Binary Outcome and Median Regression

	Binary Optimality		Median
	(1)	(2)	(3)
Exposure	0.127*	0.121**	0.300***
	(0.065)	(0.048)	(0.077)
ExposureNeg	−0.010	−0.013	−0.100
	(0.088)	(0.060)	(0.105)
Exp+F	0.120	0.122	0.400***
	(0.094)	(0.076)	(0.111)
Part 1 Optimal Share		0.704***	
		(0.040)	
Constant	0.425***	0.145***	0.500***
	(0.056)	(0.042)	(0.066)
Observations	400	400	400
Equality Tests (p-values)			
Exp = Exp+F	0.941	0.990	0.306
ExpNeg = NoExp	0.910	0.828	0.340
Exp = ExpNeg	0.073	0.014	$<$0.001

Notes: Columns (1)–(2) report OLS estimates where the dependent variable equals one if a subject makes at least 8 optimal choices out of 10 in the last 10 rounds of Part 2, with heteroskedasticity-robust (HC1) standard errors and NoExposure as the omitted category. Column (3) reports median (50th percentile) regression estimates of the subject-level share of optimal choices in the last 10 rounds of Part 2. The equality tests for Columns (1)–(2) report p-values from F-tests; Column (3) reports p-values from Wald tests based on the quantile regression variance-covariance matrix.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (3) reports median regression estimates of the subject-level share of optimal choices in the last 10 rounds of Part 2. The median regression confirms that Exposure shifts the median optimal choice share from 0.50 to 0.80 ($p < 0.001$), while Exp+F shifts it further to 0.90 ($p < 0.001$). The ExposureNeg effect on the median is negative but not significant (-0.10, $p = 0.340$). The equality test for Exposure vs. ExposureNeg is highly significant ($p < 0.001$), confirming that the distributional shift is concentrated in the positive exposure treatments.

The parametric results are corroborated by distribution-free tests.[52] The first-order stochastic dominance of Exposure over NoExposure, documented using the Barrett & Donald (2003) procedure in Section 3.2, further supports the conclusion that the treatment shifts the entire distribution of optimal behavior rather than affecting only a particular quantile.

[52] The Wilcoxon rank-sum test (Mann-Whitney U) confirms that Exposure differs significantly from NoExposure in the last 10 rounds ($p = 0.022$) and in the last round ($p = 0.001$), while ExposureNeg does not ($p = 0.914$ and $p = 0.920$, respectively). Fisher's exact test on the binary optimality indicator yields consistent results for the last 10 rounds (Exposure vs. NoExposure, $p = 0.068$; ExposureNeg vs. NoExposure, $p = 1.000$) and for the last round (Exposure vs. NoExposure, $p = 0.001$; ExposureNeg vs. NoExposure, $p = 1.000$; Exposure vs. ExposureNeg, $p = 0.011$). All pairwise comparisons between Exposure and Exp+F are insignificant under both tests.

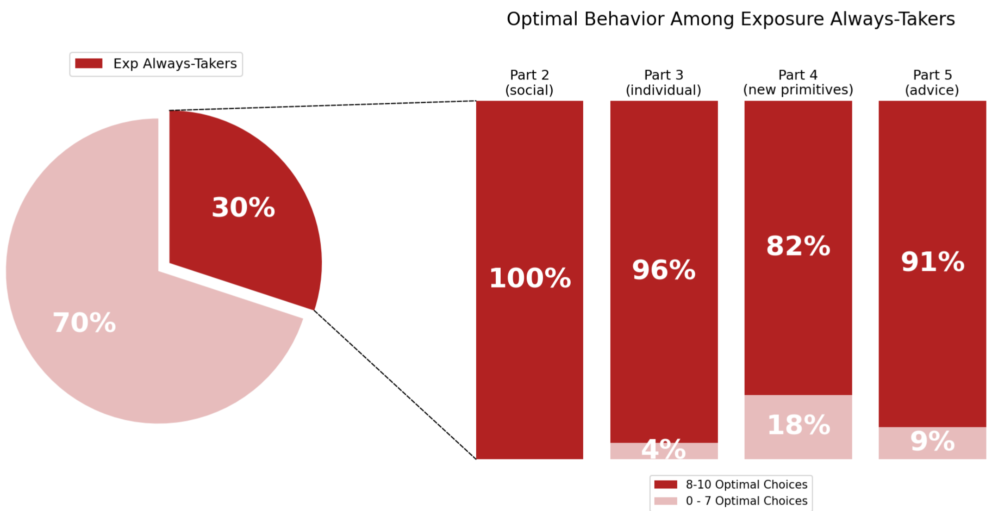
B. Additional Analysis: Always-Takers and Never-Takers

Section 3.4 classifies Exposure subjects into compliers, always-takers, never-takers, and defiers based on whether their behavior changes between the baseline and social learning phases. The main text documents complier dynamics in detail; this appendix completes the decomposition by examining always-takers and never-takers in both the Exposure ($N = 223$) and Exp+F ($N = 44$) treatments. These two types are of substantive interest because they represent the boundaries of what social exposure can achieve: always-takers reveal that a substantial fraction of subjects already behave optimally before treatment, while never-takers identify the subjects for whom passive observation is insufficient.

Always-takers. In the Exposure group, 67 subjects (30 percent) are classified as always-takers. These subjects performed optimally in both Part 1 and Part 2, meaning they had already internalized the correct decision rule prior to social exposure. Their dynamics across subsequent parts are displayed in Figure 16. The key finding is that the share of always-takers exhibiting optimal behavior remains remarkably high throughout: 96 percent in Part 3 (individual decision-making without exposure), 82 percent in Part 4 (new environment primitives), and 91 percent in Part 5 (advice). The decline to 82 percent in Part 4 is notable because it indicates that even subjects with strong prior understanding face difficulty when the task environment changes. The fact that over four-fifths of always-takers remain optimal under the new primitives is consistent with comprehension of the underlying statistical logic, though threshold-based choices alone cannot rule out alternatives such as a stable Orange-favoring heuristic combined with feedback-based relearning. The recovery to 91 percent under advice in Part 5 is likewise consistent with the Part 4 decline reflecting the difficulty of adjusting to the new environment rather than a permanent loss of understanding.

Never-takers. In the Exposure group, 94 subjects (42 percent) are classified as never-takers, the largest learning type. Despite observing participant X's optimal choices throughout Part 2, these subjects did not switch to optimal behavior. A striking pattern in their subsequent performance emerges in Figure 17. Never-takers remain largely suboptimal, with only 7 percent exhibiting optimal behavior in Part 3 and 14 percent in Part 4, confirming that they did not internalize the optimal rule during the social learning phase. However, their behavior changes dramatically in Part 5: 53 percent of never-takers achieve optimal performance once explicit written advice is provided. This finding carries two important implications. First, never-takers are not fundamentally incapable of learning the optimal rule; rather, passive observation of another agent's choices is insufficient to shift their behavior, while direct written guidance is effective. Second, the fact that nearly half of never-takers still fail to achieve optimality even under explicit advice (47 percent

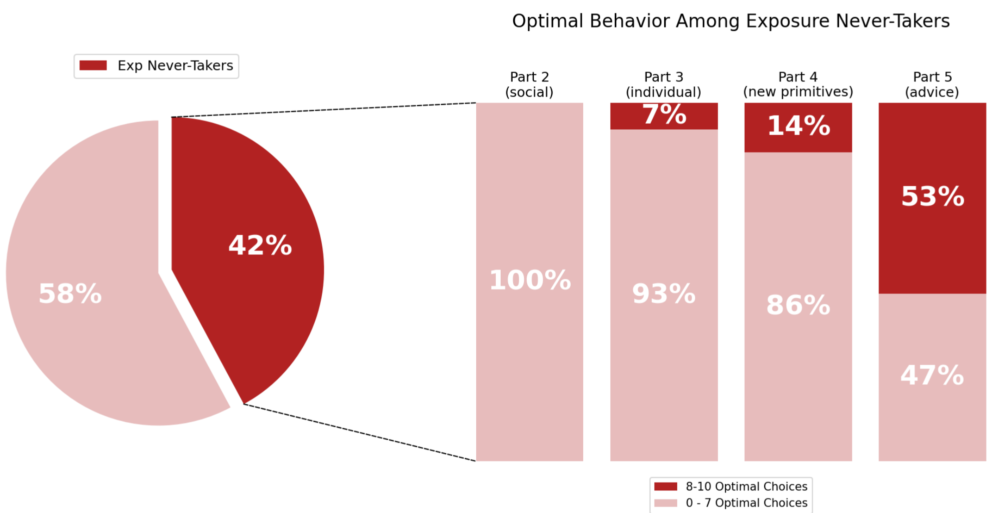
Figure 16: Exposure Always-Taker Classification and Dynamics



Notes. Always-takers in the Exposure group ($N = 67$, 30% of Exposure) are subjects who performed optimally in both Part 1 (baseline) and Part 2 (social learning), where optimality is defined as making at least 8 optimal choices in the final 10 key rounds. Each bar reports the percentage of always-takers who exhibit optimal behavior in subsequent parts.

remain suboptimal in Part 5) suggests substantial heterogeneity in the barriers to learning: for some subjects, the barrier may lie not in the mode of information transmission but in processing the statistical reasoning itself, though inattention to or distrust of the advice cannot be ruled out on this evidence.

Figure 17: Exposure Never-Taker Classification and Dynamics



Notes. Never-takers in the Exposure group ($N = 94$, 42% of Exposure) are subjects who performed suboptimally in both Part 1 and Part 2. Each bar reports the percentage who exhibit optimal behavior in subsequent parts.

C. Additional Analysis: Exposure (Full Info)

C.1. Aggregate Treatment Effects

A natural question that arises in relation to the learning mechanism in the Exposure treatment is whether learning gains are solely attributable to observing another participant's choices. Subjects never observe participant X's payoffs; however, because exposure induces some subjects to adopt the observed optimal action, the payoff feedback they generate for *themselves* becomes endogenously richer: a subject who tries the optimal action experiences its payoff distribution directly, which a suboptimal chooser learning alone encounters less often. As a result, the improvements documented in Section 3.2 could reflect a combination of social learning and this endogenously enhanced feedback, making it difficult to isolate the pure effect of exposure.

This concern relates to a broader issue studied in the literature on learning from endogenously selected samples. As Esponda & Vespa (2018) demonstrate, when agents' own choices determine the data they observe, learning can be severely hindered because the resulting feedback is not representative of what a different choice would have produced. In the present setting, subjects in Exposure face a version of this problem in reverse: because exposure shifts choices toward the optimal action, the feedback stream subjects generate for themselves is systematically more favorable than what a suboptimal chooser would encounter alone. To the extent that this enhanced feedback, rather than the act of observing optimal choices per se, drives the treatment effect, the interpretation of the main results would shift from social learning toward a form of information provision.

To disentangle these channels, the experimental design includes an additional treatment group, Exposure (Full Info) (**Exp+F**), depicted in Figure 18. Whereas subjects in Exposure receive exposure *to choices* only, subjects in Exp+F receive exposure *to choices and payoffs*: they observe the same optimal choices of participant X and are additionally provided with payoff-relevant information about the counterfactual outcome. Specifically, regardless of whether a subject's own choice is optimal or suboptimal, subjects in this treatment observe the payoff that would have resulted from the alternative hiring decision. This design eliminates the endogenous feedback channel: if the main treatment effect in Exposure is driven by richer payoff information rather than by exposure to others' choices, then providing all subjects with this information directly should lead to substantially larger effects in Exp+F relative to Exposure.

Figure 18: The Exposure (Full Info) Treatment

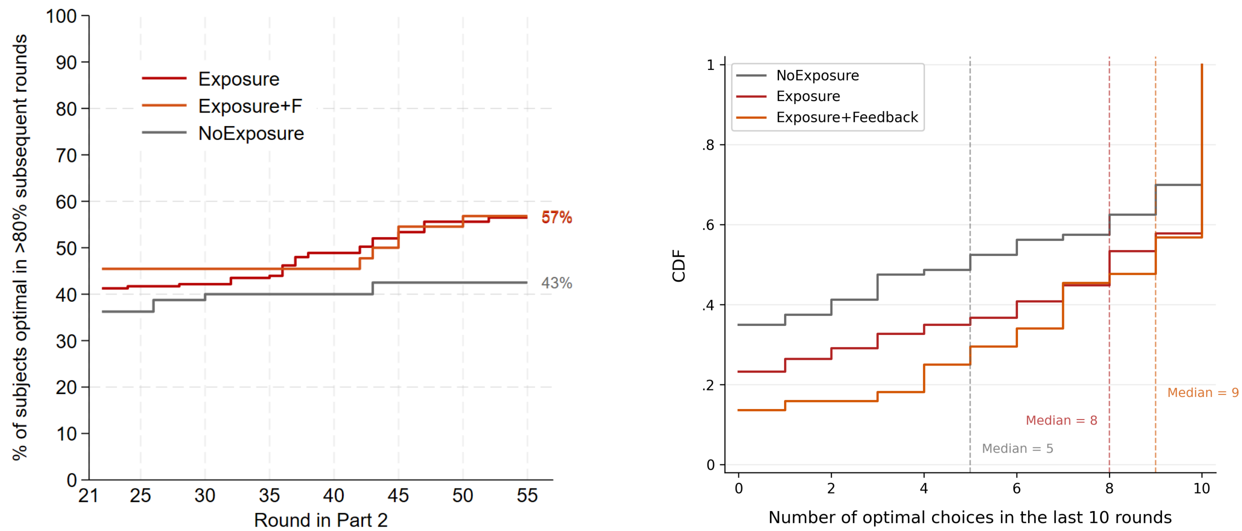
The cumulative distribution of optimal choices in the last ten rounds of Part 2 for the three relevant treatment groups is plotted in Figure 19. Over the last ten rounds of Part 2, the mean optimal choice rate in Exp+F is 70 percent, compared with 62 percent in Exposure and 49 percent in NoExposure. While Exp+F subjects perform slightly better on average, the difference between Exp+F and Exposure is not statistically significant (7.8 percentage points, 95 percent confidence interval $[-4.3, +19.9]$, $p = 0.203$); the comparison thus rules out large incremental gains from counterfactual information conditional on observation, though not moderate ones. By the final round of Part 2, the two exposure groups are virtually indistinguishable: 67 percent of Exposure subjects and 66 percent of Exp+F subjects make the optimal choice ($p = 0.863$). In contrast, both exposure treatments produce higher optimal choice rates than NoExposure, with the difference between Exp+F and NoExposure reaching significance in the last ten rounds ($p = 0.005$) and the difference between Exposure and NoExposure reaching significance in the last round ($p = 0.001$).[53]

These results suggest that the endogenous feedback channel plays a limited role in explaining the treatment effects documented in the main analysis. Providing subjects with richer counterfactual information does not substantially increase the rate of optimal choices beyond the level achieved through exposure alone. The finding that both Exposure and Exp+F produce similar improvements relative to NoExposure is consistent with the interpretation that the main mechanism underlying the treatment effect is the observation of optimal choices, rather than the informativeness of the associated feedback.

An additional piece of evidence comes from the complier classification applied to the Exp+F group. Using the same 80 percent threshold as in Section 3.4, the distribution of learning types in Exp+F closely mirrors that in Exposure. As Figure 20 shows, the share of compliers, always-

[53]The difference between Exposure and NoExposure in the last ten rounds has $p = 0.022$, while the difference between Exp+F and NoExposure in the last round has $p = 0.031$.

Figure 19: Effects of the Exposure (Full Info) Treatment



(a) Dynamics of Optimal Behavior

(b) Cumulative Distribution of Optimal Choices

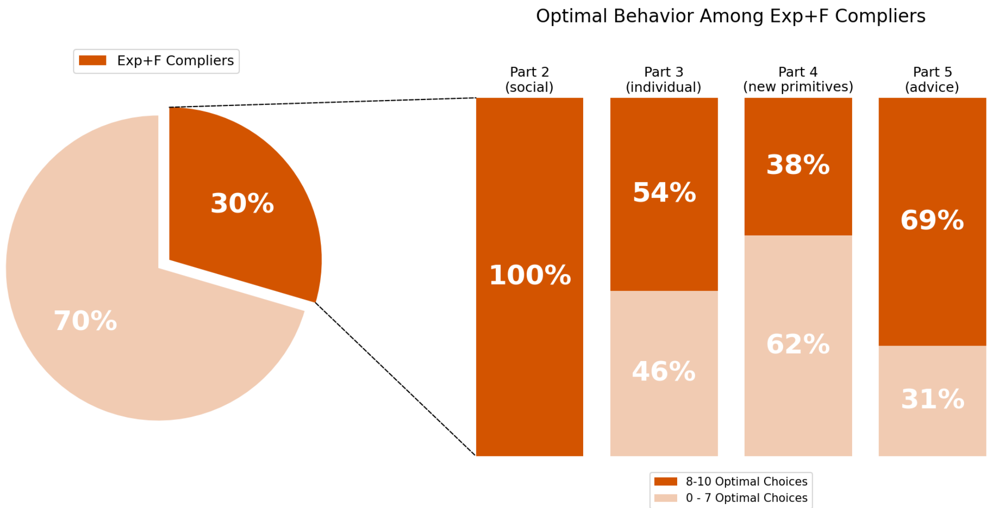
Notes: The red, orange, and gray lines correspond to Exposure ($N = 223$), Exp+F ($N = 44$), and NoExposure ($N = 80$), respectively. Panel (a) displays the percentage of subjects choosing the optimal alternative in a given round of Part 2 and then continuing to do so in at least 80% of subsequent rounds. Panel (b) plots the cumulative distribution of the number of optimal choices in the last ten rounds of Part 2 for each group. The vertical dashed lines denote the median for each group. In Exp+F, subjects receive the same social exposure as in Exposure but additionally observe the payoff that would have resulted from the alternative hiring decision.

takers, and never-takers is comparable across the two treatments, and the trajectory of complier behavior in Parts 3–5 follows a similar pattern to that documented for Exposure in Section 4: the share of compliers who retain optimal behavior declines once exposure is removed and falls further under the new primitives, before recovering under advice.[54] This parallel suggests that the type of learning induced by exposure is qualitatively similar regardless of whether counterfactual feedback is additionally provided.

The evidence from Exp+F thus serves as a useful robustness check for the main results. The fact that providing additional counterfactual feedback does not meaningfully improve performance beyond what exposure to optimal choices already achieves suggests that the learning gains in Exposure are primarily driven by the social component of the treatment, namely the observation of another participant's optimal decisions, rather than by the informativeness of the feedback environment. One caveat is in order: the design does not include a feedback-only cell in which subjects receive counterfactual payoff information *without* observing another participant's choices, so the comparison bounds the incremental effect of richer information conditional on observation

[54]The detailed classification of Exp+F always-takers and never-takers is provided in the next subsection.

Figure 20: Exp+F Complier Classification and Dynamics



Notes: Compliers in the Exp+F group ($N = 44$) are subjects who performed suboptimally in Part 1 but switched to optimal decisions in Part 2, where optimality is defined as making at least eight optimal choices in the final ten key rounds of a given part. The pie chart shows the distribution of learning types. Each bar reports the percentage of these original compliers who continue to exhibit optimal behavior in subsequent parts. In the Exp+F treatment, subjects receive the same social exposure as in Exposure but additionally observe the payoff that would have resulted from the alternative hiring decision.

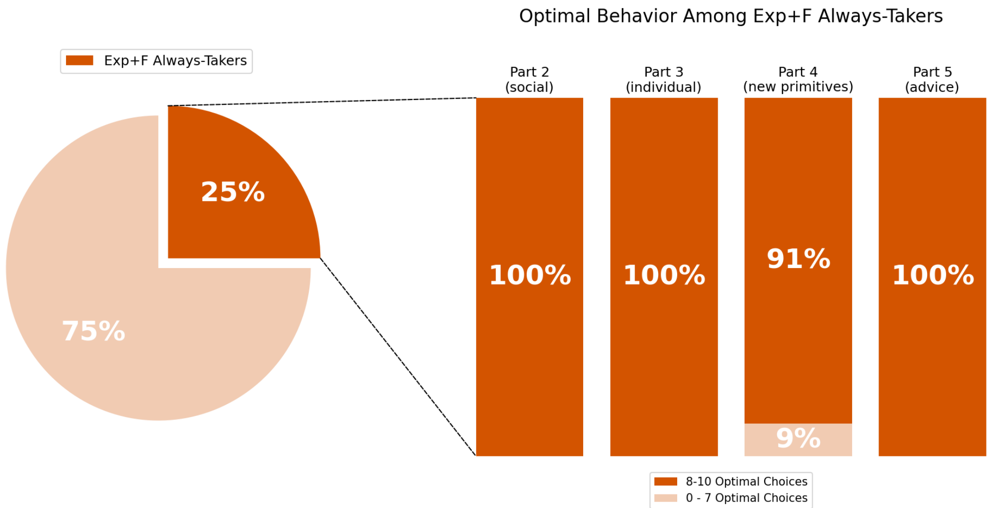
rather than separating the two channels entirely. Two features of the data nevertheless speak against a pure information-provision account. First, the observed choices are what identify the alternative worth trying: the asymmetry by baseline behavior in Table 3 shows that subjects move only when the observed behavior contradicts and outperforms their own. Second, the transition evidence shows that adoption is timed by subjects' own payoff realizations rather than by information about the alternative, which is constant across rounds.

Additional Result (Exposure Full Info): *Providing counterfactual payoffs on top of exposure does not significantly improve behavior relative to exposure alone, indicating that richer feedback adds little once observed choices are available; absent a feedback-only cell, the comparison bounds this incremental effect rather than separating observation from information entirely.*

C.2. Learning Types

The Exp+F treatment provides subjects with the same social exposure as Exposure but additionally reveals the payoff that would have resulted from the alternative hiring decision. If counterfactual feedback accelerates learning, we would expect a higher complier share and different dynamics for the remaining types. The classification of Exp+F subjects yields a distribution that is broadly similar to Exposure: 30 percent compliers ($N = 13$), 25 percent always-takers ($N = 11$), 45 percent never-takers ($N = 20$), and no defiers, although the smaller sample size ($N = 44$) limits

Figure 21: Exp+F Always-Taker Classification and Dynamics



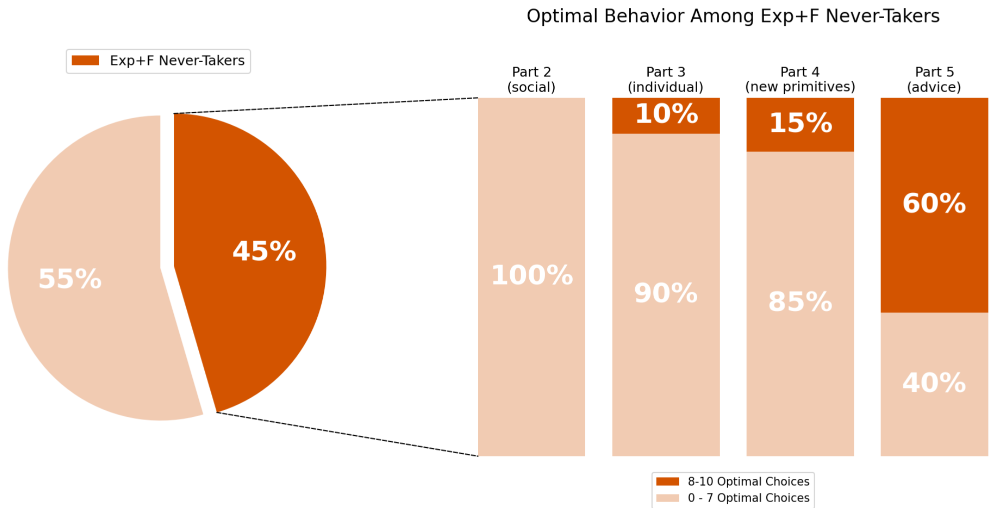
Notes. Always-takers in the Exp+F group ($N = 11$, 25% of Exp+F) are subjects who performed optimally in both Part 1 and Part 2. Each bar reports the percentage who exhibit optimal behavior in subsequent parts.

the precision of these comparisons.

Always-takers. Figure 21 shows the dynamics of the 11 always-takers in Exp+F. Their performance is remarkably stable: 100 percent remain optimal in Parts 3 and 5, with only a slight decline to 91 percent in Part 4 under the new primitives. This stability closely parallels the Exposure always-takers (96 percent, 82 percent, 91 percent), with the Exp+F group performing slightly better in each part. While the small sample size prevents strong conclusions, the pattern is consistent with the view that counterfactual feedback may reinforce the understanding of subjects who already behave optimally, helping them maintain performance even when the environment changes. Alternatively, the difference may simply reflect sampling variation in a small group.

Never-takers. Figure 22 displays the dynamics of the 20 never-takers in Exp+F. The trajectory is qualitatively similar to Exposure: low optimality in Part 3 (10 percent) and Part 4 (15 percent), rising to 60 percent in Part 5 under advice. The Part 5 recovery rate is somewhat higher than for Exposure never-takers (60 percent vs. 53 percent), which could suggest that prior exposure to counterfactual feedback primes these subjects to be more receptive to advice. However, the difference is modest and the small sample size precludes definitive conclusions. The key qualitative finding is consistent across both treatments: never-takers are not permanently resistant to learning, but they require a more direct form of information transmission than passive social observation.

Figure 22: Exp+F Never-Taker Classification and Dynamics



Notes. Never-takers in the Exp+F group ($N = 20$, 45% of Exp+F) are subjects who performed suboptimally in both Part 1 and Part 2. Each bar reports the percentage who exhibit optimal behavior in subsequent parts.

C.3. Summary

In sum, the always-taker and never-taker analyses of this and the preceding appendix reinforce and extend the main findings in several ways. First, the existence of a large always-taker group (30 percent in Exposure, 25 percent in Exp+F) confirms that a substantial fraction of the population can solve the task without any social learning, consistent with the theoretical prediction that Bayesian reasoning is accessible to some subjects even in the absence of external information. Second, always-takers' high persistence rates across Parts 3–5 are consistent with an understanding of the underlying structure: these subjects maintain optimal behavior even when the environment changes (Part 4) and advice is provided (Part 5). Third, the never-taker response to advice in Part 5 (53 percent in Exposure, 60 percent in Exp+F) demonstrates that this bundled intervention reaches many subjects whom passive observation did not; because the advice combines an explanation with explicit scenario-specific recommendations and arrives after substantial additional experience, the response does not by itself distinguish comprehension from reduced computational demands or simple compliance. Finally, the qualitative similarity of these patterns across Exposure and Exp+F reinforces the conclusion from the aggregate analysis above that additional counterfactual feedback does not fundamentally alter the nature of learning induced by social exposure.

D. Additional Analysis: Alternative Complier Specifications

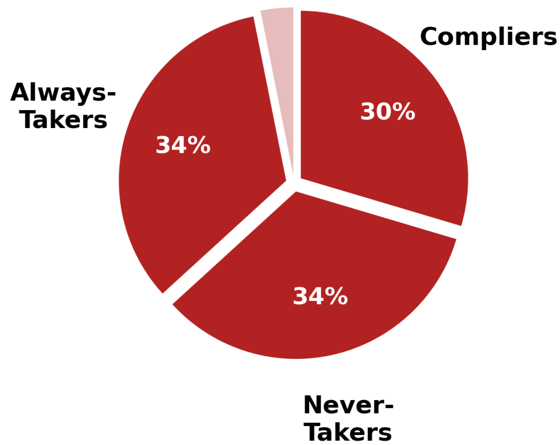
The main analysis classifies Exposure subjects as compliers, always-takers, or never-takers using a threshold of 0.8 (at least eight optimal choices in the final ten key rounds of a given part). While this threshold is standard in the experimental literature on binary repeated choices, the classification and resulting dynamics could in principle be sensitive to its exact value. This appendix examines the robustness of the complier classification under four alternative specifications: a more lenient threshold of 0.6, a stricter threshold of 1.0, and two criteria that combine a level requirement with a minimum improvement relative to baseline. The central question is whether the patterns documented in Section 4 (declining optimality when the environment changes, followed by recovery under advice) are a robust feature of complier behavior or an artifact of the particular threshold used.

D.1. Threshold 0.6

Under the 0.6 threshold, a subject is classified as exhibiting optimal behavior if at least 6 of 10 choices in the final ten key rounds are optimal. This more lenient criterion expands the set of subjects counted as having "learned" and correspondingly changes the composition of each type. The resulting distribution, displayed in Figure 23, is 30 percent compliers ($N = 66$), 34 percent never-takers ($N = 75$), 34 percent always-takers ($N = 75$), and 3 percent defiers ($N = 7$). Compared with the baseline 0.8 classification (25 percent compliers, 42 percent never-takers, 30 percent always-takers), the lower threshold reclassifies some never-takers as compliers (the complier share rises from 25 percent to 30 percent) and reclassifies some subjects from never-takers to always-takers (the always-taker share rises from 30 percent to 34 percent), reflecting subjects who achieved 6–7 optimal choices in both parts.

The critical test is whether the complier dynamics remain qualitatively similar. As Figure 24 shows, compliers under the 0.6 threshold exhibit the same U-shaped pattern documented in the main text: 74 percent remain optimal in Part 3, declining to 50 percent in Part 4, and recovering to 94 percent in Part 5 under advice. The Part 4 decline is somewhat less severe than under the 0.8 threshold (50 percent vs. 41 percent), which is expected: the broader criterion includes subjects who barely crossed the threshold in Part 2 and are therefore closer to suboptimal, but it also includes subjects who were just below the 0.8 bar and may have still partially learned. The Part 5 recovery rate (94 percent) is higher than the baseline (84 percent), though because lowering the cutoff changes both the composition of the complier group and the criterion applied to later behavior, such cross-threshold differences should not be attributed to the marginal subjects themselves.

Figure 23: Exposure Learning Type Classification under Threshold 0.6



Notes. Classification of Exposure subjects ($N = 223$) into learning types using an alternative optimality threshold of 0.6. A subject exhibits optimal behavior if at least 6 of 10 choices in the final ten key rounds are optimal. Compliers switched from suboptimal in Part 1 to optimal in Part 2; always-takers performed optimally in both; never-takers performed suboptimally in both.

The round-by-round evolution of optimal choice rates across Parts 3–5 for all learning types under the 0.6 threshold is presented in Figure 25. The intensive-margin dynamics replicate the baseline findings: always-takers maintain consistently high performance; compliers show a characteristic decline in Part 4 followed by recovery in Part 5; and never-takers start below the NoExposure control group and converge toward the complier trajectory only when advice is provided.

D.2. Threshold 1.0

Under the 1.0 threshold, a subject is classified as exhibiting optimal behavior only if all ten choices in the final ten key rounds are optimal. This is the most stringent possible criterion and admits only subjects who achieved perfect accuracy. As Figure 26 shows, the resulting distribution shifts dramatically toward never-takers: 19 percent compliers ($N = 42$), 56 percent never-takers ($N = 125$), 23 percent always-takers ($N = 52$), and 2 percent defiers ($N = 4$). The complier share falls from 25 percent to 19 percent, reflecting the fact that some subjects who made 8–9 optimal choices in Part 2 but not a perfect 10 are reclassified as never-takers. The always-taker share declines from 30 percent to 23 percent for the same reason: subjects who performed well but not perfectly in Part 1 are no longer counted as having pre-treatment mastery.

Despite the substantially smaller and more select complier group, the qualitative dynamics persist. As Figure 27 shows, compliers under the 1.0 threshold exhibit an even steeper decline than under the baseline: 83 percent optimal in Part 3, dropping to 48 percent in Part 4, and recovering to 76 percent in Part 5. The sharper Part 4 decline (a drop of 35 percentage points from Part 3, versus 29 at the baseline threshold) is consistent with the interpretation that perfect accuracy in Part 2 does not guarantee perfect transfer to the new primitives: even subjects who achieved 100 percent optimal choices under the original environment face genuine difficulty when the task parameters change. The somewhat lower Part 5 recovery (76 percent vs. 84 percent) may reflect the stringency of the threshold: achieving perfect accuracy under advice is harder than achieving 80 percent accuracy.

The intensive-margin dynamics under the 1.0 threshold, shown in Figure 28, replicate the baseline findings across all learning types. Always-takers remain near the ceiling throughout. The complier trajectory shows the same characteristic pattern of decline and recovery; because the figure plots average optimal-choice shares, the stricter selection if anything shifts the trajectory upward relative to the baseline compliers. Never-takers under the 1.0 threshold (which now includes some subjects who would have been classified as compliers or always-takers under the 0.8 threshold) show a somewhat higher average performance than under the baseline, reflecting the reclassification of partially-optimal subjects into this group.

D.3. Alternative Classification Criteria

The threshold-based classification counts any subject who crosses the optimality threshold in Part 2 but not Part 1 as a complier. A potential concern is that some subjects may cross the threshold by chance rather than in response to treatment, especially near the boundary. To address this, I consider two alternative classifications that impose a minimum improvement (Δ) requirement: a subject is classified as a complier only if she both exceeds a level requirement in Part 2 and improves by at least Δ choices relative to Part 1.

A first variant, shown in Figure 29, uses a classification requiring at least 6 optimal choices in the final ten key rounds of Part 2 with an improvement of at least 4 choices relative to Part 1. Under this criterion, the complier group comprises 61 subjects, with 75 always-takers and 87 never-takers. The improvement requirement ensures that every complier exhibited a substantial behavioral shift between the baseline and exposure phases, ruling out subjects who were close to the boundary in both parts. The resulting dynamics are qualitatively indistinguishable from the baseline: compliers show declining performance in Part 4 and recovery in Part 5, while always-takers remain stable and never-takers improve markedly only under advice.

A second variant, shown in Figure 30, uses a more stringent criterion: at least 7 optimal choices in Part 2 with an improvement of at least 5. This further restricts the complier group to 52 subjects, with 71 always-takers and 100 never-takers. Despite the more demanding definition, the learning

Table 9: Learning-Type Composition Across Classification Thresholds

Criterion	Compliers	Always-takers	Never-takers	Defiers	C1	C2	C3
≥ 6 of 10	66 (30%)	75 (34%)	75 (34%)	7	32	17	17
≥ 7 of 10	61 (27%)	71 (32%)	86 (39%)	5	27	17	17
≥ 8 of 10 (baseline)	56 (25%)	67 (30%)	94 (42%)	6	22	17	17
≥ 9 of 10	45 (20%)	59 (26%)	112 (50%)	7	21	15	9
10 of 10	42 (19%)	52 (23%)	125 (56%)	4	20	15	7

Notes: Each row classifies the 223 Exposure subjects using the stated criterion applied to the last ten key rounds of Parts 1 and 2 (y_1, y_2), and splits compliers into subtypes by applying the same criterion to Parts 3 and 4. Because y_t takes values on a ten-round grid, any threshold in $(0.7, 0.8]$ is equivalent to the baseline criterion of at least eight optimal choices.

dynamics remain qualitatively unchanged. The complier trajectory again shows the characteristic decline in Part 4 and recovery in Part 5, confirming that the pattern is not driven by marginal cases near the classification boundary. Across all four alternative specifications examined in this appendix, the core finding is robust: the decline-and-recovery trajectory of compliers (retention in the familiar environment, deterioration when primitives change, recovery under guidance) is not an artifact of any particular classification boundary. As in the main text, this trajectory characterizes behavior; on its own it does not distinguish fragile understanding from retention of a task-specific action rule.

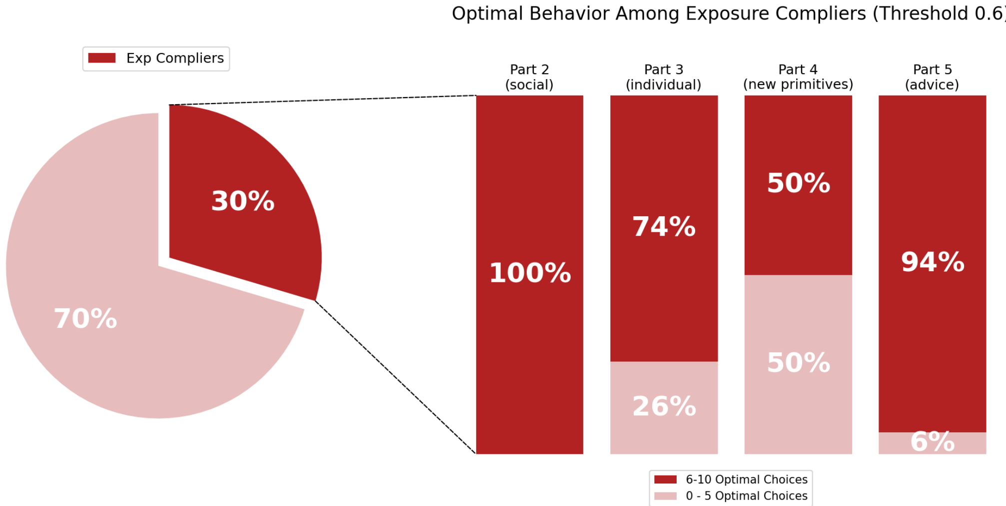
D.4. Threshold Ladder and a Sampling-Noise Benchmark

Two further checks address the role of sampling noise in the classification. Because y_t is measured over ten key rounds, the classification threshold moves on a discrete grid: any cutoff in $(0.7, 0.8]$ implies the same criterion of at least eight optimal choices, so the relevant perturbations are one round in either direction. The full ladder is reported in Table 9. The composition of the sample is stable: the complier share moves gradually from 30 percent at the most lenient cutoff to 19 percent at the strictest, and the ordering of the three subtypes is preserved throughout, with the C2 and C3 groups essentially unchanged between cutoffs of six and eight.

The second check quantifies how much of the classification could arise from chance crossings of the threshold in short windows, absent any true behavioral change. I simulate a null model in which each Exposure subject's choices are i.i.d. draws from a constant probability, set to that subject's empirical share of optimal choices over all fifteen key rounds of Part 1, and apply the classification to the simulated y_1 and y_2 windows (20,000 replications). Under this null, an average of 7.4 of the 223 subjects (3.3 percent) are labeled compliers, with 95 percent of replications producing between 3 and 13, far below the 56 compliers observed in the data. Under this null, chance threshold-crossing accounts for about 13 percent of observed compliers on average, and for about

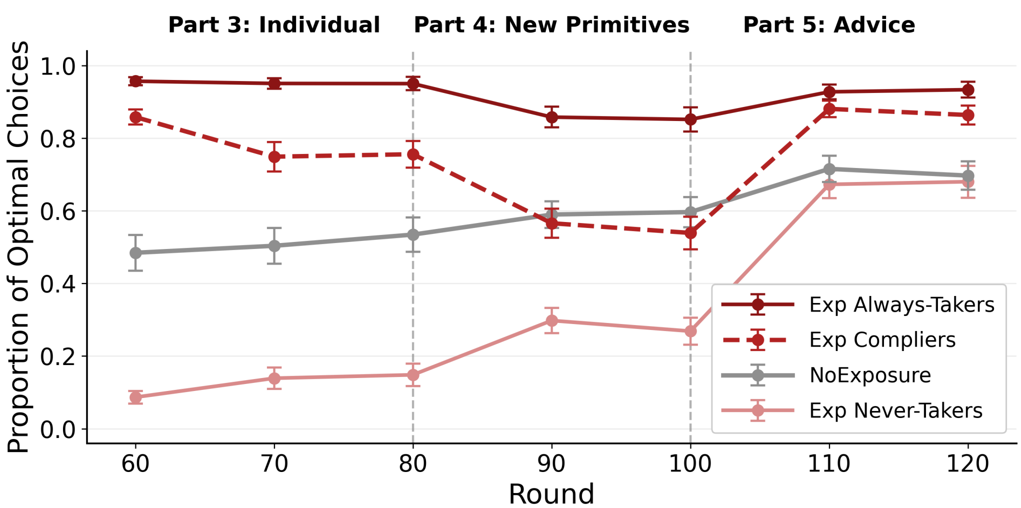
23 percent at the upper end of the simulation interval; these figures are benchmarks conditional on the fitted constant-propensity model rather than unconditional bounds, since serial dependence or nonstationarity in choices would alter both crossing and reversion rates. A variant that preserves within-subject serial dependence (resampling each subject's observed Part-1 key-round sequence in circular blocks of five) delivers nearly identical benchmarks (6.5 expected compliers on average, with 51 percent of them reverting), suggesting the constant-propensity simplification is not driving the conclusion. The same simulation also benchmarks the contribution of regression to the mean to the measured reversion: simulated null compliers "revert" in the Part 3 window 48 percent of the time, implying roughly 3.6 artifactual reverters, about one fifth of the 17 observed shallow imitators. Under this benchmark, the classification reflects genuine behavioral change for the large majority of compliers, though the size of the C3 group includes a modest noise component and should be interpreted accordingly.

Figure 24: Exposure Complier Analysis under Threshold 0.6



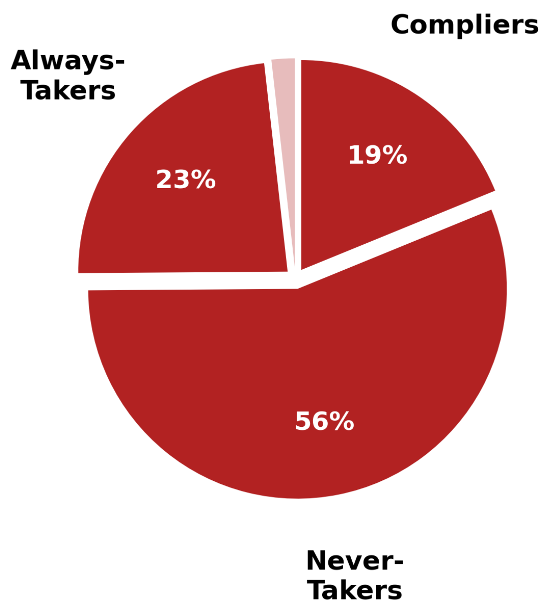
Notes. Complier dynamics using an alternative optimality threshold of 0.6 ($N = 66$, 30% of Exposure). Each bar reports the percentage of compliers who exhibit optimal behavior in subsequent parts.

Figure 25: Evolution of Optimal Behavior under Threshold 0.6



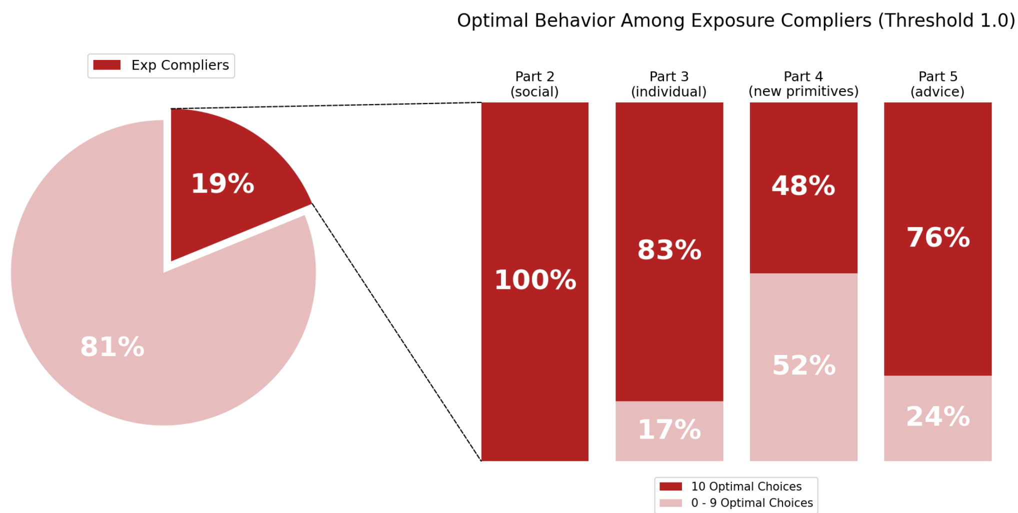
Notes. Each line tracks the average proportion of optimal choices across Parts 3–5 for a given learning type under the 0.6 threshold. Error bars indicate ±1 standard error.

Figure 26: Exposure Learning Type Classification under Threshold 1.0



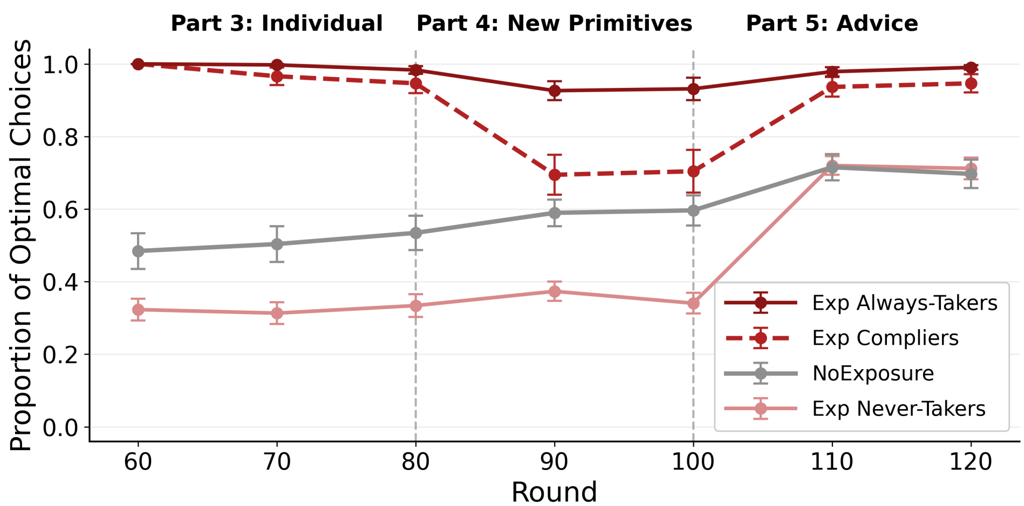
Notes. Classification of Exposure subjects ($N = 223$) into learning types using an alternative optimality threshold of 1.0. A subject exhibits optimal behavior only if all 10 choices in the final ten key rounds are optimal.

Figure 27: Exposure Complier Analysis under Threshold 1.0



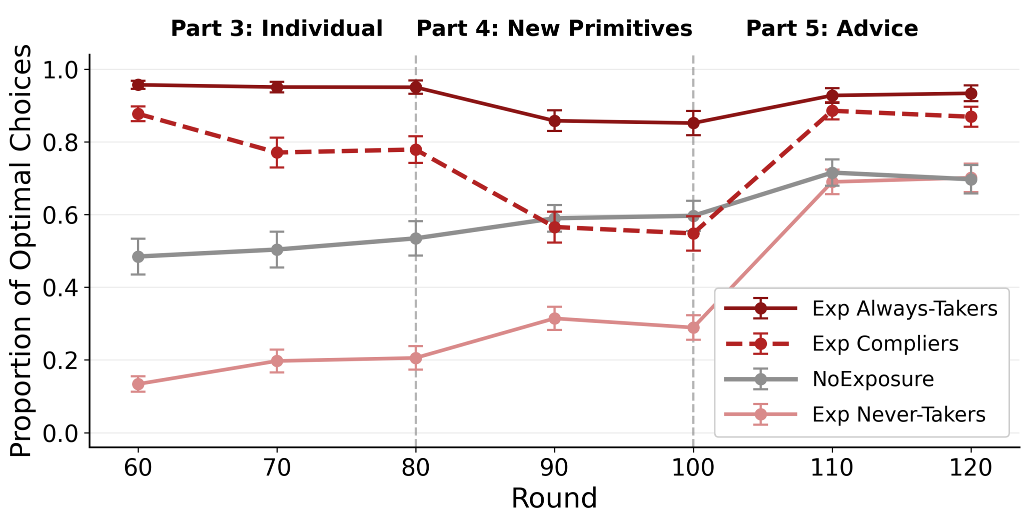
Notes. Compliers under threshold 1.0 ($N = 42$, 19% of Exposure) achieved 100% optimal choices in Part 2 but not in Part 1. Each bar reports the percentage who exhibit optimal behavior in subsequent parts.

Figure 28: Evolution of Optimal Behavior under Threshold 1.0



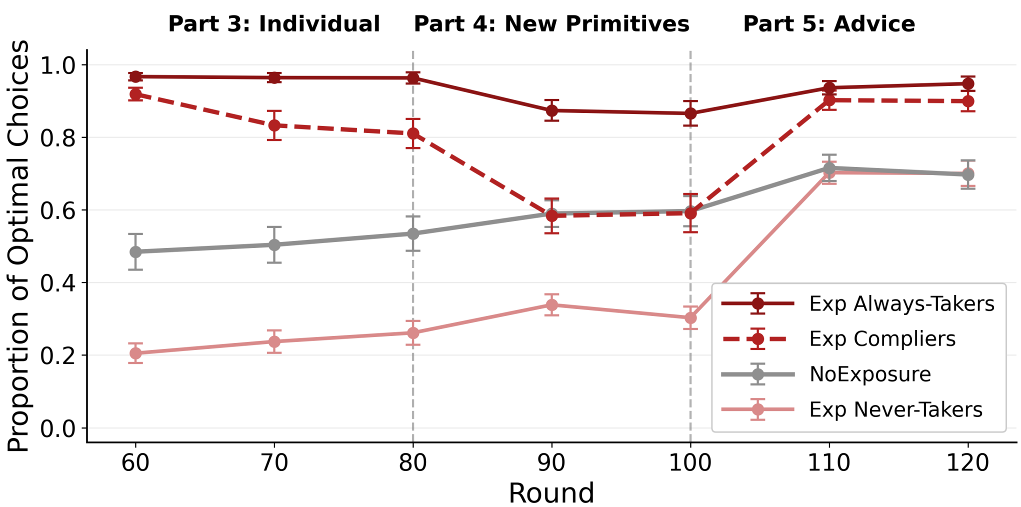
Notes. Each line tracks the average proportion of optimal choices across Parts 3–5 for a given learning type under the 1.0 threshold. Error bars indicate ± 1 standard error.

Figure 29: Learning Dynamics: Alternative Classification (≥ 6 in Part 2, $\Delta \geq 4$)



Notes. A subject in Exposure is classified as a complier ($N = 61$) if they make at least 6 optimal choices in the final ten key rounds of Part 2 and their improvement relative to Part 1 is at least 4 additional optimal choices ($\Delta \geq 4$). Always-takers ($N = 75$) meet the level requirement in both parts. Each line tracks the average proportion of optimal choices across Parts 3–5. Error bars indicate ± 1 standard error.

Figure 30: Learning Dynamics: Alternative Classification (≥ 7 in Part 2, $\Delta \geq 5$)



Notes. A subject in EXPOSURE is classified as a complier ($N = 52$) if they make at least 7 optimal choices in the final ten key rounds of Part 2 and their improvement relative to Part 1 is at least 5 additional optimal choices ($\Delta \geq 5$). Always-takers ($N = 71$) meet the level requirement in both parts. Each line tracks the average proportion of optimal choices across Parts 3–5. Error bars indicate ± 1 standard error.

E. Additional Analysis: Q-Learning Simulation

To illustrate how the observed dynamics can arise from a simple learning process in the tradition of reinforcement learning models of behavior (Erev & Roth, 1998), I simulate a Q-learning model that mirrors the main components of the NoExposure and Exposure treatments. In the simulation, agents update the value of actions based on realized payoffs and select actions probabilistically as a function of their current value estimates. The state space differs across treatments: in NoExposure, the state depends only on the agent's previous choice, while in Exposure, it additionally incorporates whether the agent's own realized payoff in the previous round was high or low. Both groups observe their own payoffs, so the difference across treatments is one of representation rather than information: the constant display of participant X's contrasting choice supplies a concrete alternative against which the agent's own payoff can be evaluated, which makes the payoff realization a behaviorally relevant component of the state. This modeling choice mirrors the selective adoption evidence in Section 3.3, where the observed choice identifies *what* to try while the agent's own payoff experience determines *when* it is tried and whether it is kept. In NoExposure, with no alternative on display, behavior conditions on the previous action alone. The simulation uses a learning rate of $\alpha = 0.05$, a discount factor of $\delta = 0.9$, and a softmax sensitivity parameter of $\lambda = 0.1$; these values are chosen for illustration rather than estimated from the data, and the exercise is accordingly a demonstration that the qualitative patterns are reproducible by a minimal payoff-gated learning process, not a fitted model. Estimating the parameters from round-level choice sequences and testing the specification against parsimonious alternatives, such as a win-stay/lose-shift rule, is a natural next step.

Formally, each agent maintains a Q-value $Q(a, s)$ for each action $a \in \{\text{Green}, \text{Orange}\}$ and state s, representing the expected value of taking action a in state s. After each round, Q-values are updated according to the standard reinforcement learning rule

$$Q(a, s) \leftarrow (1 - \alpha)Q(a, s) + \alpha \left[r + \delta \max_{a' \in \mathcal{A}} Q(a', s') \right],$$

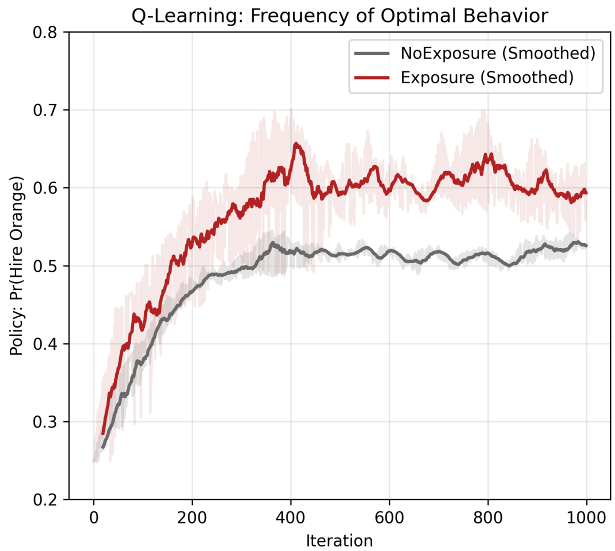
where r denotes the payoff from choosing action a in state s, s' is the subsequent state, $\alpha \in (0, 1)$ is the learning rate, and $\delta \in (0, 1)$ is the discount factor. Given these value estimates, actions are selected using a softmax rule,

$$\mathbb{P}(a \mid s) = \frac{e^{\lambda Q(a,s)}}{\sum_{a' \in \mathcal{A}} e^{\lambda Q(a',s)}},$$

where $\lambda > 0$ governs the sensitivity of choice probabilities to differences in Q-values.

The simulated paths in Figure 31 plot the evolution of the probability of choosing the optimal

Figure 31: Q-Learning Simulation for NoExposure and Exposure



Notes: This figure plots simulated behavior from a Q-learning model. In the NoExposure condition, the state depends only on the agent's previous choice. In the Exposure condition, the state additionally incorporates whether the agent's own realized payoff in the previous round was high or low. The simulated paths show the evolution of the probability of choosing the optimal action over time.

action over time and closely mirror the empirical patterns observed in the experiment. In particular, exposure to optimal behavior leads to faster increases in optimal choices early on, while agents learning only from their own feedback improve more gradually. The Exposure path reaches high optimality rates earlier because the richer state representation allows agents to learn faster from payoff-relevant signals; the simulation holds each treatment's representation fixed throughout, so it speaks to the speed of learning under exposure rather than to behavior after exposure is withdrawn.

The Q-learning framework also suggests a conjecture, not simulated here, about the fragility of learning gains documented in Section 4. Under reinforcement learning, action values are updated incrementally based on recent experience, and the weight placed on any given piece of information decays as new evidence accumulates. If exposure accelerates learning primarily by providing additional states that facilitate faster discrimination between actions, removing exposure collapses the state space back to its baseline size. Agents who have not yet accumulated sufficient experience to sustain their new behavior through the reduced state representation may gradually drift back toward suboptimal choices. This mechanism is consistent with the empirical observation that a subset of compliers revert once exposure is removed, while those who have had more rounds of reinforcement tend to retain their gains.

Overall, the transition evidence and the Q-learning simulation point to the same mechanism:

learning from exposure operates through short-run, payoff-driven selective adoption, in which a low realized payoff from one's own choice triggers experimentation with the observed alternative and favorable payoff realizations sustain the new behavior once adopted. This helps explain why exposure to optimal behavior can generate rapid improvements in performance while also producing learning gains that are sensitive to recent outcomes and may fail to persist or transfer once the environment changes.

Additional Result (Q-Learning): *Simulating a learning model in which observing another subject's choices expands the learner's state representation generates dynamic patterns consistent with the empirical ones documented in the experiment.*

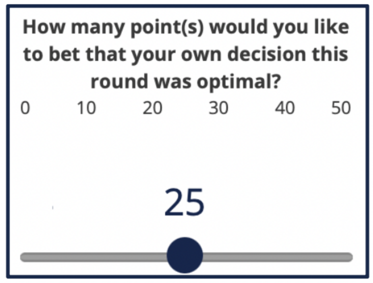
F. Additional Analysis: The Role of Confidence

Section 3.3 established that exposure to suboptimal choices has no detectable average effect on behavior. A natural question, consistent with evidence that confidence governs how strongly decisions respond to fundamentals (Enke et al., 2026), is whether this null result masks heterogeneity across subjects with different levels of confidence in their own strategy. Intuitively, subjects who are less certain about the optimality of their approach may be more susceptible to the influence of observed choices, regardless of their quality, whereas more confident subjects may be better able to evaluate and filter the information they receive. This appendix examines how baseline confidence moderates the treatment effects documented above.

Confidence elicitation. Confidence is elicited at six points during the experiment (rounds 10, 30, 50, 70, 90, and 110). In each elicitation round, after making a hiring choice but before observing the payoff, subjects are asked how many points out of 50 they wish to bet on the event that their current choice is optimal (Figure 32).[55] This betting task is incentive-compatible: if the subject's choice turns out not to be optimal, all bet points are lost and the final number of points equals the points not bet. If the choice is optimal, each bet point is multiplied by 1.5, and the final number of points equals the points not bet plus 1.5 times the bet points. The final point total is then converted into a probability of winning a \$2 bonus at a rate of one point per percentage point. Under risk neutrality, this rule makes the optimal bet a threshold report (zero points when the believed probability of an optimal choice is below 2/3 and all fifty points when it is above), so interior bets are not cardinally interpretable as beliefs; for risk-averse subjects the optimal bet instead rises smoothly with confidence. Accordingly, the analysis treats the 0–50 wager as an ordinal behavioral proxy for confidence rather than as an exact belief measure, which is how it enters the specifications below (as a standardized moderator).

[55] Subjects are informed that *optimal* means that the worker they hired would generate a payoff at least as high as that of the worker they did not hire. The bet is resolved using the realized abilities of both drawn workers: the choice counts as optimal if the hired worker's payoff equals or exceeds the payoff the other worker would have delivered. Because the benchmark action maximizes the probability of this event conditional on the observed signals, the outcome-based criterion used to resolve the bets and the ex-ante benchmark rule used throughout the paper coincide in expectation.

Figure 32: Confidence Elicitation Interface



Notes: Screenshot of the confidence elicitation interface. Subjects use a slider to indicate how many points out of 50 they wish to bet on the event that their current hiring choice is optimal. The bet is incentive-compatible as described in the text.

The baseline confidence measure, elicited in round 10 of Part 1 before any treatment assignment, has a mean of 25.1 and a standard deviation of 15.7 across the 356 subjects in the main sample. Regressing baseline confidence on treatment dummies confirms that randomization successfully balanced this measure across groups ($p = 0.622$ for Exposure, $p = 0.214$ for ExposureNeg). Throughout this appendix, confidence is standardized to have mean zero and unit variance in the estimation sample.

Confidence interactions. To study whether confidence moderates the learning effects from exposure, I estimate specifications that add the standardized confidence measure and its interactions with the treatment dummies. The results, reported in Table 10, cover three outcome measures: the share of optimal choices in the last ten rounds (columns 1–2), the optimal choice indicator in the last round (columns 3–4), and a binary indicator for whether a subject achieves at least 8 optimal choices out of 10 in the last ten key rounds (columns 5–6). Odd-numbered columns include confidence as a control; even-numbered columns add the interaction terms. Confidence is a small and insignificant predictor of optimal choices on its own (-0.007, $p = 0.782$ in column 1; -0.022, $p = 0.392$ in column 3; 0.005, $p = 0.853$ in column 5), and the treatment effect estimates are virtually unaffected by its inclusion.

The key result emerges in the even-numbered columns, which add the interaction terms. The interaction between Exposure and confidence is essentially zero across all three outcome measures (-0.002, $p = 0.970$ in column 2; 0.000, $p = 0.997$ in column 4; 0.024, $p = 0.720$ in column 6), indicating that the beneficial effect of observing optimal choices operates with similar strength regardless of the subject's baseline confidence level. In contrast, the interaction between ExposureNeg and confidence is large and statistically significant in every specification (0.172, SE $= 0.076$, $p = 0.024$ in column 2; 0.170, SE $= 0.083$, $p = 0.040$ in column 4; 0.188, SE $= 0.082$, $p = 0.023$ in column 6).

Table 10: The Role of Confidence in Part 2

	Optimal Choices in Last 10 Rounds		Optimal Choice in Last Round		Binary Optimality ($\geq$ 8 of 10)	
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure	0.129**	0.131**	0.212***	0.213***	0.126*	0.129**
	(0.056)	(0.056)	(0.064)	(0.064)	(0.065)	(0.065)
ExposureNeg	−0.005	−0.025	0.014	−0.005	−0.011	−0.029
	(0.079)	(0.075)	(0.090)	(0.087)	(0.088)	(0.085)
Confidence (std)	−0.007	−0.033	−0.022	−0.050	0.005	−0.041
	(0.024)	(0.052)	(0.026)	(0.058)	(0.026)	(0.057)
Exp × Confidence		−0.002		0.000		0.024
		(0.060)		(0.066)		(0.066)
ExpNeg × Confidence		0.172**		0.170**		0.188**
		(0.076)		(0.083)		(0.082)
NoExposure Mean	0.491	0.491	0.463	0.463	0.425	0.425
Observations	3,560	3,560	356	356	356	356
R^2	0.017	0.033	0.043	0.060	0.016	0.032

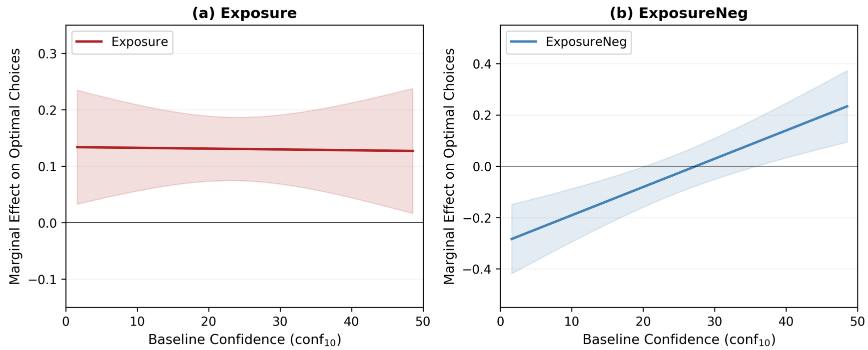
Notes: Each column reports a separate OLS regression of an optimality measure on treatment dummies, standardized baseline confidence, and their interactions. The sample includes NoExposure (N = 80), Exposure (N = 223), and ExposureNeg (N = 53). Columns (1)–(2) use a round-level optimal choice indicator in the last ten rounds of Part 2 with standard errors clustered at the subject level. Columns (3)–(4) use the optimal choice indicator in the last round of Part 2 with HC1 robust standard errors. Columns (5)–(6) use a binary indicator equal to one if a subject makes at least 8 optimal choices in the last 10 key rounds of Part 2, with HC1 robust standard errors. Odd-numbered columns control for confidence; even-numbered columns add the interaction terms. Confidence is the standardized baseline elicitation from round 10 of Part 1 (raw mean = 25.1, SD = 15.7, range 0–50). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

This implies that the effect of exposure to suboptimal choices varies substantially with baseline confidence: point estimates indicate adverse effects on the quality of own choices concentrated among less confident subjects, while the choices of more confident subjects appear unaffected. The consistency of this pattern across the intensive margin (columns 1–2), the last-round indicator (columns 3–4), and the extensive margin of the 80 percent optimality threshold (columns 5–6) confirms that the heterogeneity is not an artifact of any particular outcome measure.

Marginal treatment effects. To visualize how the treatment effects vary across the confidence distribution, Figure 33 plots the marginal effects of Exposure and ExposureNeg from column (2) of Table 10 as a function of baseline confidence. Because the interaction model is linear, each marginal effect is a linear function of confidence: for Exposure, the marginal effect at confidence level z equals $\hat{\beta}_{\text{Exp}} + \hat{\beta}_{\text{Exp}\times\text{Conf}} \cdot z$, with an analogous expression for ExposureNeg.

Panel (a) confirms that the Exposure effect is remarkably stable: the marginal effect hovers

Figure 33: Marginal Treatment Effects by Baseline Confidence



Notes: Each panel plots the estimated marginal treatment effect (relative to NoExposure) on the share of optimal choices in the last ten rounds of Part 2 as a function of baseline confidence. Panel (a) shows the Exposure effect; panel (b) shows the ExposureNeg effect. The marginal effects are computed from the interaction model in column (2) of Table 10: for each treatment T, the marginal effect at standardized confidence z equals $\hat{\beta}_T + \hat{\beta}_{T\times\text{Conf}} \cdot z$. The horizontal axis is expressed in raw confidence units (0–50). Shaded regions represent ± 1 standard error bands, computed from the variance-covariance matrix of the estimated coefficients.

around 0.13 throughout the confidence distribution, and the standard error band excludes zero across a wide range. Exposure to optimal choices benefits subjects of all confidence levels. Panel (b) reveals the stark heterogeneity in the ExposureNeg effect. At the bottom quartile of confidence (corresponding to a raw score of roughly 8 out of 50), the marginal effect of ExposureNeg is -0.214 ($p = 0.053$), a marginally significant estimate suggesting that low-confidence subjects exposed to suboptimal choices perform worse than those who learn independently. At the sample mean, the effect is indistinguishable from zero (-0.025, $p = 0.742$). At the top quartile of confidence (a raw score of roughly 40), the point estimate reverses sign to $+0.147$ ($p = 0.177$), suggesting that high-confidence subjects may even benefit from observing suboptimal behavior, though this positive estimate is imprecisely measured.

Splitting the sample into confidence quartiles and estimating separate regressions within each group provides a complementary, nonparametric view of this heterogeneity. In the bottom quartile ($N = 95$, confidence scores 0–10), the ExposureNeg coefficient is -0.146 ($p = 0.322$), while in the top quartile ($N = 70$, confidence scores 41–50), it is $+0.216$ ($p = 0.244$). Although neither estimate is individually significant given the reduced sample sizes, the sign reversal from the bottom to the top of the confidence distribution is consistent with the pattern identified by the interaction specification, though it should be read as descriptive corroboration rather than independent confirmation.[56] This pattern is consistent with a model in which less confident subjects are more

[56]Baseline confidence does not predict whether a subject will respond to exposure in general: within the Exposure group, mean confidence is virtually identical across compliers (24.95), always-takers (24.00), and never-takers (25.14), and a joint test of equality fails to reject the null ($p = 0.911$). Thus, the heterogeneity by confidence documented

willing to adopt observed strategies regardless of their quality, making them vulnerable when the observed choices are suboptimal, while more confident subjects evaluate the quality of what they observe and potentially recognize the inferiority of the suboptimal strategy, reinforcing their own approach.

Finally, while treatment assignment substantially affects behavior, it leaves no detectable imprint on subjects' stated confidence in their own strategy. Regressing later confidence elicitations (in rounds 50, 70, 90, and 110) on treatment dummies, controlling for baseline confidence, yields no significant treatment effects in any part of the experiment (all $p > 0.12$). These null results do not, absent formal equivalence bounds, rule out moderate effects; what can be said is that subjects who shift to optimal behavior under exposure do not report detectably greater certainty that their choices are correct, and subjects exposed to suboptimal behavior report no detectably lower confidence.

Additional Result (Confidence): *The beneficial effect of exposure to optimal choices does not vary with baseline confidence, whereas the response to suboptimal choices declines sharply with confidence, with point estimates indicating adverse effects concentrated among the least confident subjects; treatment assignment leaves no detectable imprint on subsequent confidence elicitations.*

here, which concerns susceptibility to the *quality* of observed choices, operates through a different channel than the learning type classification in Section 3.4, which concerns whether subjects respond to exposure at all.

G. Theoretical Framework

This appendix develops a conceptual framework for the central inference task in the experiment and for learning through exposure to another agent's optimal decisions. The first subsection formalizes the decision environment, including the signal structure and the Bayesian benchmark. The second subsection adapts an imitation learning model to the experimental setting. The third subsection distinguishes imitation from genuine belief updating.

G.1. Decision Setting

Consider a decision-maker who must choose between hiring a worker from group $g \in \{\text{Green}, \text{Orange}\}$. Each group consists of four workers with ability $a \in \{l, m, h\}$ drawn from group-specific distributions. Let k_a^g denote the proportion of workers with ability a in group g. Before making a hiring decision, the decision-maker observes a binary education signal for each worker, generated by the conditional probabilities $P(e \mid h) = 1$, $P(e \mid m) = p_m \in (0, 1)$, and $P(e \mid l) = 0$. This signal structure ensures that education is perfectly informative about the absence of high ability among the uneducated (since $P(e \mid l) = 0$ and $P(e \mid h) = 1$, an uneducated worker cannot be of high ability) and partially informative among the educated (since $P(e \mid m) = p_m > 0$, an educated worker may be of either high or medium ability).

Payoff structure. The decision-maker's payoff depends on the hired worker's education status and ability. In the experiment, hiring an educated worker yields \$18 if the worker is of high ability and \$13 if the worker is of medium ability; hiring an uneducated worker yields \$10 if the worker is of medium ability and \$5 if the worker is of low ability. The maximum possible payoff difference is therefore $\bar{\pi} = 18 - 5 = 13$.

Bayesian benchmark. When both workers are educated, low-ability workers are screened out ($P(e \mid l) = 0$), so the relevant comparison is the posterior probability of high ability conditional on being educated. By Bayes' rule:

$$P(h \mid e, g) = \frac{P(e \mid h) \cdot k_h^g}{P(e \mid h) \cdot k_h^g + P(e \mid m) \cdot k_m^g} = \frac{k_h^g}{k_h^g + p_m \cdot k_m^g}. \quad (1)$$

The optimal choice is to hire the worker from the group with the higher posterior $P(h \mid e, g)$, which reduces to comparing the ratio k_h^g / k_m^g across groups, since p_m is common.

Under the main primitives (Parts 1–3), the ability proportions are $(k_h^G, k_m^G, k_l^G) = (0.25, 0.75, 0)$ for the Green group and $(k_h^O, k_m^O, k_l^O) = (0.25, 0.50, 0.25)$ for the Orange group, with $p_m = 0.9$.

Applying equation (1):[57]

$$P(h \mid e, O) \approx 0.357 > 0.270 \approx P(h \mid e, G).$$

The posterior probability that the educated Orange worker is of high ability exceeds that of the educated Green worker, making hiring Orange optimal despite the Green group having higher average ability. The key insight is that education screening operates differently across groups: because the Orange group has a higher share of low-ability workers who are screened out, the surviving educated pool is of higher quality. A decision-maker who neglects the education signal and relies instead on unconditional group averages will suboptimally prefer the Green worker. The full derivation is in Section 2.

G.2. Learning Through Imitation

Setup. Consider an agent i who faces the binary choice problem over T rounds. In each round t, the agent selects an action $a_i^t \in \{G, O\}$ and receives a payoff π_i^t. Between rounds, agent i observes the action a_j^t chosen by another agent j who faces the same environment and uses a learning rule to determine her action in round $t+1$.

A learning rule is *improving* if, for any payoff distributions with $\mu_O > \mu_G$, the expected share of agents choosing the optimal action O weakly increases. Among all improving rules, the learning rule that maximizes the rate of convergence toward the optimal action is one in which the probability of switching to the observed action is proportional to the payoff advantage.[58] Under such a rule,

[57]For the main primitives: $P(h \mid e, G) = \frac{0.25}{0.25+0.9\times0.75} = \frac{0.25}{0.925} \approx 0.270$ and $P(h \mid e, O) = \frac{0.25}{0.25+0.9\times0.50} = \frac{0.25}{0.70} \approx 0.357$. For the new primitives (Parts 4–5), where $(k_h^G, k_m^G, k_l^G) = (0.25, 0.25, 0.50)$, $(k_h^O, k_m^O, k_l^O) = (0, 0.50, 0.50)$, and $p_m = 0.1$, the relevant case is both workers being uneducated. Then, since a medium-ability worker is uneducated with probability $1 - p_m = 0.9$, $P(m \mid \neg e, G) = \frac{(1-p_m)\cdot k_m^G}{(1-p_m)\cdot k_m^G + k_l^G} = \frac{0.9\times0.25}{0.9\times0.25+0.50} \approx 0.31$ and $P(m \mid \neg e, O) = \frac{0.9\times0.50}{0.9\times0.50+0.50} \approx 0.47$. In both cases, the Orange worker has a higher posterior probability of being the better type, making hiring Orange optimal.

[58]This is the Proportional Imitation Rule (PIR) introduced by Schlag (1998): $\Pr(a_i^{t+1} = a_j^t \mid a_j^t \neq a_i^t) = \max\{\pi_j^t - \pi_i^t, 0\}/\bar{\pi}$, where $\bar{\pi}$ is the maximum payoff difference. The PIR is optimal in a general class of environments where the agent does not know the underlying payoff distributions. In our experimental setting, the payoff structure is simpler: when both workers are educated, payoffs take one of two values ($18 for high ability, $13 for medium ability), and the payoff difference between a high-ability and a medium-ability hire is fixed at $5. The PIR reduces to a rule in which switching occurs with a fixed probability whenever the observed action yields a higher realized payoff. Since subjects observe participant X's action but not X's payoff, the rule cannot operate on round-level payoff comparisons; an implementable adaptation must substitute experience-based action values accumulated in Part 1, when virtually all subjects tried both actions repeatedly. This adaptation is a distinct learning rule rather than the literal PIR, and

the expected change in the share p^t choosing the optimal action follows:

$$\mathbb{E}[p^{t+1} - p^t] = \frac{\Delta\mu}{\bar{\pi}} \cdot p^t(1 - p^t), \tag{2}$$

where $\Delta\mu = \mu_O - \mu_G > 0$ is the expected payoff advantage and $\bar{\pi}$ is the maximum payoff difference. This is the discrete-time analogue of the replicator dynamic from evolutionary game theory. The term $p^t(1-p^t)$ reflects the fact that switching can only occur when the two agents choose different actions.

Adaptation to the experimental setting. In the experiment, agent j is participant X, who always chooses optimally ($a_j^t = O$ for all t). This simplifies the dynamics: agent i can only switch *toward* the optimal action, never away from it. The per-round switching probability for a suboptimal agent is:

$$\Pr(a_i^{t+1} = O \mid a_i^t = G) \;\equiv\; \lambda > 0, \tag{3}$$

and the probability that agent i remains suboptimal after T rounds of exposure decays geometrically:

$$q^T = q^0 \cdot (1 - \lambda)^T. \tag{4}$$

With $T = 40$ rounds in Part 2, even a modest per-round switching probability generates substantial cumulative learning. Because switching is probabilistic rather than deterministic, the model implies that subjects do not switch simultaneously; instead, the cumulative share of switchers rises gradually, approaching its limit at a geometrically decaying rate, consistent with the gradual divergence between Exposure and NoExposure documented in Section 3.2.

This framework also clarifies why exposure to suboptimal choices has no effect. Under the learning rule, an agent never switches to an action that yielded a weakly lower payoff. When agent j always chooses suboptimally (the ExposureNeg treatment), a subject currently choosing Green observes $a_j^t = G = a_i^t$ and no switching occurs; a subject currently choosing Orange observes $a_j^t = G \neq a_i^t$ but since Orange yields a higher expected payoff, the switching probability toward Green is zero. The asymmetric response to optimal versus suboptimal exposure documented in Section 3.3 is therefore an implication of the model under the adaptation above. Similarly, because switching operates only when $a_j^t \neq a_i^t$, subjects who already choose optimally in Part 1 (always-takers) are unaffected by the treatment, while the largest impact falls on subjects who start suboptimally and are exposed to a superior alternative (Section 3.4).

it does not by itself guarantee that subjects rank the observed action correctly; the PIR serves here as a theoretical benchmark for payoff-proportional imitation, not as a description of the implemented information structure.

The role of own-payoff feedback. A key feature of the experiment is that subjects observe only participant X's action, not participant X's payoff. However, subjects receive feedback on their own payoff after each round. This allows a modified learning rule in which the agent compares her realized payoff π_i^t with an inferred counterfactual payoff $\hat{\pi}_j^t$ for the action she did not take:

$$\Pr(a_i^{t+1} = O \mid a_i^t = G) = f\big(\pi_i^t,\ \hat{\pi}_j^t\big), \tag{5}$$

where f is increasing in $\hat{\pi}_j^t - \pi_i^t$. This implies that switching should be more likely in rounds where the subject's own payoff is low (e.g., hiring a medium-ability Green worker at \$13 rather than a high-ability one at \$18), a pattern consistent with the payoff-driven selective adoption documented in Section 3.3.

G.3. Imitation vs. Learning

The framework above describes *action imitation*: agents copy the observed action without necessarily understanding why it is optimal. Under pure imitation, a subject who switches to hiring the Orange worker in Part 2 would revert to suboptimal behavior once the environment changes (Part 4) or exposure is removed (Part 3), because the copied action may no longer be correct or the copying stimulus is absent.

A richer model allows for *belief updating*: through repeated exposure to optimal actions and the associated payoff feedback, agents may come to understand the underlying principle that education signals carry useful information about worker quality. Formally, let $\beta_i^t \in [0, 1]$ denote agent i's belief that the orange action is optimal. Under belief updating, each round of exposure shifts β_i^t upward:

$$\beta_i^{t+1} = \beta_i^t + \alpha \cdot \big(\mathbf{1}\{\text{evidence favoring } O \text{ in round } t\} - \beta_i^t\big), \tag{6}$$

where $\alpha \in (0, 1)$ is a learning rate. Under this specification, the probability of choosing optimally depends on the accumulated belief rather than on the presence of the external signal, so behavior persists even when exposure is removed.

Parts 3–5 of the experiment are designed to distinguish these channels. For most subjects, the evidence favors the action-based end of the spectrum: only about 40 percent of compliers continue to behave optimally under the new environment primitives in Part 4 (Section 4.2), and the group-level optimal choice rate becomes statistically indistinguishable from the control group's. For this minority of persisting subjects, behavior is consistent with belief updating: they have internalized the principle that the education signal carries useful information and apply it in a novel setting where the observed choice stream is no longer displayed and cannot simply be copied.

For the majority, however, gains acquired under exposure do not survive the change in primitives, consistent with adoption of the observed action rather than revision of the underlying model.

H. Experimental Instructions and Procedures

This section reproduces the complete experimental instructions and procedures as presented to participants in the Exposure treatment. It covers the overview of the task, the hiring decision interface, the payoff structure, the confidence elicitation mechanism, and the detailed instructions for each of the five parts of the experiment. Participants in the NoExposure treatment faced the exact same interface and instructions, except they did not observe the hiring choices of another participant in Part 2 and were not provided with that participant's written advice in Part 5.

Experiment Instructions

Welcome to the study!

This is an experiment in decision-making. Your final payment will be based on your performance and there will be an element of random chance. This will be paid to you privately at the end of the experiment. From now on, we ask that you do not communicate with each other. If you have a question, please raise your hand.

The experiment will consist of short, independent rounds. In each round, you will be asked to make one choice.

Next, we will review the main task description, followed by a series of comprehension questions. You need to answer these questions correctly to continue with the experiment. After that, you will proceed to the main part of the study, which will determine your final payoff.

On each page, please read all the information carefully, as you will **NOT** be able to return to any of the previous pages. During the task description part, please do not move to the next page until we have reviewed all the instructions on a given page and you are instructed to proceed. Do not exit the browser window at any point.

Before we proceed, please write down your desk number below:

Description

Overview

This study will consist of **120 independent short rounds**.

In each round, the interface will display two groups of workers: one with 4 Green workers and one with 4 Orange workers.

Each worker can have a **HIGH, MEDIUM or LOW ABILITY** level

- You will be able to see the number of workers at each ability level in both groups
- Rounds 1–80 will each randomly use one of the <u>same two distributions of abilities</u>
- Rounds 81–120 will use a <u>different distribution of abilities</u>

In each round, one worker from each group will be **RANDOMLY** selected — we will call these two workers **SELECTED WORKERS**

- You will NOT be able to see the ability level of these selected workers. You will instead see the **EDUCATION** status (Educated or Not educated) of the Green selected worker and the Orange selected worker

Regardless of the group that the worker belongs to, the following information will always be true:

- A high ability worker is educated with 100% chance (for sure)
- A medium ability worker is educated with 90% chance for rounds 1–80 and 10% chance for rounds 81–120
- A low ability worker is educated with 0% chance (for sure not)

After each round, the selected workers will be replaced back into their respective pools, so that any individual worker can be selected more than once across multiple rounds.

In rounds 21–60, you will also be provided with the **HIRING CHOICE OF ANOTHER STUDY PARTICIPANT** who took part in the exact same study in one of the past sessions, we will call this participant **PARTICIPANT X**

- In each round, participant X **saw the exact same information** as you will (distribution of abilities and education statuses of the selected workers)
- For example, if in round 30 you see that the abilities draw is Draw #1, the selected Green worker is educated, and the selected Orange worker is educated, it means that participant X in round 30 saw that the abilities draw is Draw #1, the selected Green worker is educated, and the selected Orange worker is educated
- All of the hiring choices you will see were made by **the same participant**

• In rounds 21–60, you will be provided with this hiring choice and the interface will, for instance, say "**Participant X chose to hire the Green worker**" or "**Participant X chose to hire the Orange worker**"

Below is an example of the interface that you will see in each round:

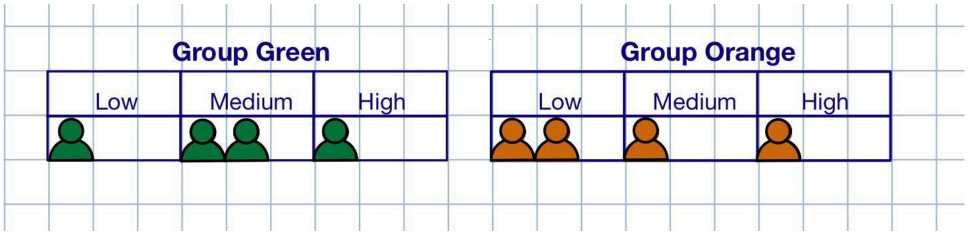


Figure: Example interface showing ability distributions for the two worker groups

In this example, among the workers in Green group:

• 1 has low ability, 2 have medium ability and 1 has high ability

In this example, among the workers in Orange group:

• 2 have low ability, 1 has medium ability and 1 has high ability

The interface will then **randomly** select one worker from the Green group and one worker from the Orange group — you will not see the ability levels of these selected workers

• You will instead see their education statuses

• For example, it could be that the Green worker is **Educated** and the Orange worker is **Not educated**

Your Task

In each round, your task is to hire one of the two selected workers. You will choose **one** of the following:

• Hire **GREEN** worker

• Hire **ORANGE** worker

After each round, you will learn the **ability of the worker that you hired.** You will then move on to the next round.

Let us know if you have any questions at this moment.

Your Payoff

Your final payoff will be determined based on your payoff from a **randomly selected round.**

In other words, after you complete all 120 rounds, the computer will randomly select one round and your final earnings in this experiment will be equal to your earnings in that randomly selected round.

Your earnings will be equal to:

Worker Type	Educated	Not Educated
High ability	$18	$11
Medium ability	$13	$10
Low ability	$12	$5

Therefore, you will get a higher payoff if you hire an Educated worker compared to a Not educated worker.

Also, within the same education status you will get a higher payoff if you hire a worker of Higher ability when compared to a worker of Lower ability.

You will be reminded of the payoff information on each decision page.

In the previous example, suppose that a randomly chosen worker from the Green group is Educated and of Medium ability, while a randomly chosen worker from the Orange group is Not educated and of Medium ability.

Then, if you hire a Green worker you will get $13 and if you hire an Orange worker you will get $10.

Bonus

Six times during the experiment, in rounds 10, 30, 50, 70, 90 and 110, you will make choices which can lead to a $2 bonus in addition to the payoff described earlier.

To determine whether you will win the bonus or not, we will randomly choose one out of your six bonus choices and the outcome of only that one choice will count.

Here is the description of the additional task you will face:

- You will be given a budget of 50 points
- You will use the slider to decide **how many points out of 50 to bet that your decision in a given round is optimal**
- "Optimal" here means that the worker you hired will bring you a higher or the same payoff as the worker you did not hire

- Every point you don't bet you get to keep.

Based on whether your decision in that round was actually optimal or not, we will determine your bonus as follows:

- If your decision was not optimal, i.e. if the worker you **did not hire** is associated with a higher payoff, every point you bet will be <u>lost</u> and the final amount of points will be:

Final amount of points = Points you did not bet

- If your decision was optimal, i.e. if the worker you **hired** is associated with a higher payoff, every point you bet will be <u>multiplied by a factor of 1.5</u> and the final amount of points will be:

Final amount of points = Points you did not bet + 1.5 × Points you bet

All points will then be transformed to a percent chance to win a $2 bonus at the rate 1 point = 1 percent.

Example 1:

Suppose you chose to bet all 50 points. Then:

- If your decision this round was not optimal, you will have 0 points as a final amount, and 0% chance of winning a bonus
- If your decision this round was optimal, you will have 0 + 1.5×50 = 75 points as a final amount, and 75% chance of winning a bonus

Example 2:

Suppose you chose to bet 20 points. Then:

- If your decision this round was not optimal, you will have 30 points as a final amount, and 30% chance of winning a bonus
- If your decision this round was optimal, you will have 30 + 1.5×20 = 60 points as a final amount, and 60% chance of winning a bonus

Example 3:

Suppose you chose to bet 0 points. Then:

- Regardless of whether your decision this round was optimal or not, you will have 50 points as a final amount, and 50% chance of winning a bonus

Therefore, if you are certain that your decision in a given round is optimal, then betting more points is associated with a higher chance of winning a bonus.

And if you are not certain that your decision in a given round is optimal, then betting less points is associated with a higher chance of winning a bonus.

Summary

The study consists of 120 rounds.

In each round, the interface will show one group of 4 Green workers and one group of 4 Orange workers.

- Each of the 8 workers can be of **high, medium, or low ability**, and can be **educated or Not educated**
- Rounds 1–80 will each randomly use one of the same two distributions of abilities • Rounds 81–120 will use a different distribution of abilities
- In rounds 21–60, you will also be provided with the **hiring choice of another study participant** (participant X) from one of the past study sessions

One worker from each group will be **randomly** selected, and their education status will be shown:

- A high ability worker has a 100% probability of being educated
- A medium ability worker has a 90% probability of being educated for rounds 1–80 and a 10% probability of being educated for rounds 81–120
- A low ability worker has a 0% probability of being educated

You will **choose** to either hire the selected worker from the Green group or hire the selected worker from the Orange group.

Your **payoff** will depend on the Ability level and Education status of the worker you hired.

At the end of each round, you will learn the **ability of the worker you hired.**

After each round, the selected workers will be replaced back into their respective pools, so that any individual worker can be selected more than once across multiple rounds.

In six different rounds you will also bet on whether your choice in a given round is a choice that maximizes your payoff.

We will pay you based on the hiring outcome in **one randomly selected round**.

After you finish, please do not leave as we will wait for everyone to complete the experiment.

Comprehension

Comprehension Questions

Next, you will see 6 comprehension questions, which you need to answer **correctly** in order to proceed to the main part. Similar to the main part of the study, when answering these questions you will be provided with the ability distributions for the Green and Orange groups.

Q1. In the example above, how many workers in the Green group are of high ability?

A. 2
B. 0
C. 4
D. 1

Q2. In the example above, how many workers in the Orange group are of low ability?

A. 3
B. 2
C. 4
D. 0

Q3. From which group(s) will the two randomly selected workers come from?

A. **Both workers from the Green group**
B. **Both workers from the Orange group**
C. **One worker from each group**
D. **All of the above are possible**

Q4. What information will you be provided with about the randomly selected worker from each group?

A. **Both their education status and ability level**
B. **Only their ability level**
C. **Only their education status**
D. **Neither their education status nor ability level**

Q5. Which choice out of four presented below will result in the highest payoff?

A. **Hiring an educated worker of medium ability**
B. **Hiring a not educated worker of high ability**
C. **Hiring a not educated worker of medium ability**
D. **Hiring an educated worker of low ability**

Q6. Suppose you are certain that you made the optimal hiring decision in round 50. Which of the following options will result in the highest chance of receiving the $2 bonus?

A. **Betting 35 out of 50 points**
B. **Betting 25 out of 50 points**
C. **Betting 45 out of 50 points**
D. **Betting 5 out of 50 points**

Pause

Now, you will begin the main part of the study, which includes 120 independent rounds of the task with which you just became familiar.

In each round, please make your hiring choice when instructed to do so and click ‘Next’ to go to the next page. DO NOT hit ‘Back’ or exit the full screen interface. If you have a question at any point, raise your hand.

By the end of 120 rounds, you will answer a few questions before learning about your earnings.

Remember that once you are done, you should wait for others to finish before receiving your payment and leaving.

Go to the next page and begin whenever you are ready.

Tasks — Part 1

```
Click next to begin Round 1
```

In Rounds 21–60 you will additionally see which worker Participant X chose to hire.

In each round, participant X saw the exact same information as you will (distribution of abilities and education statuses of the selected workers)

- For example, if in round 30 you see that the selected Green worker is educated and the selected Orange worker is educated, it means that participant X in round 30 saw that the selected Green worker is educated and the selected Orange worker is educated

Example: Round 1 (Individual)

Distribution of abilities based on Draw 1.

The randomly drawn worker from the Green group is **Educated**
The randomly drawn worker from the Orange group is **Educated**

Which worker do you want to hire?

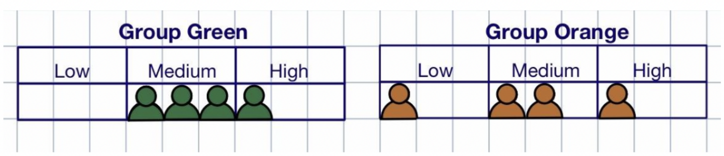
Hire GREEN worker ◯

Hire ORANGE worker ◯

Figure: Decision screen without Participant X (Rounds 1–20)

Example: Round 21 (Social Learning)

Round 22.



Distribution of abilities based on Draw 1.

The randomly drawn worker from the Green group is **Educated**
The randomly drawn worker from the Orange group is **Educated**

In this round, Participant X chose to hire the **Orange** worker.

Which worker do you want to hire?

Hire GREEN worker ◯

Hire ORANGE worker ◯

Figure: Decision screen with Participant X's choice (Rounds 21–60)

Reminder

Reminder:

Regardless of the group that the worker belongs to:

- A high ability worker is educated with 100% chance (for sure)
- **A medium ability worker is educated with 90% chance**
- A low ability worker is educated with 0% chance (for sure not)

Your earnings will be equal to:

- $18 if you hire an Educated and High-ability worker
- $13 if you hire an Educated and Medium-ability worker
- $12 if you hire an Educated and Low-ability worker
- $11 if you hire a Not educated and High-ability worker
- $10 if you hire a Not educated and Medium-ability worker
- $5 if you hire a Not educated and Low-ability worker

Figure: Reminder panel shown on each decision page

Feedback

In this round, you chose to hire a Green Educated worker.
The worker you hired is of **MEDIUM** ability.

If this round is chosen for payment, your earnings would be $13.

Figure: Example feedback after a hiring decision

Betting Interface

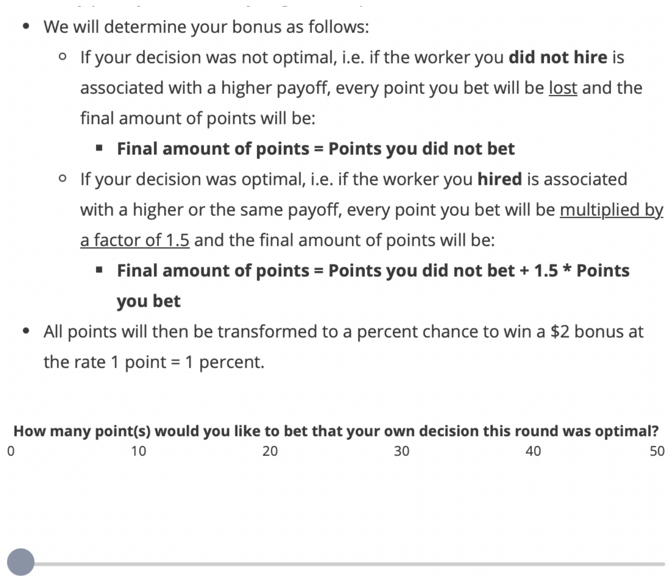


Figure: Confidence elicitation (betting) interface

Pause

You have completed rounds 1–80. The final 40 rounds will use a different distribution of abilities than rounds 1–80.

In addition, medium ability workers will now be educated with 10% chance instead of 90% chance.

Go to the next page and begin whenever you are ready.

Tasks — Part 2

For the remaining rounds of the study, you will also see the hiring recommendations with explanations that participant X wrote for you.

In each round, participant X saw the exact same information as you will (distribution of abilities and education statuses of the selected workers)

- For example, if in round 110 you see that the abilities draw is Draw #3, the selected Green worker is educated, and the selected Orange worker is educated, it means that participant X in round 110 saw that the abilities draw is Draw #3, the selected Green worker is educated, and the selected Orange worker is educated

```
Click next to begin Round 101
```

Participant X’s Advice

Scenario 1. Green is Educated and Orange is Educated

Recommendation: Hire **GREEN** worker

Explanation: Both guarantee at least a medium skill worker, but green has a high chance (10/11) of being high skilled.

Scenario 2. Green is Educated and Orange is Not educated

Recommendation: Hire **GREEN** worker

Explanation: Educated has higher payoff than non-educated and guarantees at least medium skill worker with a high chance (10/11) of being high skilled.

Scenario 3. Green is Not educated and Orange is Educated

Recommendation: Hire **ORANGE** worker

Explanation: Educated has higher payoff than non-educated and guarantees medium skill worker. Since green is not educated, it cannot be the high skill worker, so this choice is strictly better.

Scenario 4. Green is Not educated and Orange is Not educated

Recommendation: Hire **ORANGE** worker

Explanation: Both are uneducated, so the worker cannot be high skilled. There are 2 medium workers in orange as opposed to only 1 in green, so orange has a 50% chance of being medium while green only has a 33% chance.

Pause

You have completed the main part of the study. Click next to continue.

Reflection

Reflection on Your Experience

As we conclude this experiment, we would greatly appreciate your insights into how you approached the task at hand. Your responses will be invaluable in enhancing our understanding of decision-making processes.

1. Strategy Reflection:

- Could you please describe the strategy you employed in making your hiring decision in the study?
- Did your approach evolve or change over the course of the 120 rounds? If so, how?

2. Use of Participant X:

- How did you use participant X's hiring choices in Rounds 21–60 when making your own hiring choices?
- How did you use participant X's recommendations and explanations in Rounds 101–120 when making your own hiring choices?

Your candid and thoughtful reflections on these aspects of the experiment are crucial for our study. We appreciate the time and effort you have invested in participating in this research. Note that your response in this question does not affect your final payment.

Final Payoff

Before learning your final payoff, please answer the demographic questions below.

What is your age?

What is your gender identity?

■ Female ■ Male ■ Non-binary ■ Prefer not to say

What is your major? If you have multiple majors, please choose one.

■ Economics ■ Business Economics ■ Joint Mathematics-Economics ■ Other (specify below)

Click next to view your final payoff

Final Payoff

The round that was randomly chosen for the main payment is Round 47

- You chose to hire the Green worker in Round 47, who turned out to be of Medium ability
- Your payoff in Round 47 was $13, so your total earnings in the main part of the study are $13

The round that was randomly chosen for bonus payment is Round 50

- Your bet on your own choice was 34 points
- You chose to hire the Orange worker, who turned out to be of High ability
- Your earnings in Round 50 were $18
- You did not choose to hire the Green worker, who turned out to be of Low ability
- Your earnings in Round 50 would have been $5 were you to choose the Green worker
- Therefore, the final amount of points you have earned from betting is 67, so you have 67 per cent chance of winning a bonus
- As an outcome of the lottery with 67 per cent chance of win, your final bonus in the study is $2

This is the end of the experiment, please remain seated and wait for other participants in the session to finish.

In the meantime, please record the amount you earned on the paper form on your desk — take that form with you as you exit the lab, and then return it back to the experimenter once you have received the payment.

Your earnings from the main part: $13

Your earnings from the bonus: $2

Response record code (for experimenter use only, do not input anything here):

If you have any questions, feel free to email abatmanov@ucsd.edu.

This is the end of the experiment, please remain seated and do NOT leave your desk!

Thank you for participating!